

8

7

6

5

4

3

2

1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD
DATE

<REV>

<ECN>

<ECO_DESCRIPTION>

<ECODATE>

SCHEM, WHITE_ARROW, MLB, K18

02/01/10

Page

(.csa)

Contents

Sync

Date

1

1

Table of Contents

MASTER

MASTER

2

2

System Block Diagram

K17_REF

06/30/2009

3

3

Power Block Diagram

K17_REF

06/30/2009

4

4

Revision History

MASTER

MASTER

5

5

BOM Configuration

K17_REF

05/28/2009

6

7

Functional / ICT Test

MASTER

MASTER

7

8

Power Aliases

MASTER

MASTER

8

9

Signal Aliases

K17_REF

06/11/2009

9

10

CPU DMI/PEG/FDI/RSVD

K17_REF

06/15/2009

10

11

CPU Clock/Misc/JTAG

K17_REF

06/15/2009

11

12

CPU DDR3 Interfaces

K17_REF

06/15/2009

12

13

CPU Power (1 of 2)

K17_REF

06/15/2009

13

14

CPU Power (2 of 2)

K17_REF

06/15/2009

14

15

CPU Grounds

K17_REF

06/15/2009

15

16

CPU Non-GFX Decoupling (1 of 2)

K17_REF

06/15/2009

16

17

CPU Non-GFX Decoupling (2 of 2)

K17_REF

08/24/2009

17

18

PCH SATA/PCIE/CLK/LPC/SPI

K17_REF

06/15/2009

18

19

PCH DMI/FDI/Graphics

K17_REF

10/07/2009

19

20

PCH PCI/FlashCache/USB

K18_MLB

06/15/2009

20

21

PCH MISC

K17_REF

06/15/2009

21

22

PCH Power

K17_REF

06/15/2009

22

23

PCH Grounds

K17_REF

06/15/2009

23

24

PCH Non-GFX Decoupling

K17_REF

06/15/2009

24

25

CPU/PCH GFX Decoupling

K17_REF

06/15/2009

25

26

eXtended Debug Port (XDP)

K17_REF

06/23/2009

26

27

Clock (CK505)

K17_MLB

06/15/2009

27

28

Chipset Support

K17_REF

06/15/2009

28

29

DDR3 SO-DIMM Connector A

MASTER

MASTER

29

30

DDR3 Byte/Bit Swaps

MASTER

MASTER

30

31

DDR3 SO-DIMM Connector B

MASTER

MASTER

31

32

CPU Memory S3 Support

K17_REF

06/15/2009

32

33

FSB/DDR3/FRAMEBUF Vref Margining

K17_REF

06/15/2009

33

34

X16/ALS/CAMERA CONNECTOR

K18_COMMS

08/26/2009

34

35

SecureDigital Card Reader

T27_REF

10/07/2009

35

36

USB HUB 1

K18_MLB

10/06/2009

36

37

USB HUB 2

K23F

08/20/2009

37

39

Ethernet PHY (Caesar II/IV)

T27_REF

06/15/2009

38

40

Ethernet Connector

K17_REF

05/29/2009

39

41

FireWire LLC/PHY (FW643)

K19_MLB

05/29/2009

40

42

FireWire Port Power

K19_MLB

05/29/2009

41

43

FireWire Ports

K19_MLB

10/01/2009

42

45

SATA Connectors

T27_REF

06/15/2009

43

46

External USB Connectors

K17_REF

05/29/2009

44

48

Front Flex Support

K19_MLB

06/15/2009

45

49

SMC

K17_REF

Page

(.csa)

Contents

Sync

Date

46

50

SMC Support

K18_SENSORS

06/29/2009

47

51

LPC+SPI Debug Connector

K17_MLB

06/23/2009

48

52

K18 SMBus Connections

K18_SENSORS

06/18/2009

49

53

Current & Voltage Sensing

K18_SENSORS

06/29/2009

50

54

Current Sensing

K18_SENSORS

07/02/2009

51

55

Thermal Sensors

K18_SENSORS

06/18/2009

52

56

Fan Connectors

K19_MLB

05/29/2009

53

57

WELLSPRING 1

K19_MLB

05/29/2009

54

58

WELLSPRING 2

K19_MLB

05/29/2009

55

59

Sudden Motion Sensor (SMS)

K19_MLB

05/29/2009

56

60

DEBUG SENSORS AND ADC

K18_SENSORS

07/07/2009

57

61

SPI ROM

K17_REF

06/15/2009

58

62

AUDIO: CODEC/REGULATOR

K18_AUDIO

09/21/2009

59

63

AUDIO: LINE INPUT FILTER

K18_AUDIO

07/29/2009

60

65

AUDIO: HEADPHONE FILTER

K18_AUDIO

07/29/2009

61

66

AUDIO: SPEAKER AMP

K18_AUDIO

07/29/2009

62

67

AUDIO: JACKS

K18_AUDIO

07/29/2009

63

68

AUDIO: JACK TRANSLATORS

K18_AUDIO

07/29/2009

64

69

DC-In & Battery Connectors

K18_POWER

06/30/2009

65

70

PBus Supply & Battery Charger

K18_POWER

06/30/2009

66

72

5V / 3.3V Power Supply

K18_POWER

07/13/2009

67

73

1.5V DDR3 Supply

K18_POWER

07/14/2009

68

74

CPU IMVP VCore Regulator

K18_POWER

06/29/2009

69

75

GFX IMVP VCore Regulator

K18_POWER

07/08/2009

70

76

CPUVTT (1.05V) Power Supply

K18_POWER

07/14/2009

71

77

Misc Power Supplies

K18_POWER

06/29/2009

72

78

Power FETs

K18_POWER

06/10/2009

73

79

Power Control

K17_REF

06/15/2009

74

80

NV GT216 PCI-E

K17_REF

06/15/2009

75

81

NV GT216 CORE/FB POWER

K17_REF

06/15/2009

76

82

NV GT216 FRAME BUFFER I/F

K17_REF

06/15/2009

77

84

GDDR3 Frame Buffer A (Top)

K17_REF

06/15/2009

78

85

GDDR3 Frame Buffer B (Top)

K17_REF

06/15/2009

79

86

NV GT216 GPIO/MIO/MISC

K17_REF

06/15/2009

80

87

GT216 GPIOs & STRAPS

K17_REF

06/15/2009

81

88

NV GT216 VIDEO INTERFACES

K17_REF

06/15/2009

82

89

GPU (GT216) CORE SUPPLY

K18_POWER

07/14/2009

83

90

LVDS Display Connector

K19_MLB

05/29/2009

84

93

Muxed Graphics Support

K17_REF

06/15/2009

85

94

DisplayPort Connector

K17_REF

06/15/2009

86

95

1V8 / 1V55 FB Power Supply

K18_POWER

06/26/2009

87

96

Graphics MUX (GMUX)

K17_REF

06/15/2009

88

97

LCD BACKLIGHT DRIVER

K18_BKLT

07/29/2009

89

98

LCD Backlight Support

K19_MLB

05/29/2009

90

99

Misc Power Supplies

K18_POWER

06/10/2009

Page

(.csa)

Contents

Sync

Date

91

100

CPU Constraints

K17_REF

06/15/2009

92

101

Memory Constraints

K17_REF

06/15/2009

93

102

PCH Constraints 1

K17_REF

06/15/2009

94

103

PCH Constraints 2

K17_REF

06/15/2009

95

104

Ethernet Constraints

K17_REF

06/15/2009

96

105

FireWire Constraints

K17_REF

06/15/2009

97

106

SMC Constraints

K17_REF

06/15/2009

98

107

GPU (GT216) CONSTRAINTS

K17_REF

06/15/2009

99

108

Project Specific Constraints

K17_REF

06/15/2009

100

109

PCB Rule Definitions

K17_REF

06/15/2009

101

132

BluRay Decrypter Card Connector

K17_REF

06/15/2009

ALIASES

RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8504	1	SCHEM, WHITE_ARROW, MLB, K18	SCH	CRITICAL	
820-2850	1	PCBF, WHITE_ARROW, MLB, K18	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED=Mon Feb 1 10:13:48 2010

SCHEM, WHITE_ARROW, MLB, K18

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

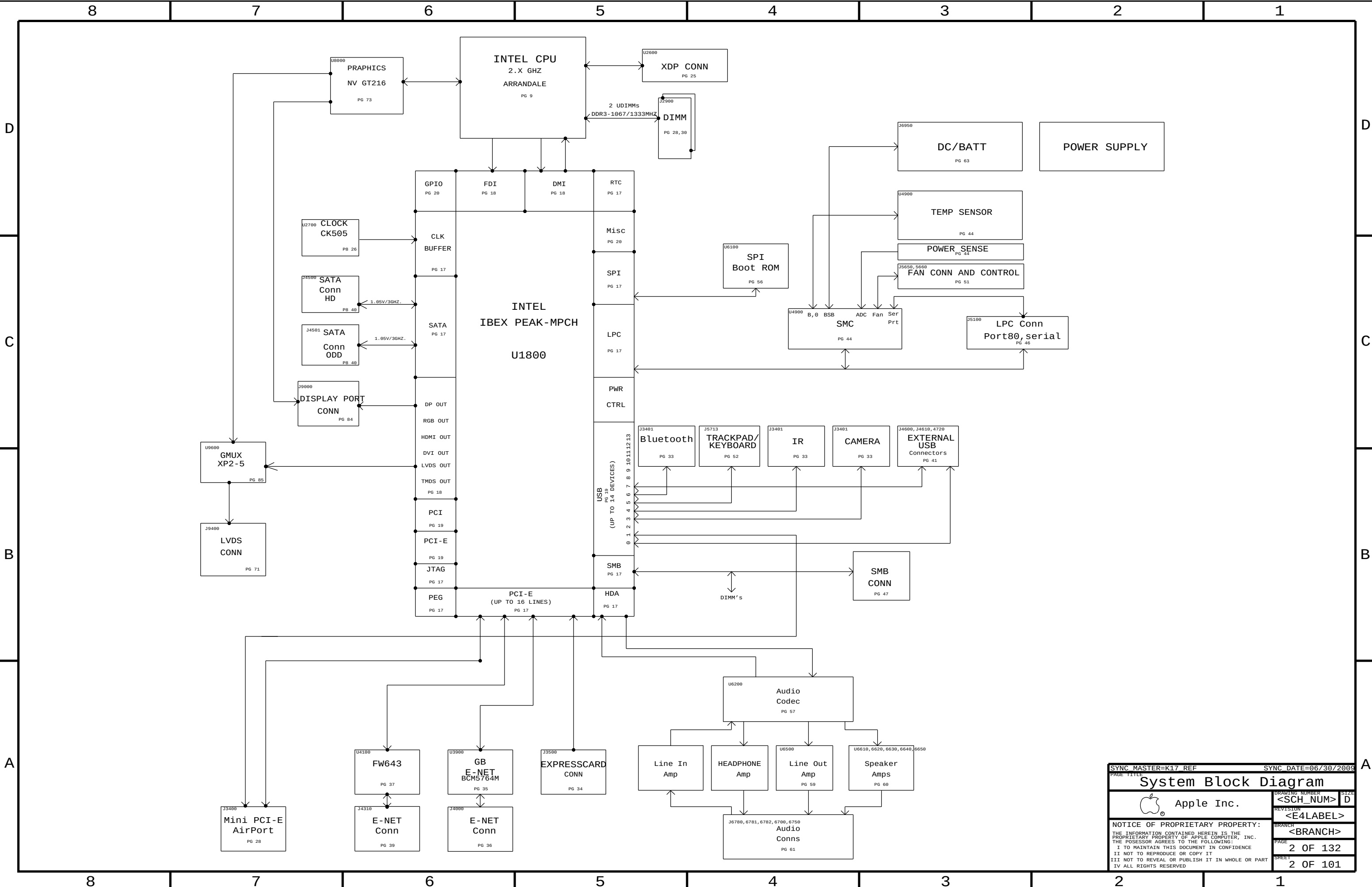
DRAWING NUMBER
<SCH_NUM>

REVISION
<E4LABEL>

BRANCH
<BRANCH>

PAGE
1 OF 132

SHEET
1 OF 101



SYNC MASTER=K17 REF

SYNC DATE=06/30/2009

System Block Diagram

 Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

<SCH_NUM>

REVISION

<E4LABEL>

BRANCH

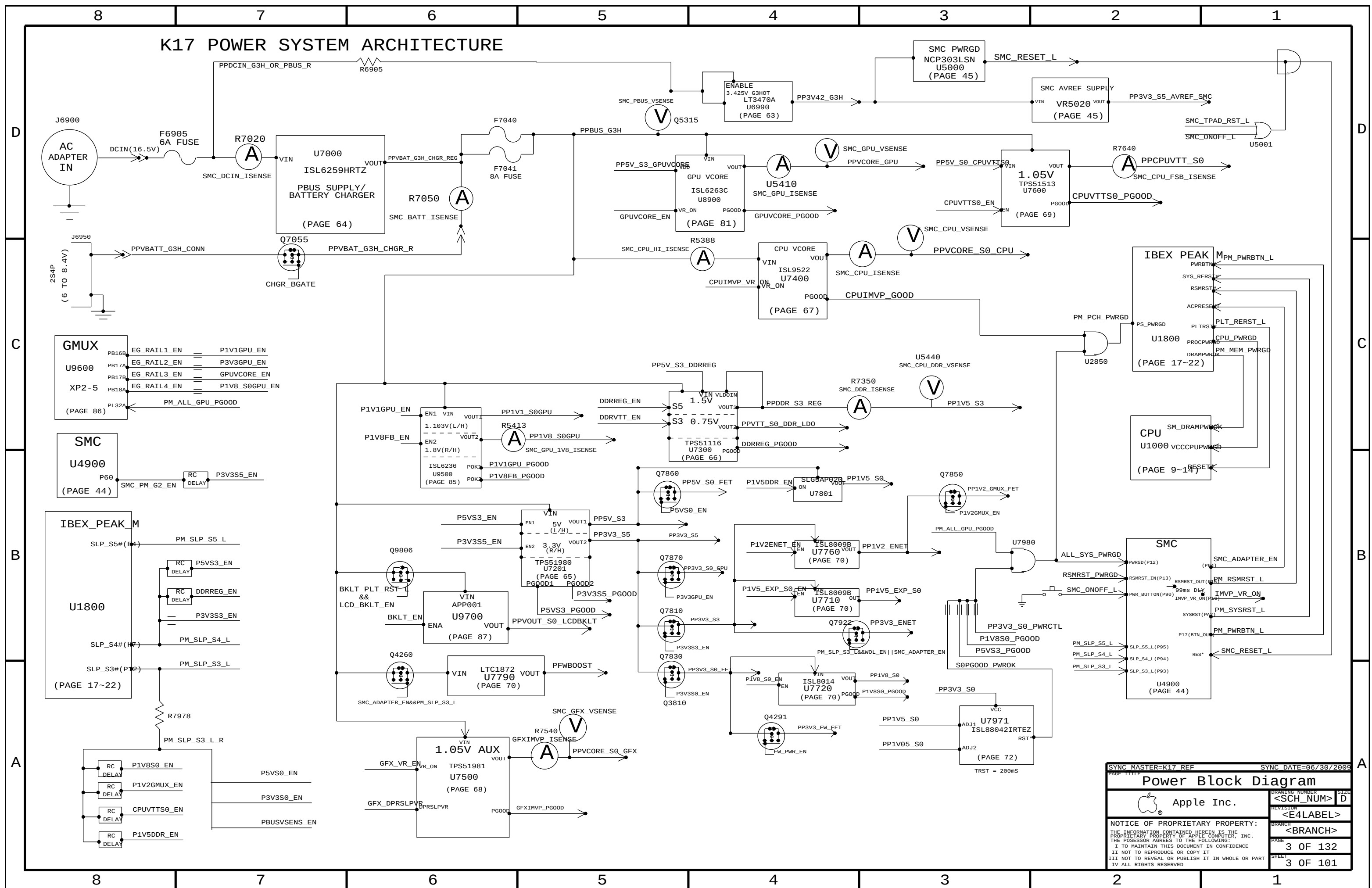
<BRANCH>

PAGE

2 OF 132

SHEET

2 OF 101



8

7

6

5

4

3

2

1

PROTO:

D

D |

C

C

B

B

A

A

SYNC_MASTER=MASTER		SYNC_DATE=MASTER	
PAGE TITLE			
Revision History			
 Apple Inc.	DRAWING NUMBER	SIZE	
	<SCH_NUM>		D
	REVISION	<E4LABEL>	
	BRANCH	<BRANCH>	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE	4 OF 132		
SHEET	4 OF 101		

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-0952	PCBA, 2.0G, 512SAM_VRAM, K18	K18_COMMON, CPU_2_4GHZ, FB_256_SAMSUNG, K18_PVT, EEEE_DCJ7
639-0953	PCBA, 2.0G, 512HYN_VRAM, K18	K18_COMMON, CPU_2_4GHZ, FB_256_HYNIX, K18_PVT, EEEE_DCJ8
639-0954	PCBA, 2.13G, 512SAM_VRAM, K18	K18_COMMON, CPU_2_53GHZ, FB_512_SAMSUNG, K18_PVT, EEEE_DCJ9
639-0955	PCBA, 2.13G, 512HYN_VRAM, K18	K18_COMMON, CPU_2_53GHZ, FB_512_HYNIX, K18_PVT, EEEE_DCJC
639-0956	PCBA, 2.4G, 512SAM_VRAM, K18	K18_COMMON, CPU_2_66GHZ, FB_512_SAMSUNG, K18_PVT, EEEE_DCJD
639-0957	PCBA, 2.4G, 512HYN_VRAM, K18	K18_COMMON, CPU_2_66GHZ, FB_512_HYNIX, K18_PVT, EEEE_DCJF
085-1404	K18 DEVELOPMENT BOM	

K18 BOM GROUPS

BOM GROUP	BOM OPTIONS
K18_COMMON	ALTERNATE, COMMON, K18_COMMON1, K18_COMMON2, K18_PROGPARTS, USBHUB_2061, RDRV:8515A2, DCI
K18_COMMON1	BATT_3S, BCM5764M, GL137, CPUPOC_IMAX_40_50, CPUMEM_S0, SMC_EXCARD_NOT, SMC_DEBUG_YES, HUB1_2NONREM, HUB2_3NONREM
K18_COMMON2	GMUXPLL_3V3, GPU_SS_INT, MIKEY, GPUVID_0P90V, DPMUX_EN_PLD, DP_CA_DET_EG_PLD, DP_ESD, VFRQ_SLPS3, SMC_OSC_YES, RAIL_MON
K18_PVT	BMON_PROD, VREFMRGN_NOT, XDP, XDP_NORMAL, XDP_CPU_BPM
K18_PROGPARTS	GMUX_PROG, BOOTROM_PROG, SMC_PROG, TPAD_PROG

BOM GROUP	BOM OPTIONS
FB_256_SAMSUNG	VRAM4, VRAM_256_SAMSUNG, FB1V55
FB_256_HYNIX	VRAM4, VRAM_256_HYNIX, FB1V55
FB_512_SAMSUNG	VRAM4, VRAM_512_SAMSUNG, FB1V35
FB_512_HYNIX	VRAM4, VRAM_512_HYNIX, FB1V35

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJ7]	CRITICAL	EEEE_DCJ7
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJ8]	CRITICAL	EEEE_DCJ8
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJ9]	CRITICAL	EEEE_DCJ9
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJC]	CRITICAL	EEEE_DCJC
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJD]	CRITICAL	EEEE_DCJD
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DCJF]	CRITICAL	EEEE_DCJF

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3848	1	ARD, SLBPE, PRQ, 2.66G, 35W, C2, 3M, BGA	U1000	CRITICAL	CPU_2_66GHZ
337S3847	1	ARD, SLBPF, PRQ, 2.53G, 35W, C2, 3M, BGA	U1000	CRITICAL	CPU_2_53GHZ
337S3846	1	ARD, SLBNA, PRQ, 2.4G, 35W, C2, 4M, BGA	U1000	CRITICAL	CPU_2_4GHZ
337S3849	1	IC, PCH, IBEX PEAK-M, SLGZS, PRQ, B3, BGA	U1800	CRITICAL	
337S3839	1	IC, GPU, NV GT126 LP++, 969BGA, 40NM, A03	U8000	CRITICAL	
343S0493	1	IC, ASIC, BCM5764M, ENET CONTROLLER, 8x8, 64 QFN	U3900	CRITICAL	BCM5764M
341S2731	1	IC, 1MBIT, SPI FLASH, K17/K18	U3990	CRITICAL	
338S0753	1	IC, FWB43-E2, 1394B PHY/OHCI LINK/PCI-E, 12	U4100	CRITICAL	
338S0563	1	IC, SMC, HS8/2117, 9MMX9MM, TLP	U4900	CRITICAL	SMC_BLANK
341T0233	1	IC, SMC, K18	U4900	CRITICAL	SMC_PROG
335S0610	1	IC, FLASH, SPI, 32MBIT, 3.3V, 86MHZ, 8-SOP	U6100	CRITICAL	BOOTROM_BLANK
341S2562	1	IC, EFI ROM, DEVELOPMENT, K18	U6100	CRITICAL	BOOTROM_PROG
341S2384	1	IR, ENCORE II, CY76C3833-LFXC	U4800	CRITICAL	
341S2616	1	IC, PSOC +W/USB, 56PIN, MLF, K18	U5701	CRITICAL	TPAD_PROG
336S0025	1	IC, XP2-5, HF, CPLD, BLANK	U9600	CRITICAL	GMUX_5K_BLANK
341S2566	1	IC, CPLD, LATTICE, 132CSBGA, K18	U9600	CRITICAL	GMUX_PROG
333S0507	4	IC, SGRAM, GDDR3, 16MX32, 1000MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_256_SAMSUNG
333S0483	4	IC, SDRAM, GDDR3, 16MX32, 900MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_256_HYNIX
333S0533	4	IC, SGRAM, GDDR3, 32MX32, 1000MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_SAMSUNG
333S0535	4	IC, SDRAM, GDDR3, 32MX32, 1000MHZ, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_HYNIX

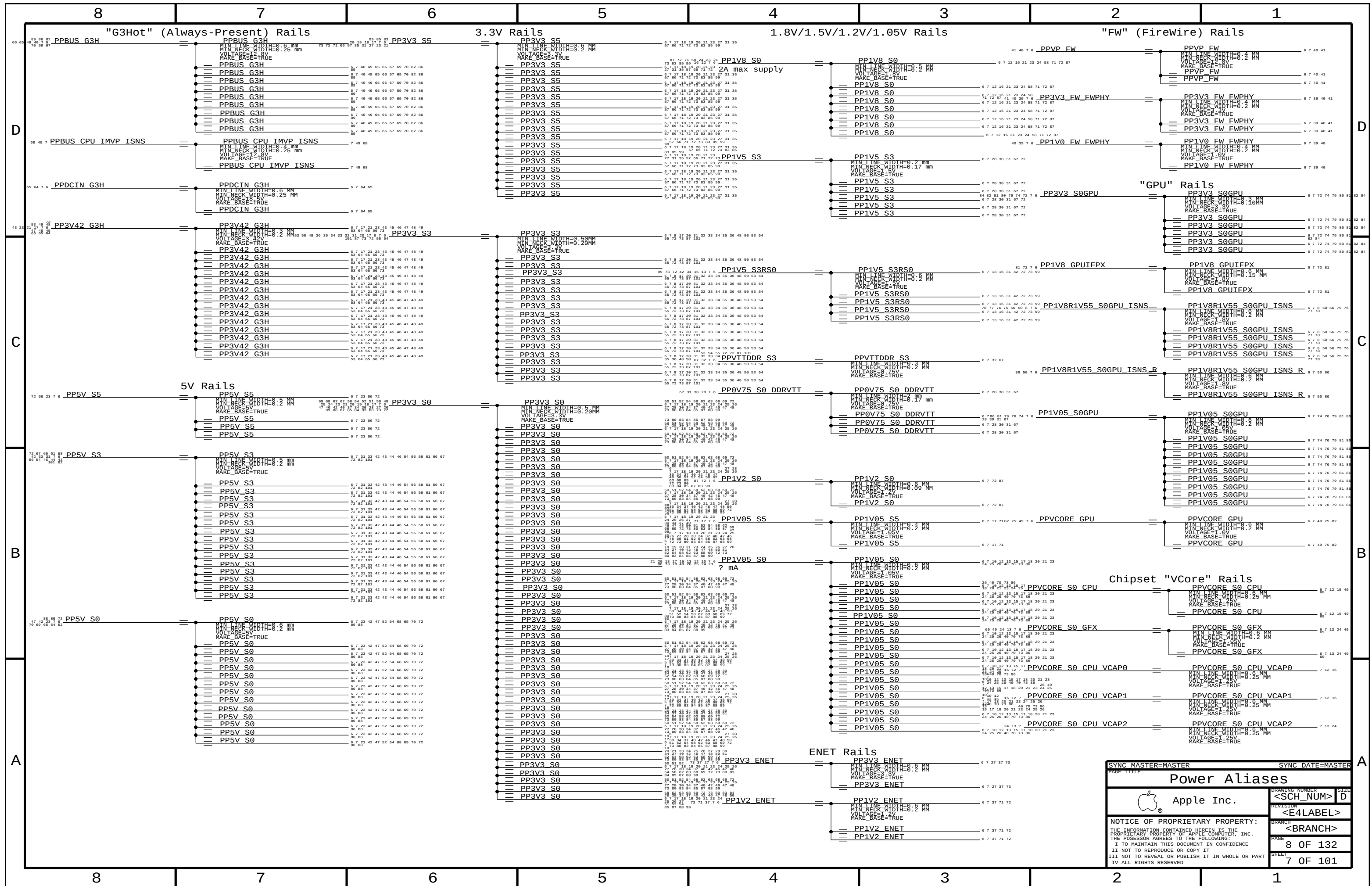
Development BOM

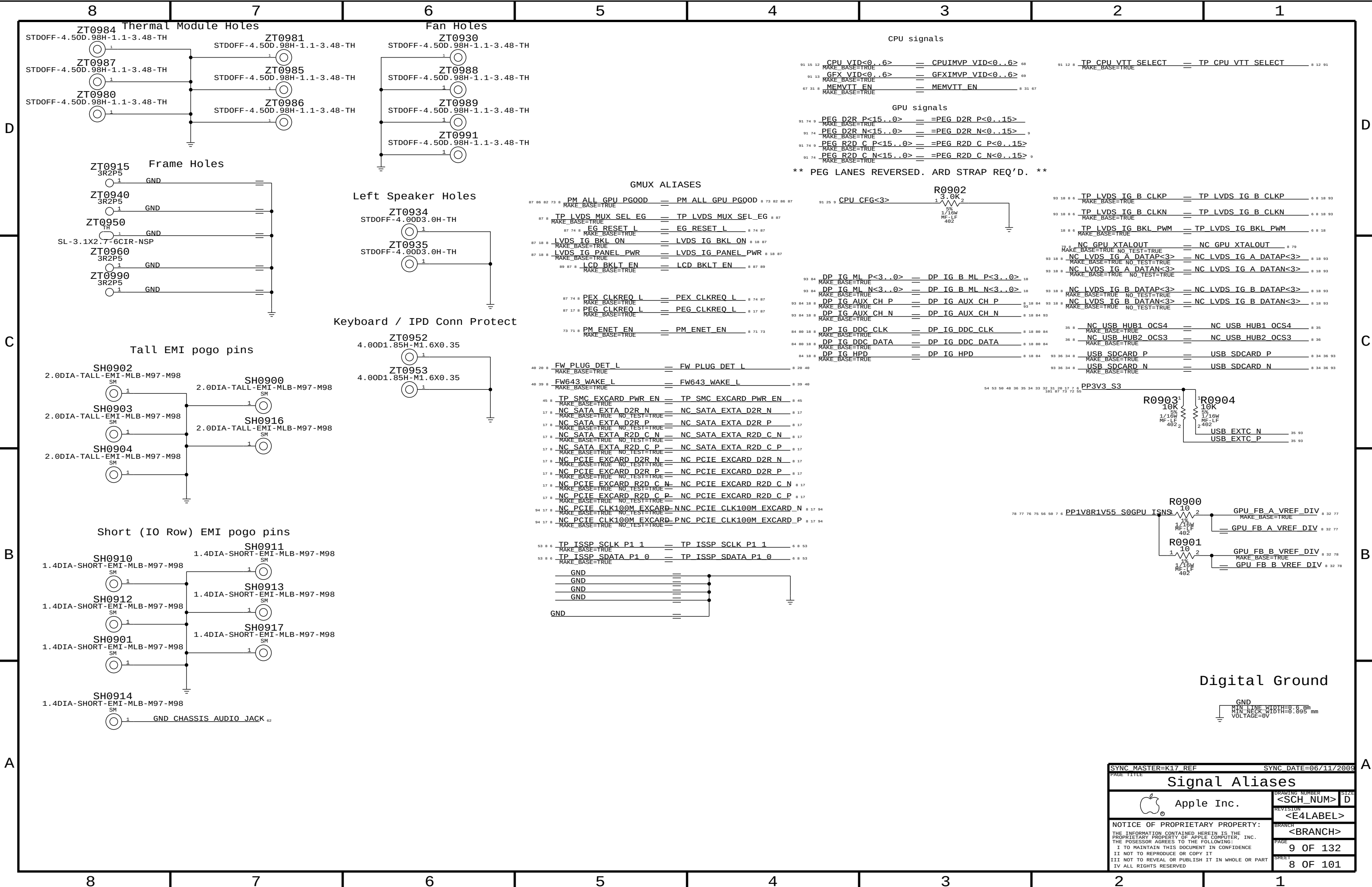
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-1404	1	K18 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM

Alternate Parts

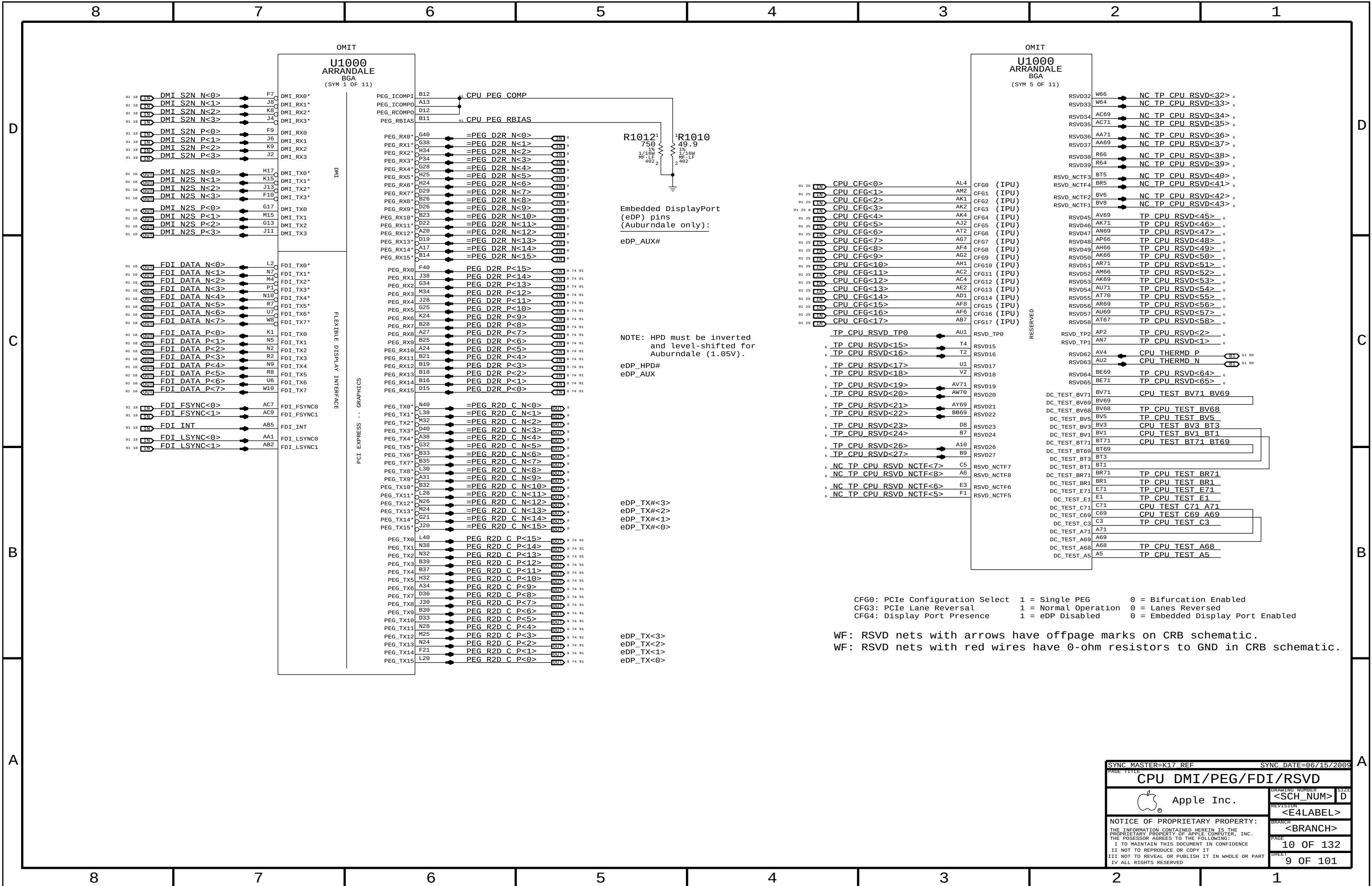
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
152S0896	152S0518		ALL	MAG LAYERS ALT TO CYNTEC
155S0457	155S0329		ALL	MAG LAYERS ALT TO MURATA
333S0506	333S0535		ALL	Hynix 900M alt to 1000M
516S0805	516S0806		ALL	Molex alt to Foxconn
152S1102	152S1088		ALL	Mag layer alt to Vishay
353S2805	353S2603		ALL	Fairchild wafer option
333S0542	333S0507		ALL	Samsung I die alt to H
128S0264	128S0257		ALL	Sanyo alt to Kemet
128S0303	128S0282		ALL	Panasonic alt to Sanyo
337S3808	337S3839		ALL	A02 alt to A03 GPU
128S0305	128S0294		ALL	6.3V alt to 11V Sanyo

SYNC MASTER-K17 REF		SYNC DATE=05/28/2009	
PAGE TITLE			
BOM Configuration			
	Apple Inc.	DRAWING NUMBER	SIZE
		<SCH_NUM>	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	5 OF 132
		SHEET	5 OF 101



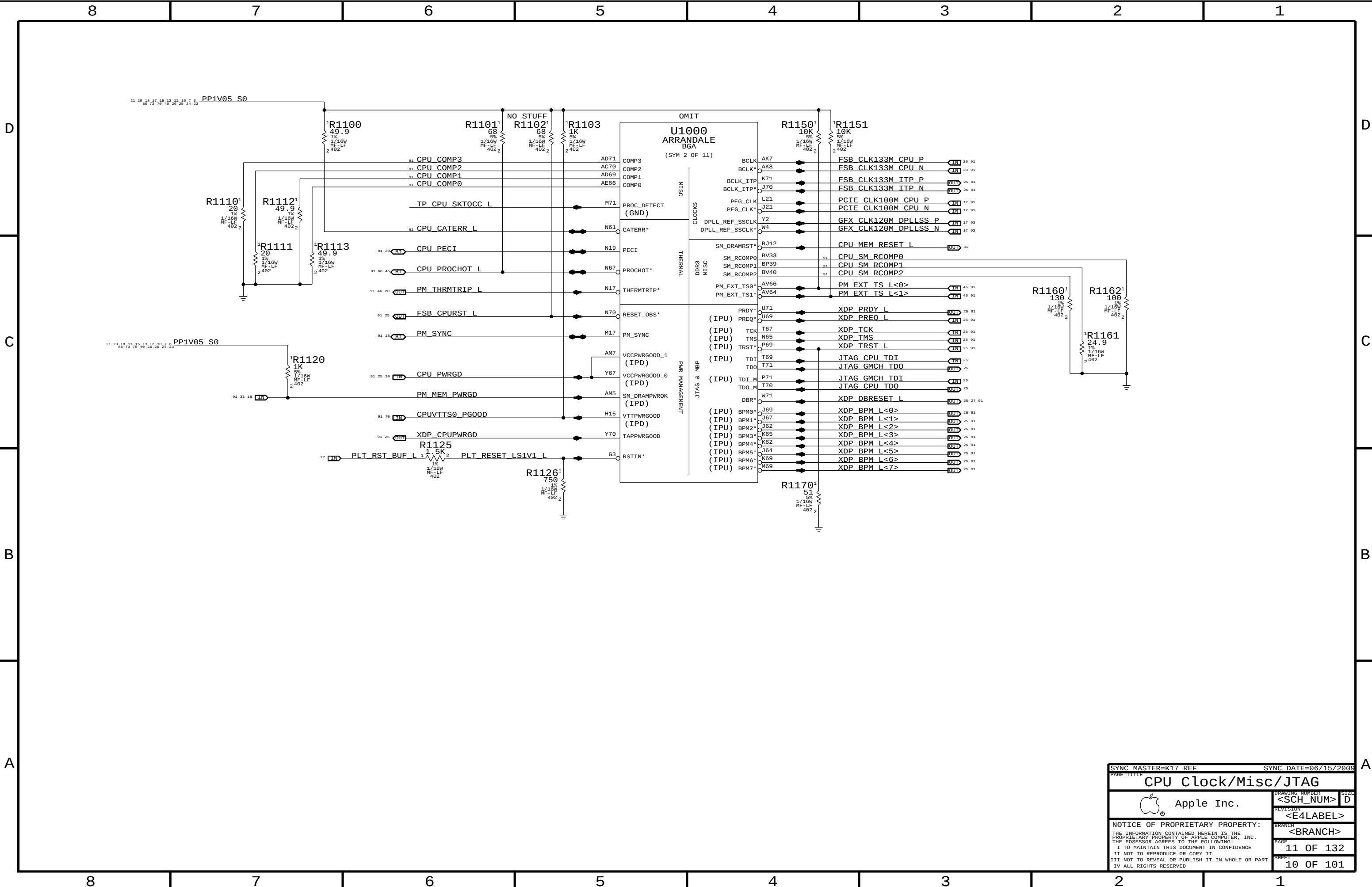


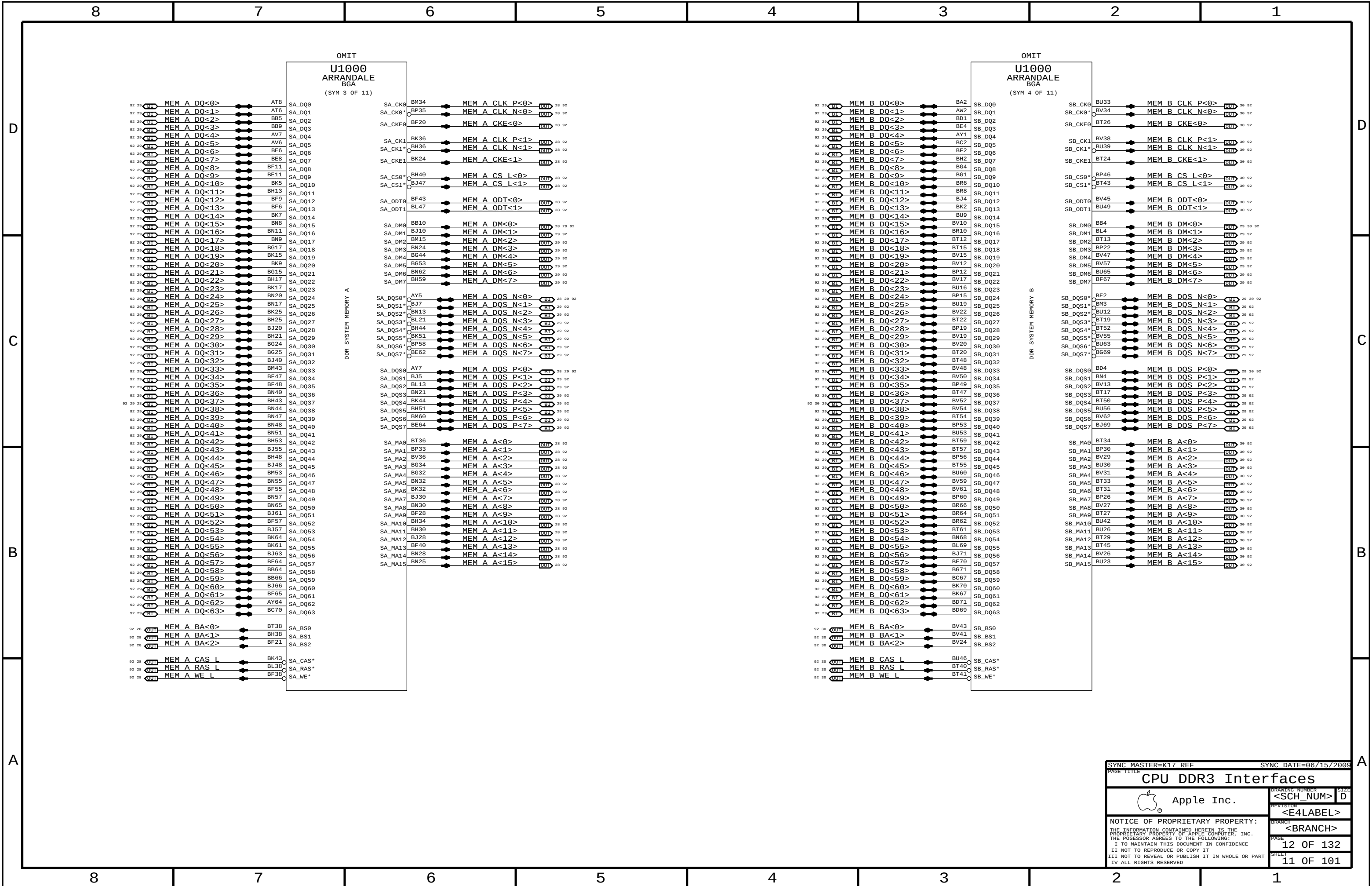
SYNC MASTER=K17_REF		SYNC DATE=06/11/2009	
PAGE TITLE			
Signal Aliases			
	Apple Inc.		DRAWING NUMBER
			<SCH_NUM> D
			REVISION
		<E4LABEL>	
		BRANCH	<BRANCH>
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		9 OF 132	
SHEET		8 OF 101	

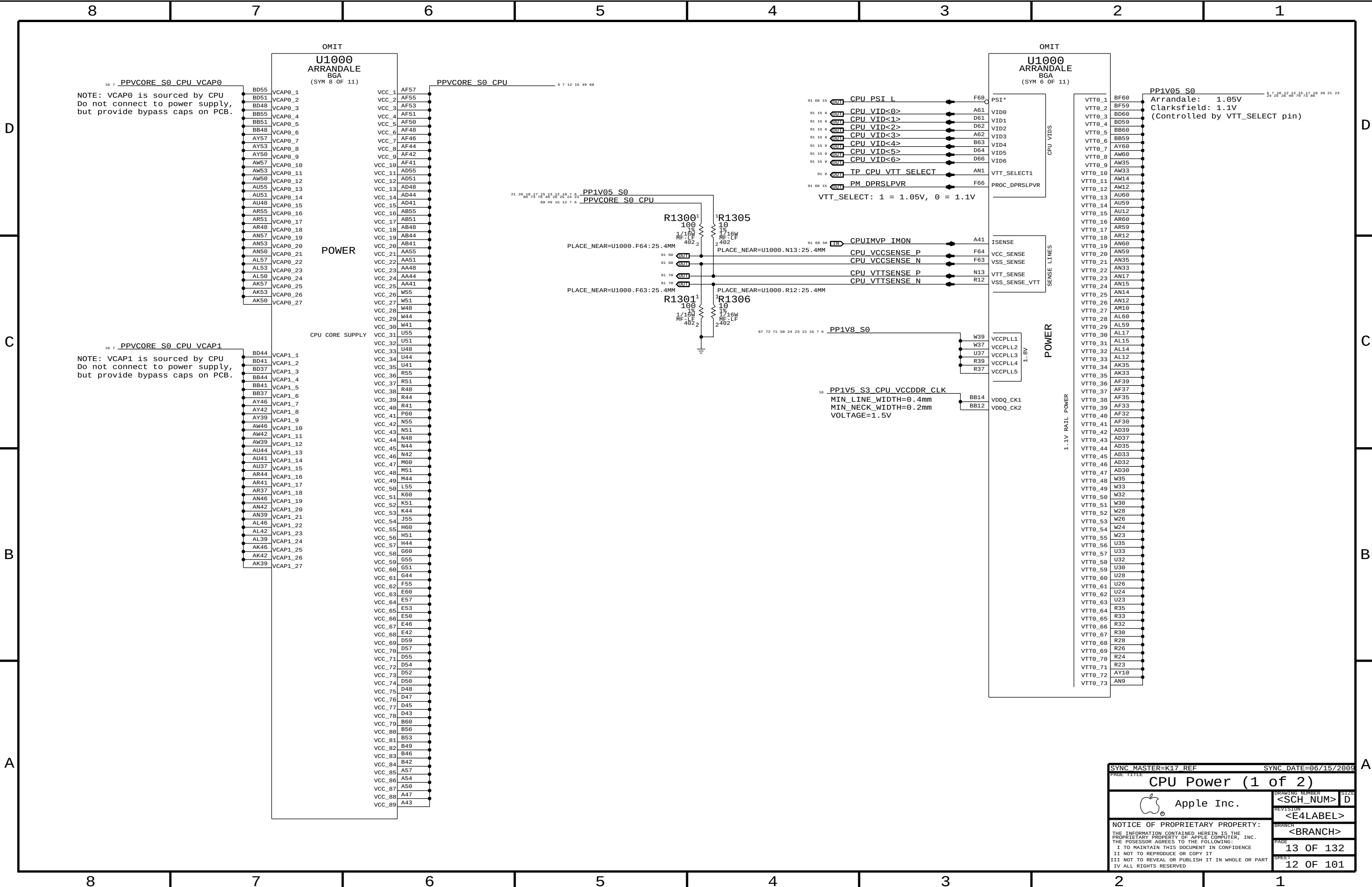


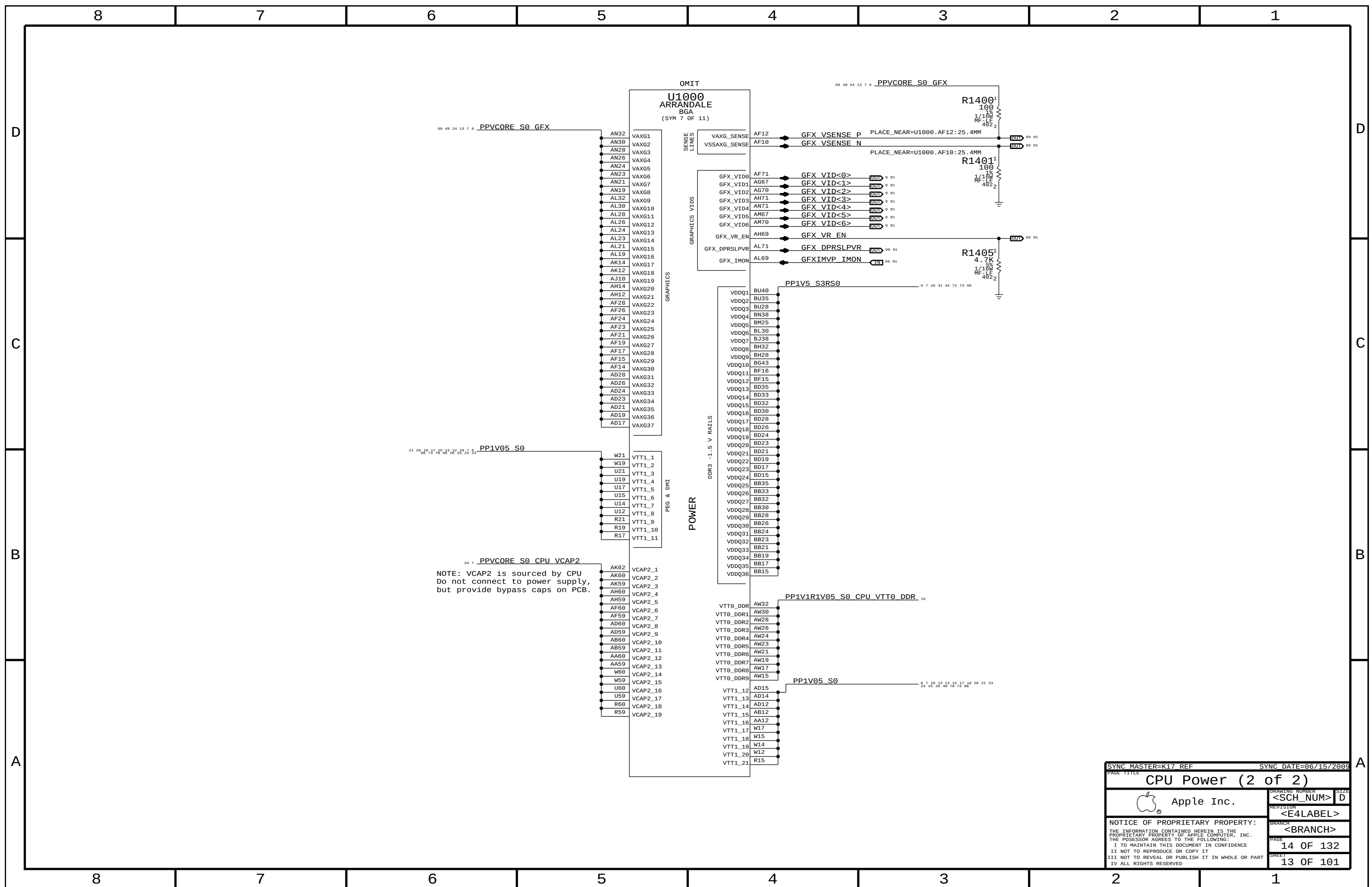
CFG0: PCIe Configuration Select 1 = Single PEG 0 = Bifurcation Enabled
CFG3: PCIe Lane Reversal 1 = Normal Operation 0 = Lanes Reversed
CFG4: Display Port Presence 1 = eDP Disabled 0 = Embedded Display Port Enabled

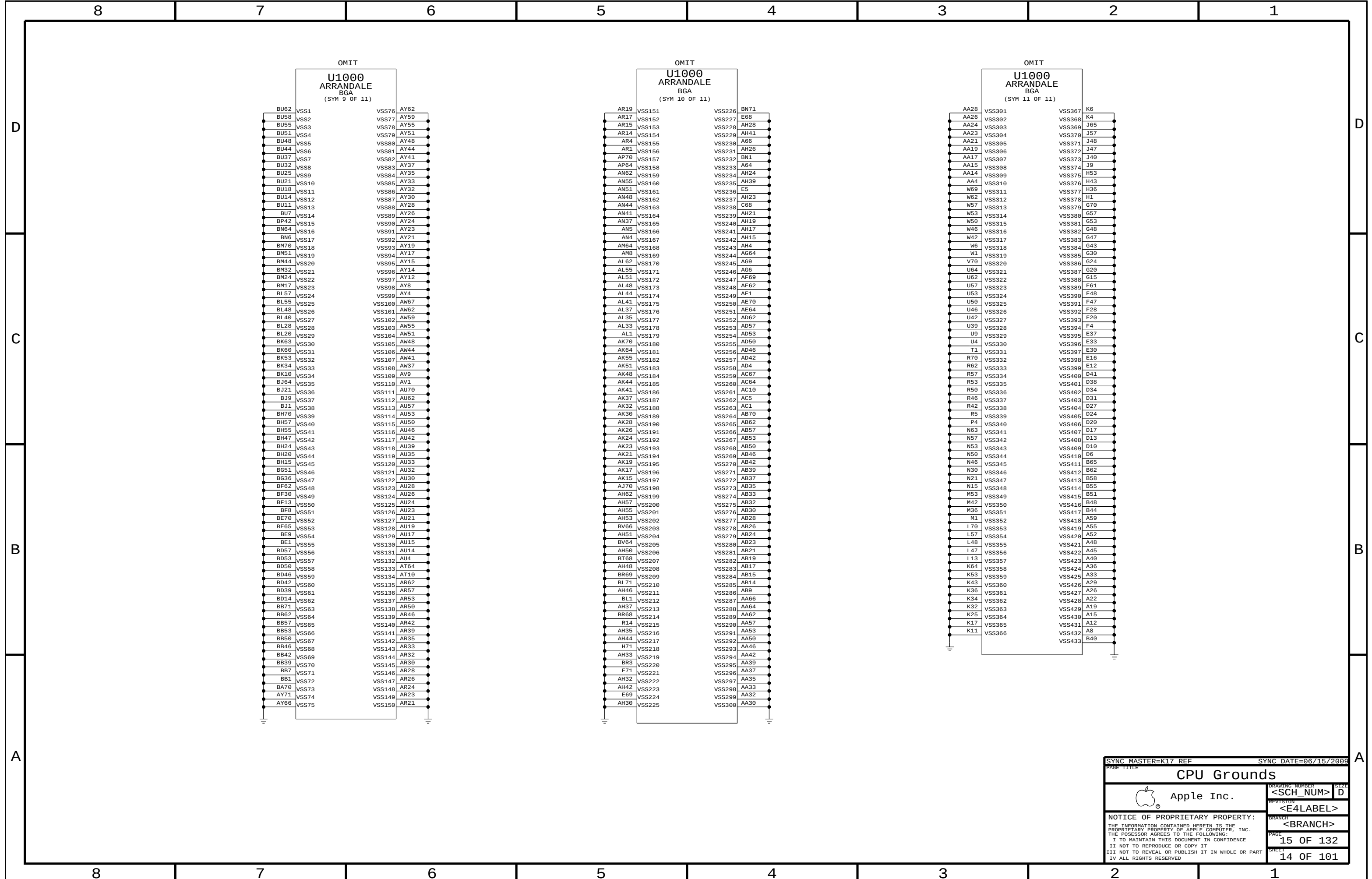
WF: RSVD nets with arrows have offpage marks on CRB schematic.
WF: RSVD nets with red wires have 0-ohm resistors to GND in CRB schematic.









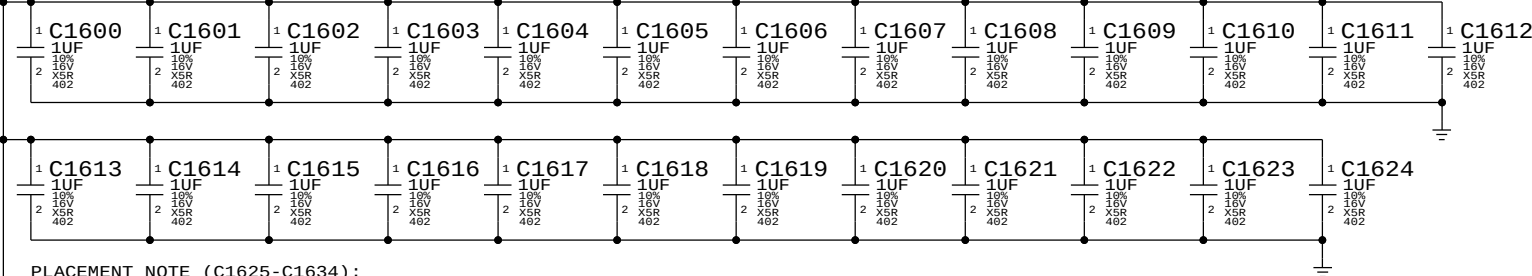


CPU VCore HF and Bulk Decoupling

4x 470uF 4.5mOhm, 3x 62uF B2, 10x 22uF 0603, 25x 1uF 0402

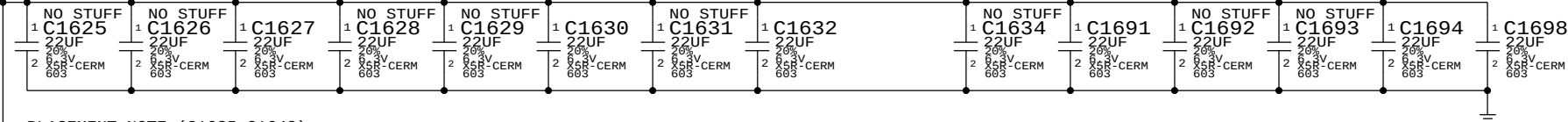
PLACEMENT_NOTE (C1600-C1624):

Place on bottom side of U1000..



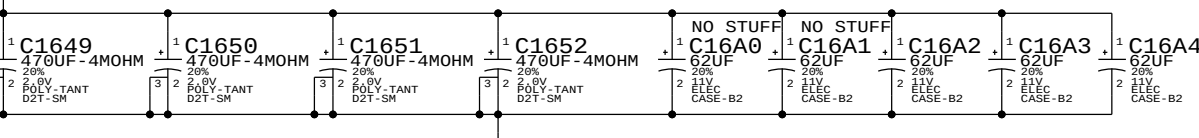
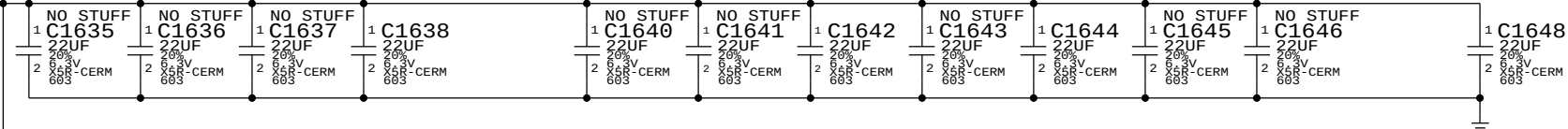
PLACEMENT_NOTE (C1625-C1634):

Place near U1000 on bottom side.



PLACEMENT_NOTE (C1635-C1648):

Place near inductors on bottom side.

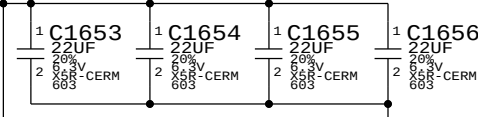


VTT (CPU Uncore) DECOUPLING

3x 330uF 6 mOhm, 4x 22uF 0805, 7x 10uF 0603, 24x 1uF 0402

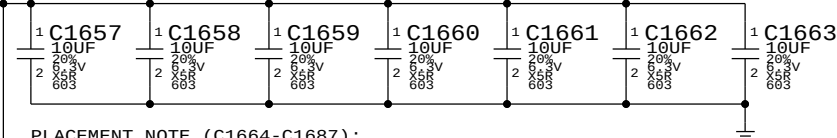
PLACEMENT_NOTE (C1653-C1656):

Place on bottom side of U1000..



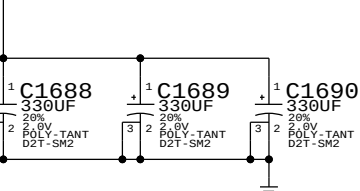
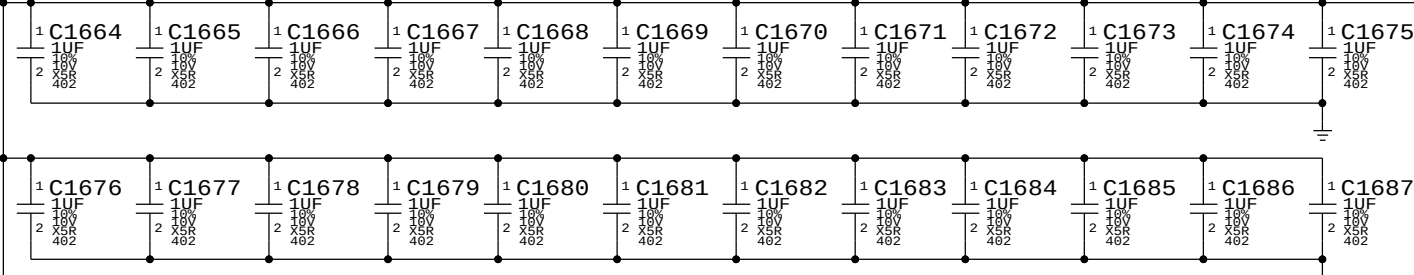
PLACEMENT_NOTE (C1657-C1663):

Place on bottom side of U1000..



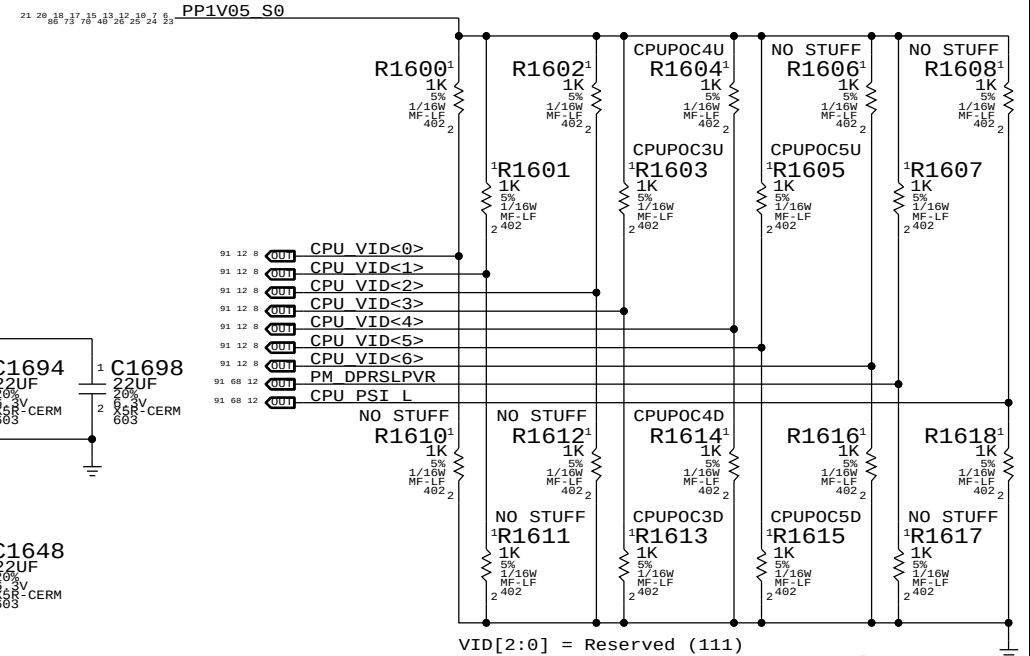
PLACEMENT_NOTE (C1664-C1687):

Place on bottom side of U1000..



CPU Power On Configuration (POC) Straps

Intel recommends all option straps should be provided in layout



VID[2:0] = Reserved (111)
VID[5:3] = GPU Gain Setting (See below)
VID[6] = Reserved (0)
DPRSLPVR = 1 - IMVP-6.5 compliant controller
PSI# = Reserved (0)

BOM GROUP	IMAX @ 900mV	CPU Gain Setting	BOM OPTIONS	Equivalent Gain
CPUPOC_IMAX_DIS		000	CPUPOC3D, CPUPOC4D, CPUPOC5D	
CPUPOC_IMAX_0_20	20A	001	CPUPOC3D, CPUPOC4D, CPUPOC5U	45
CPUPOC_IMAX_20_30	30A	010	CPUPOC3D, CPUPOC4U, CPUPOC5D	30
CPUPOC_IMAX_30_40	40A	011	CPUPOC3D, CPUPOC4U, CPUPOC5U	22.5
CPUPOC_IMAX_40_50	50A	100	CPUPOC3U, CPUPOC4D, CPUPOC5D	18
CPUPOC_IMAX_50_60	60A	101	CPUPOC3U, CPUPOC4D, CPUPOC5U	15
CPUPOC_IMAX_60_70	70A	110	CPUPOC3U, CPUPOC4U, CPUPOC5D	12.857
CPUPOC_IMAX_70_90	90A	111	CPUPOC3U, CPUPOC4U, CPUPOC5U	10

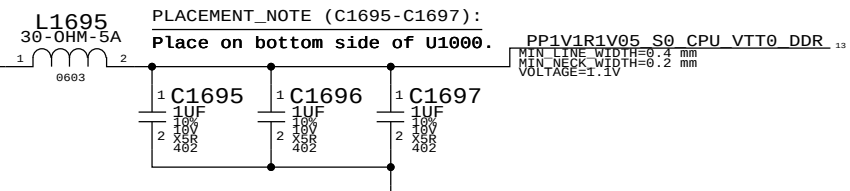
NOTE: BOM Configurations should not call out CPUPOCnU/D BOMOPTIONS directly.
Instead call out appropriate BOM GROUP defined in tables above.

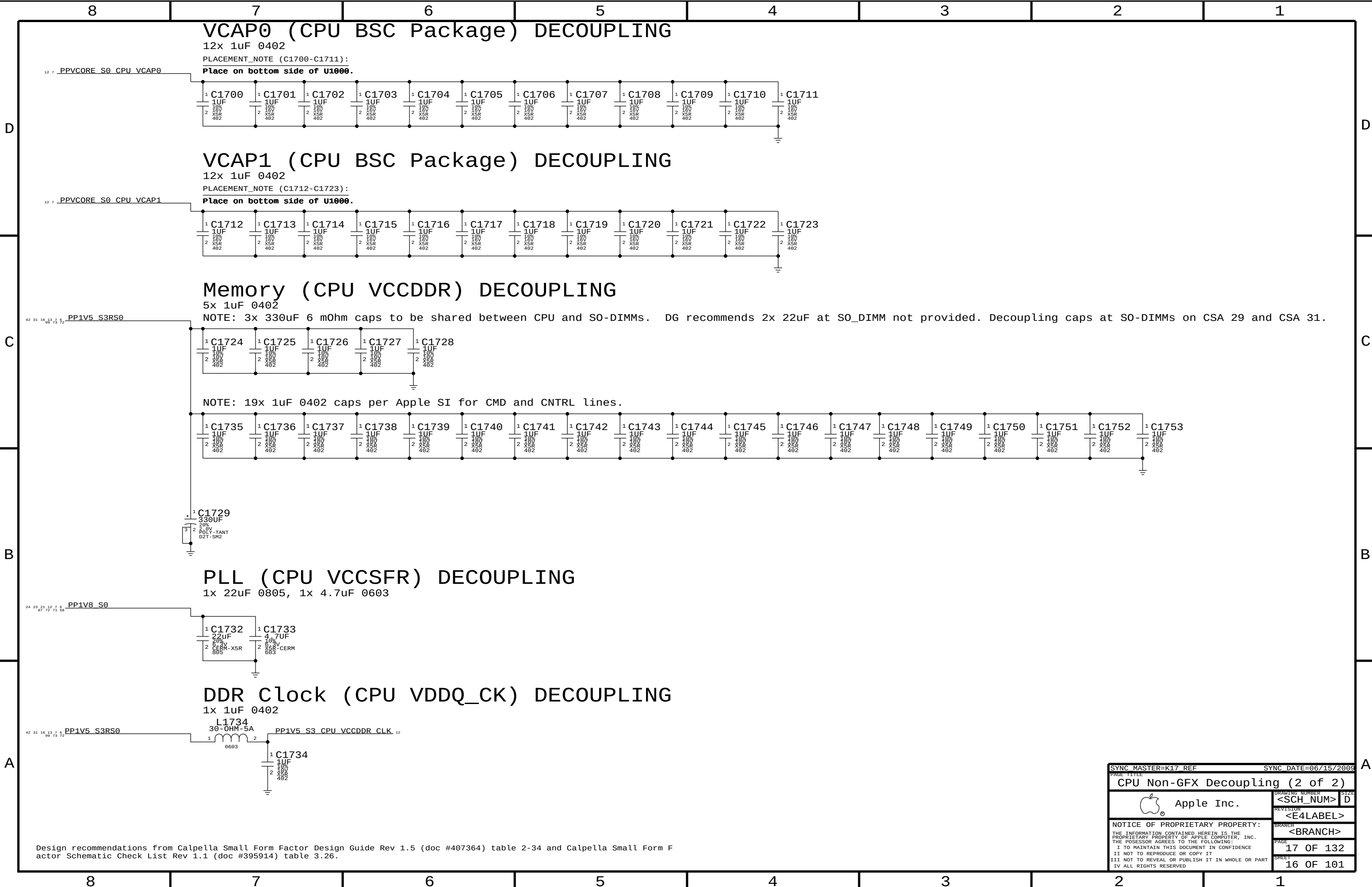
VTT0_DDR DECOUPLING

3x 1uF 0402

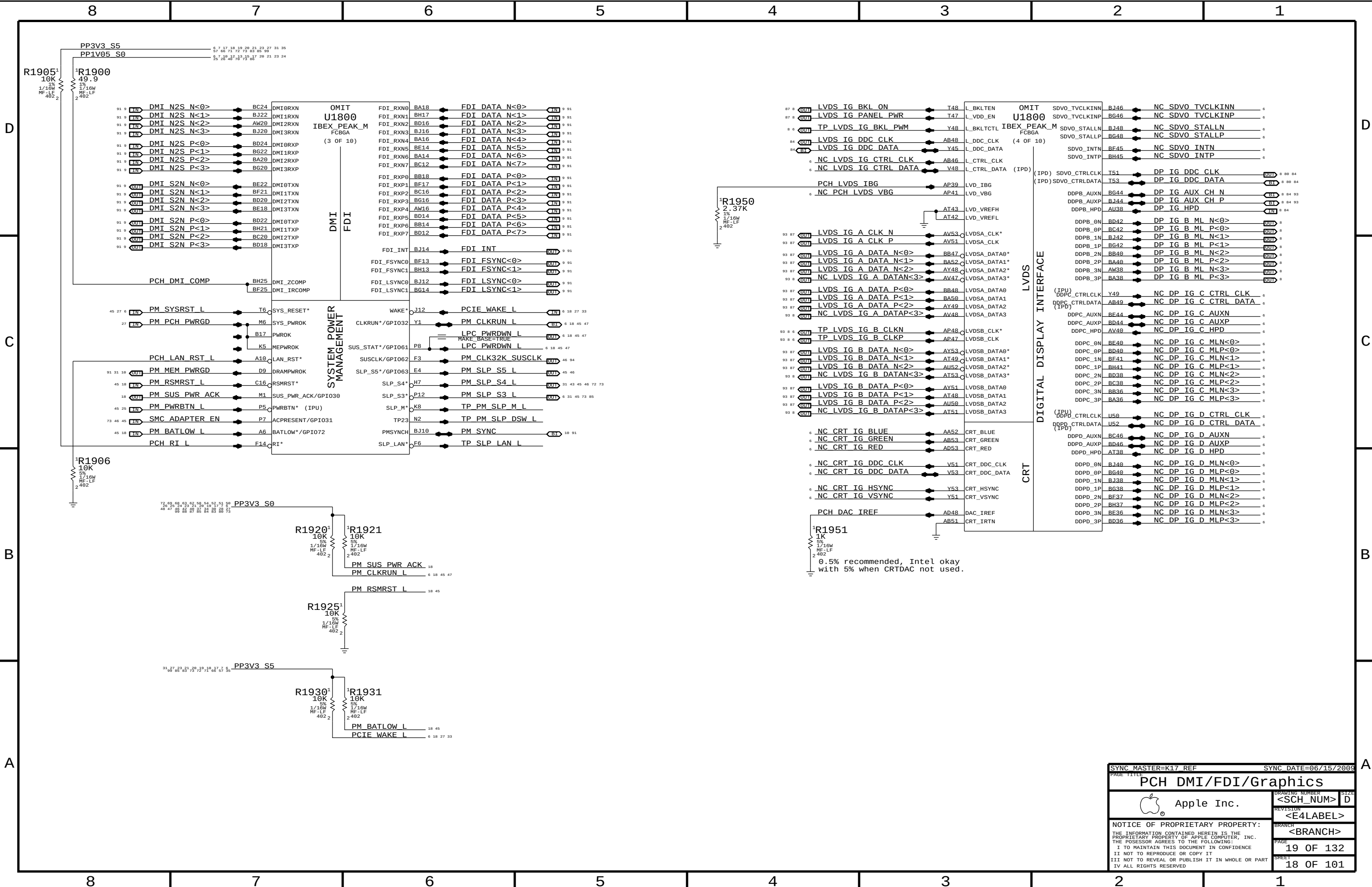
PLACEMENT_NOTE (C1695-C1697):

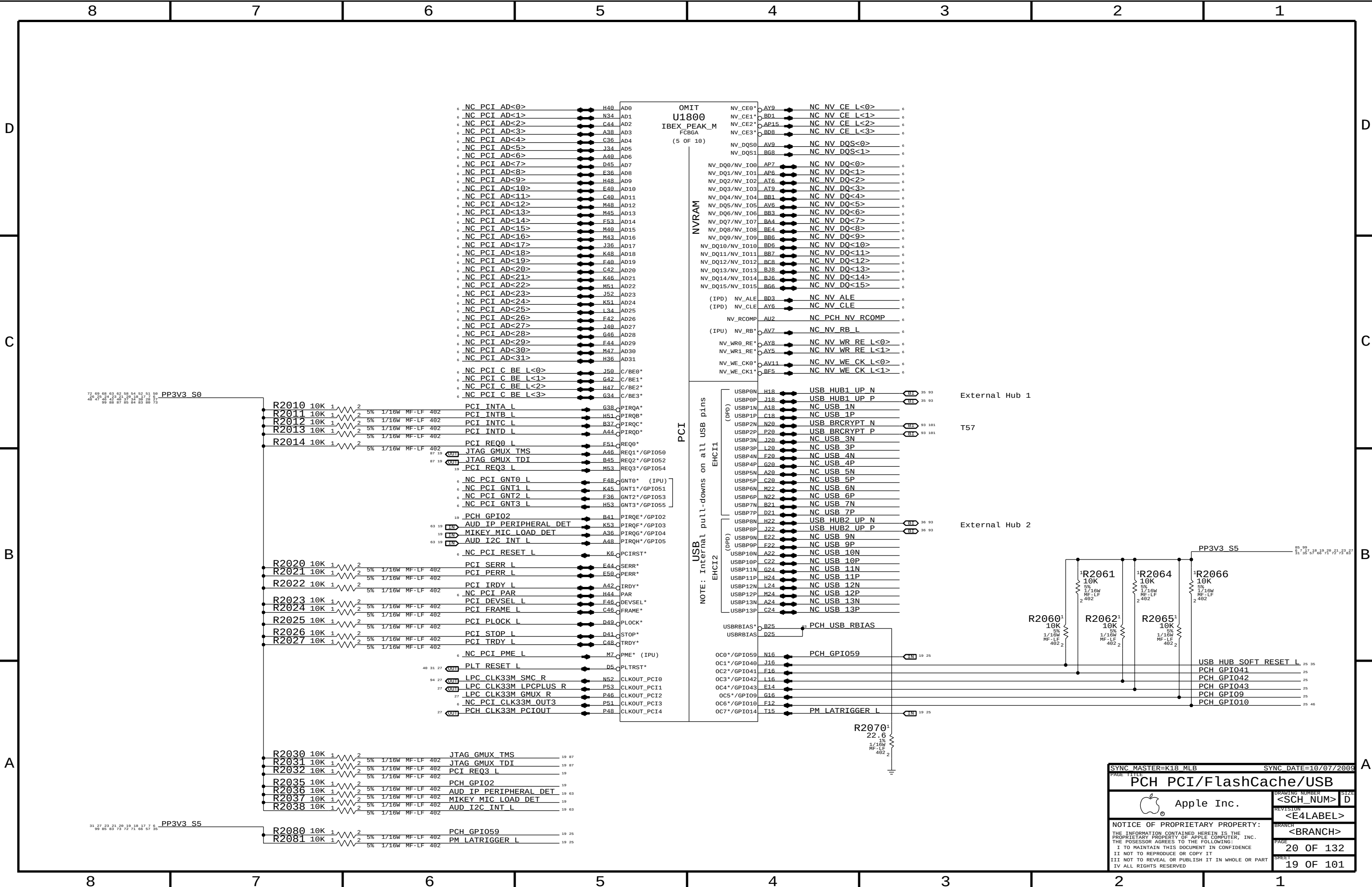
Place on bottom side of U1000..

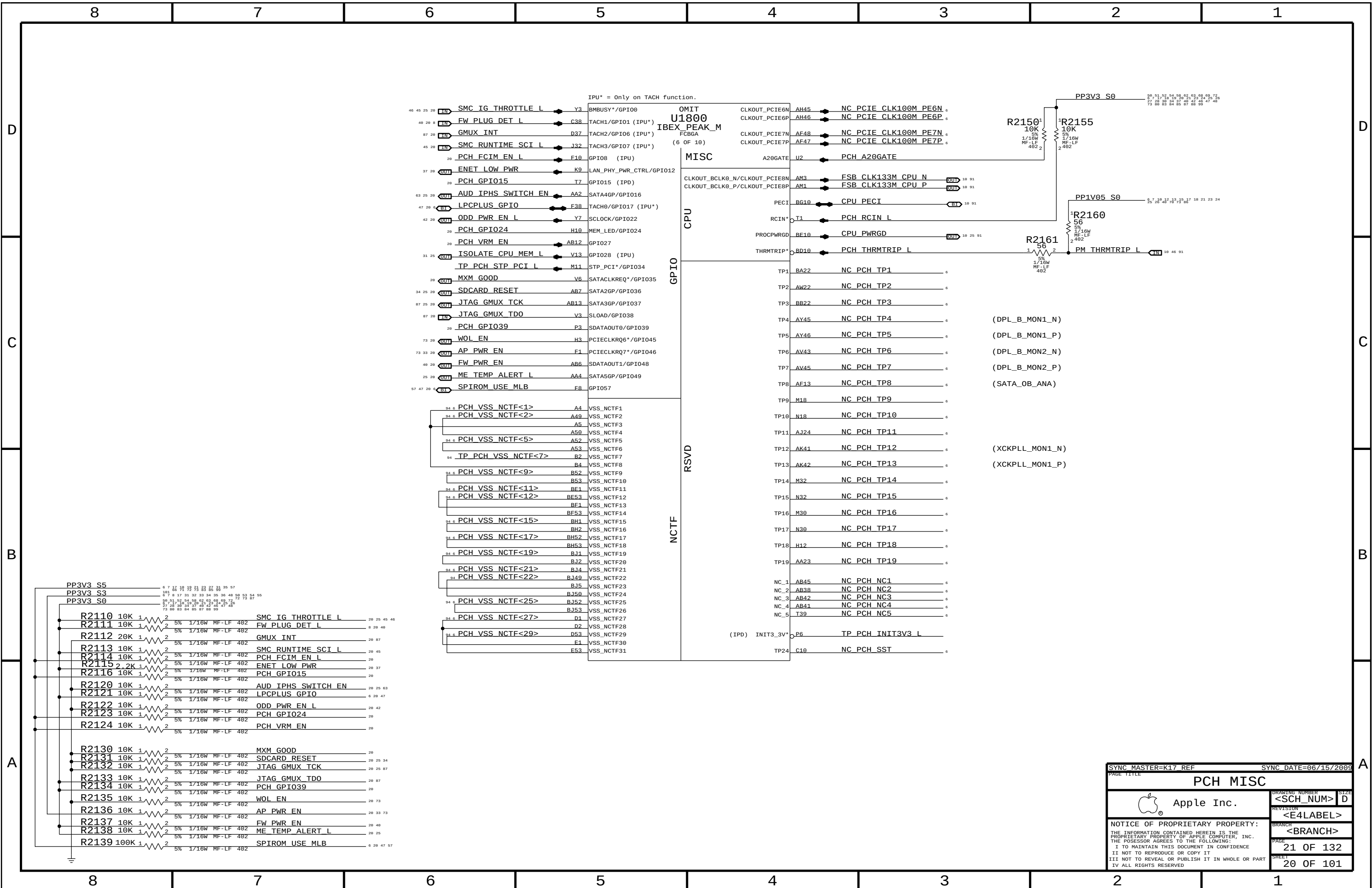


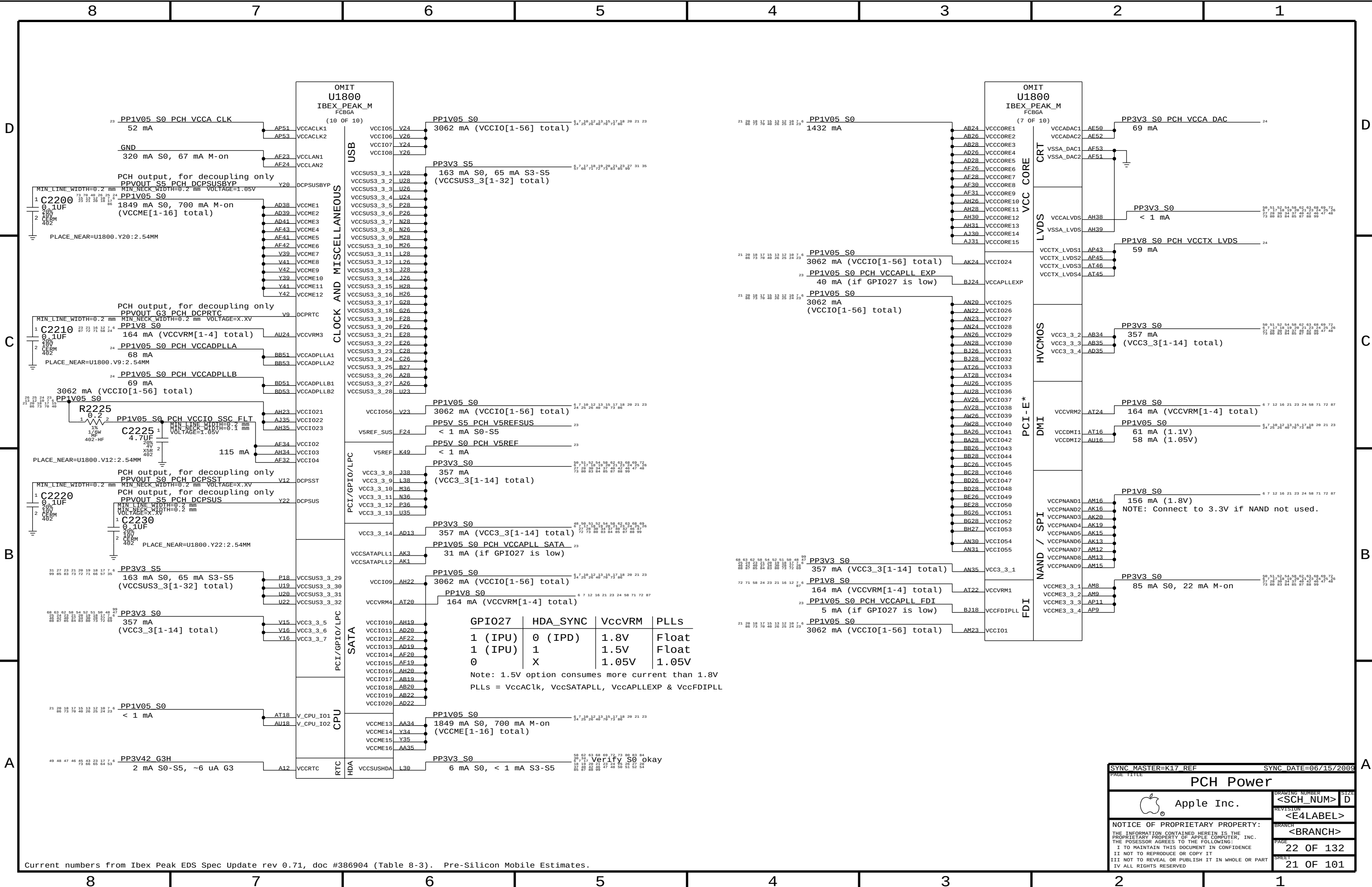












GPI027	HDA_SYNC	VccVRM	PLLs
1 (IPU)	0 (IPD)	1.8V	Float
1 (IPU)	1	1.5V	Float
0	X	1.05V	1.05V
Note: 1.5V option consumes more current than 1.8V			
PLLs = VccAClk, VccSATAPLL, VccAPLLEXP & VccFDIPLL			

SYNC MASTER=K17 REF

SYNC DATE=06/15/2009

PAGE TITLE

PCH Power

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

<SCH_NUM>

REVISION

<E4LABEL>

BRANCH

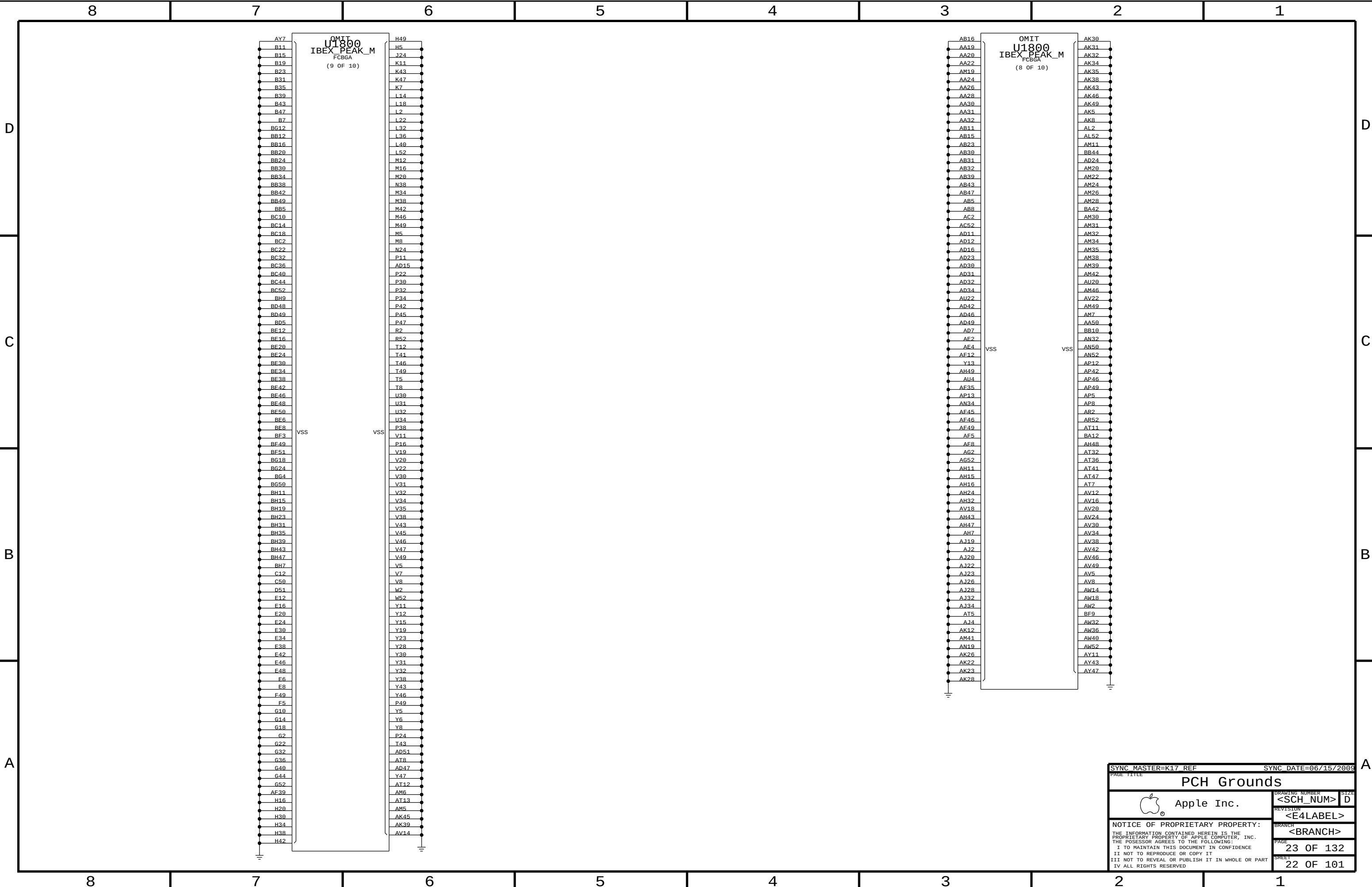
<BRANCH>

PAGE

22 OF 132

SHEET

21 OF 101



SYNC_MASTER=K17_REF

SYNC_DATE=06/15/2009

PAGE_TITLE

PCH Grounds



Apple Inc.

DRAWING_NUMBER

<SCH_NUM>

SIZE

D

REVISION

<E4LABEL>

BRANCH

<BRANCH>

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

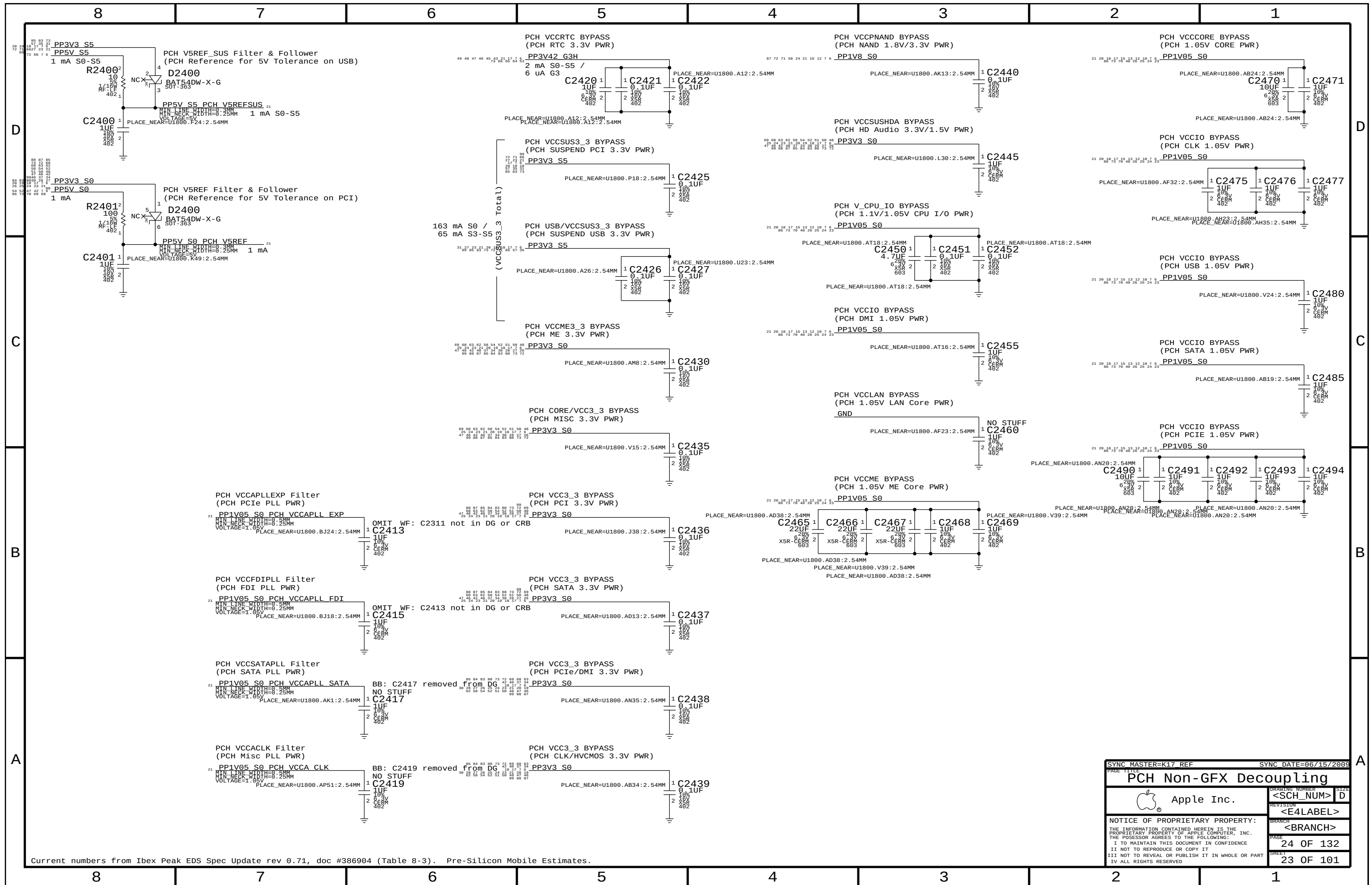
IV ALL RIGHTS RESERVED

PAGE

23 OF 132

SHEET

22 OF 101

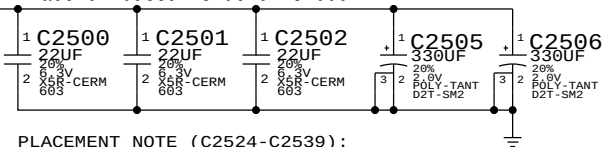


GFX (CPU VCCAXG) DECOUPLING

3x 330uF 6 m0hm (2 stuffed), 3x 22uF 0603, 16x 1uF 0402

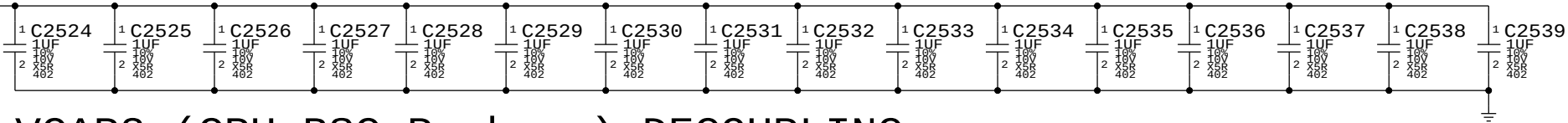
PLACEMENT_NOTE (C2500-C2506):

Place on bottom side of U1000.



PLACEMENT_NOTE (C2524-C2539):

Place on bottom side of U1000.

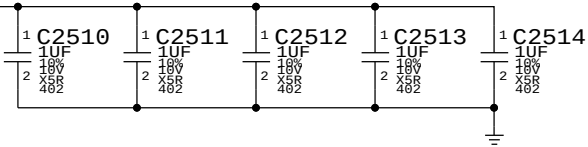


VCAP2 (CPU BSC Package) DECOUPLING

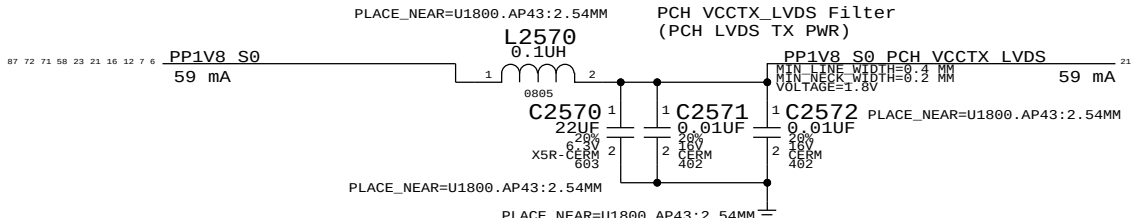
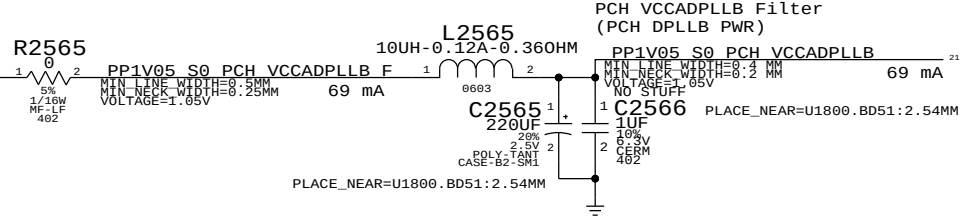
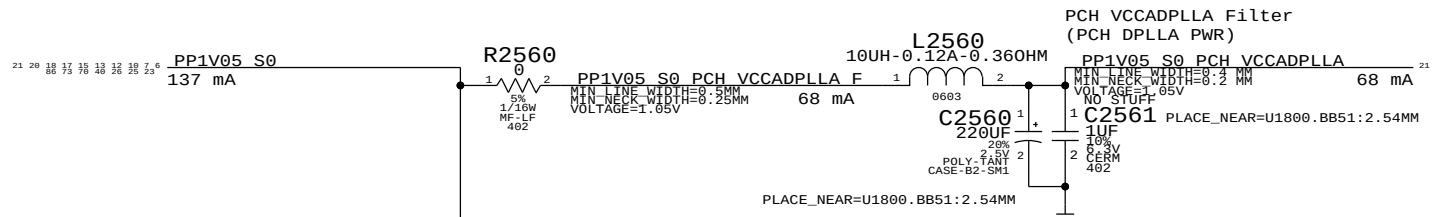
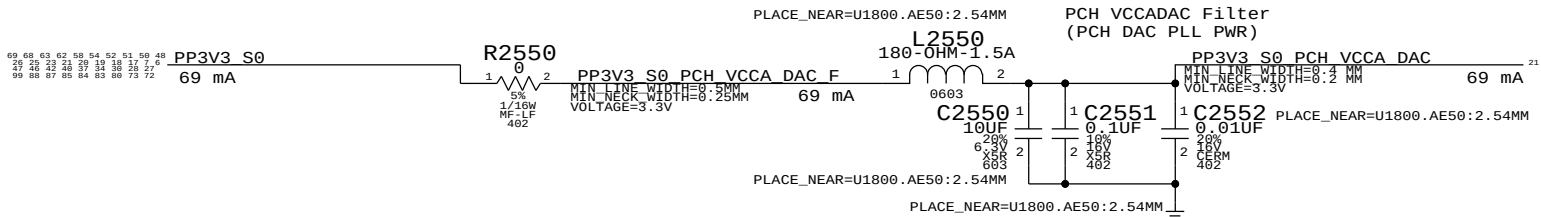
5x 1uF 0402

PLACEMENT_NOTE (C2510-C2514):

Place on bottom side of U1000.



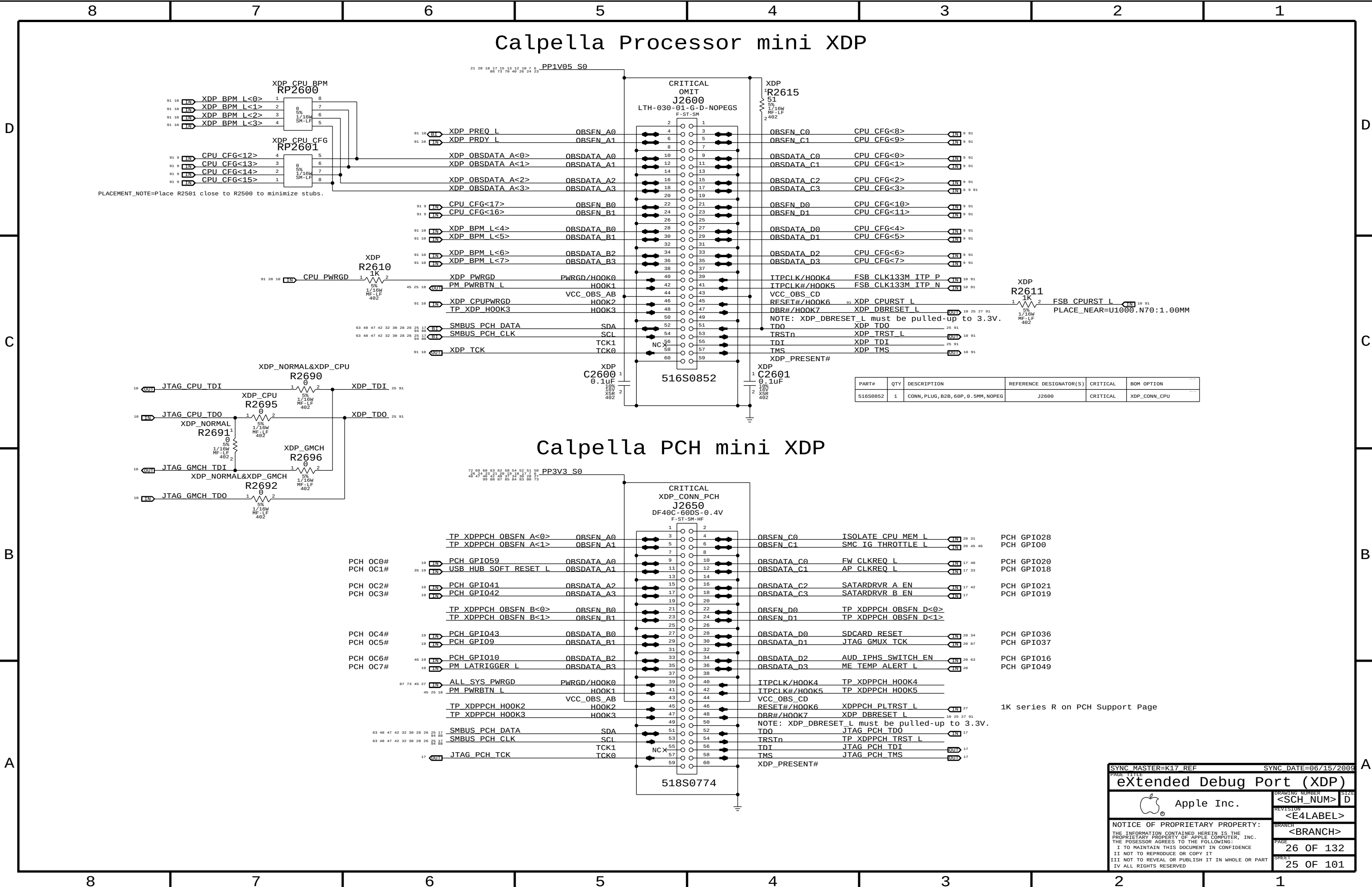
Design recommendations from Calpella Small Form Factor Design Guide Rev 1.5 (doc #407364) table 2-34 and Calpella Small Form F actor Schematic Check List Rev 1.1 (doc #395914) table 3.26.



Design recommendations from Calpella Design Guide Rev 1.5 (doc #398905) Section 3.25.3 tables 161 and 162.

Current numbers from Ibex Peak EDS Spec Update rev 0.71, doc #386904 (Table 8-3). Pre-Silicon Mobile Estimates.

SYNC MASTER=K17_REF		SYNC DATE=06/15/2009	
PAGE TITLE			
CPU/PCH GFX Decoupling			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	25 OF 132
		SHEET	24 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			



Calpella Processor mini XDP

Calpella PCH mini XDP

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
516S8852	1	CONN, PLUG, B2B, 60P, 0.5MM, NOPEG	J2600	CRITICAL	XDP_CONN_CPU

SYNC MASTER=K17 REF

SYNC DATE=06/15/2009

extended Debug Port (XDP)

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

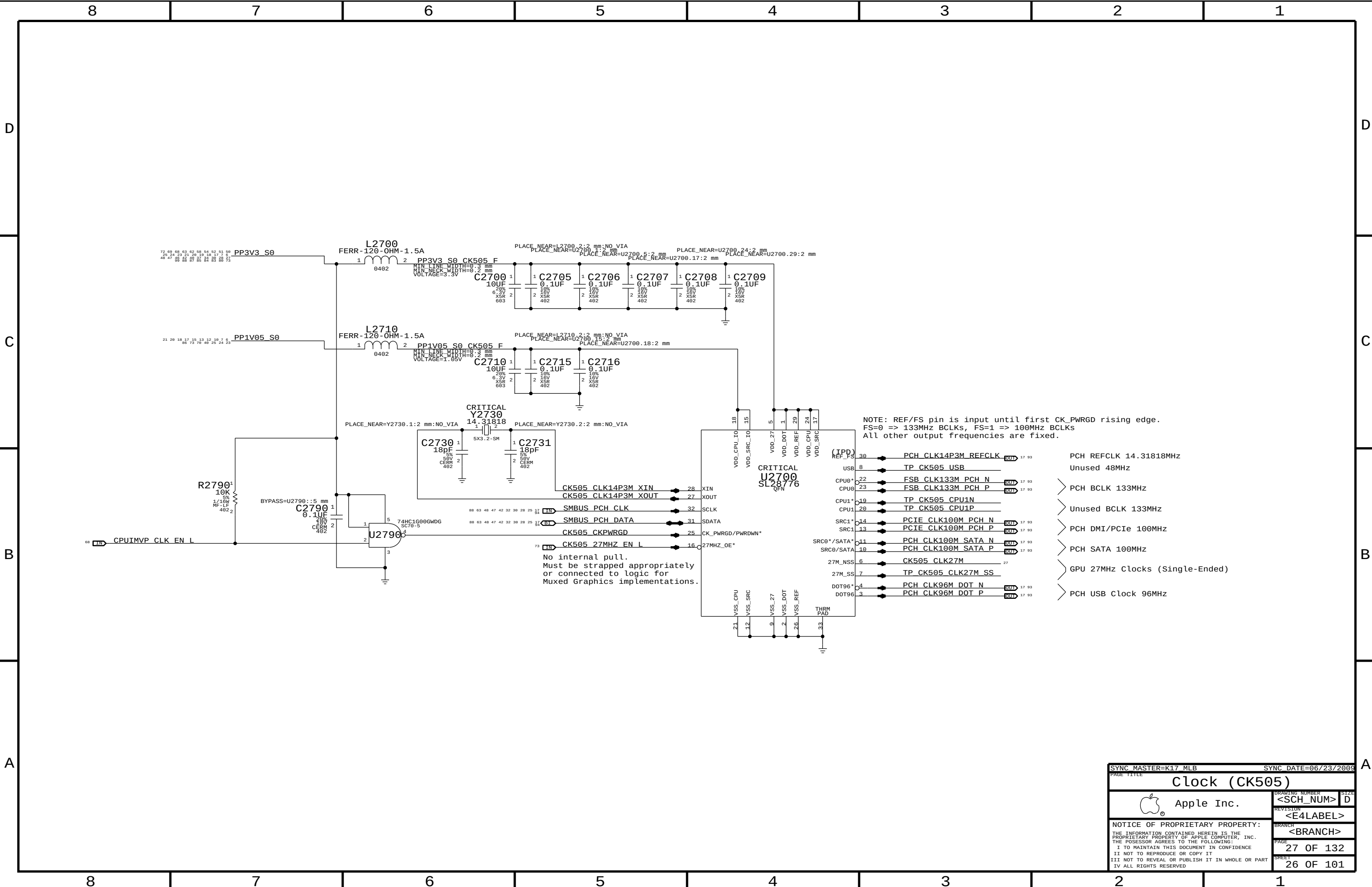
DRAWING NUMBER
<SCH_NUM>

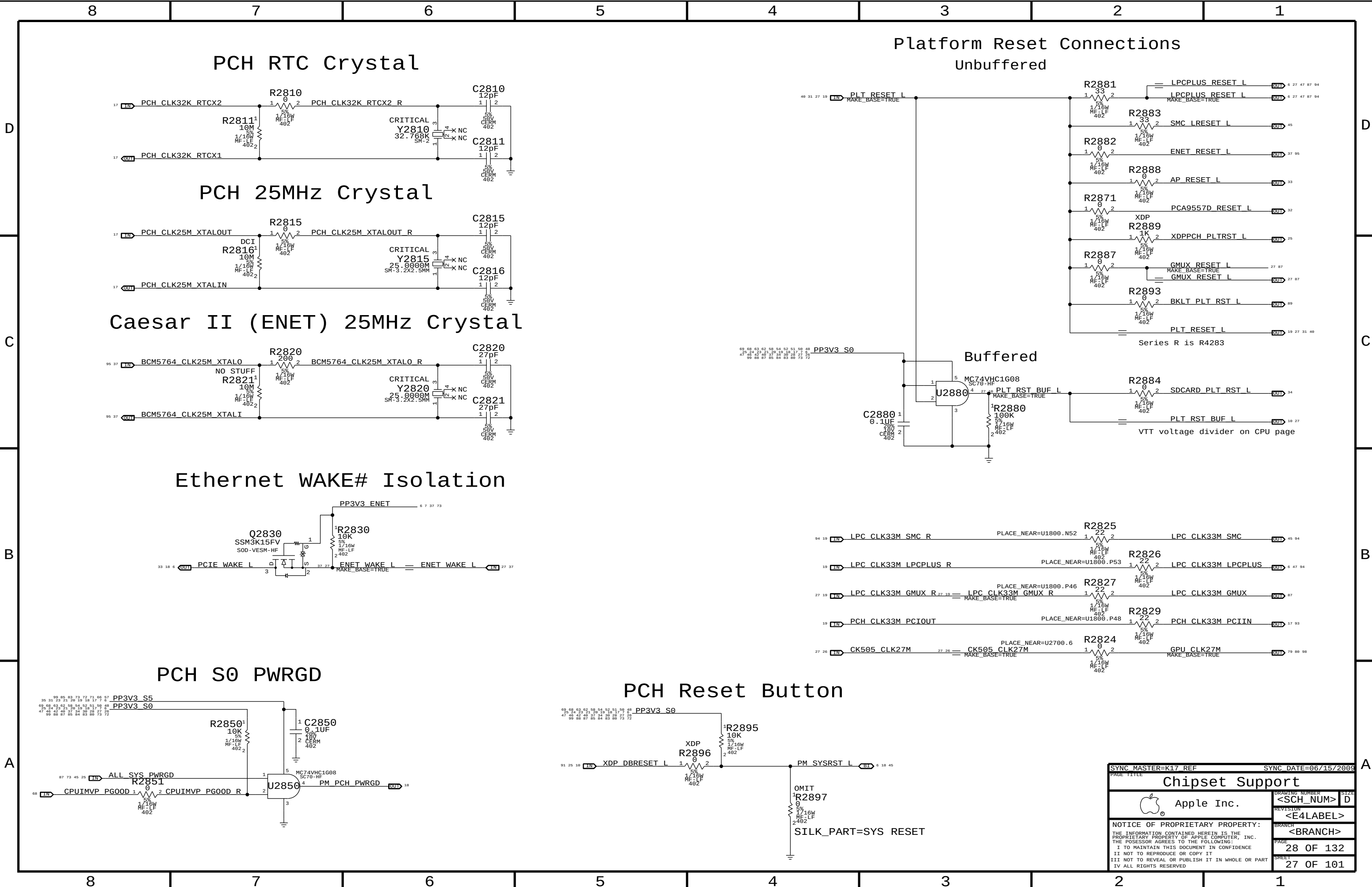
REVISION
<E4LABEL>

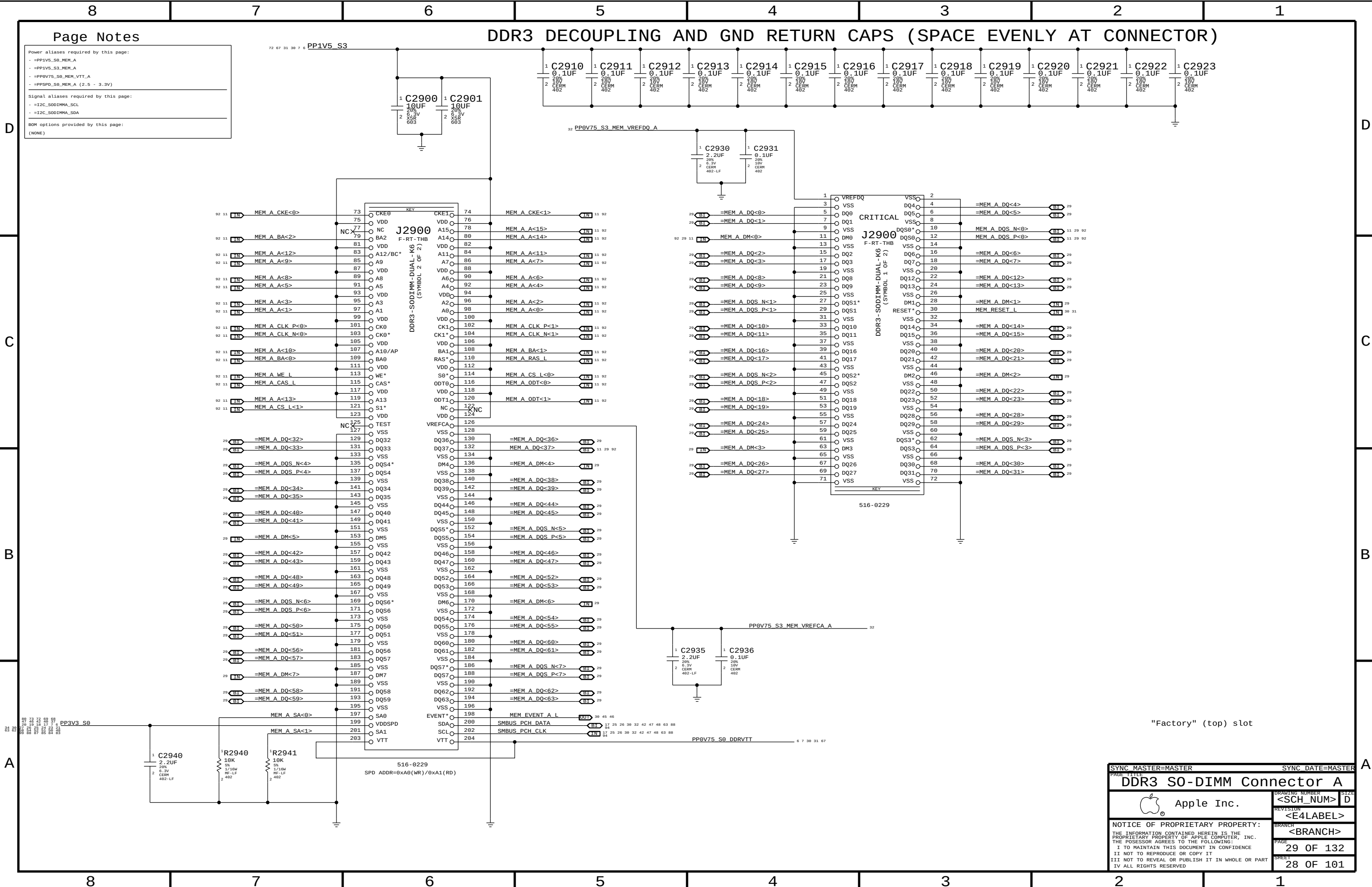
BRANCH
<BRANCH>

PAGE
26 OF 132

SHEET
25 OF 101







Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
DDR3 S0-DIMM Connector A			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	29 OF 132
		SHEET	28 OF 101

		8		7		6		5		4		3		2		1			
		CPU CHANNEL A DQS 0 -> DIMM A DQS 0		CPU CHANNEL B DQS 0 -> DIMM B DQS 0															
		MEM A DQS N<0>		MEM A DQS N<0>		MEM B DQS N<0>		MEM B DQS N<0>											
		MEM A DQS P<0>		MEM A DQS P<0>		MEM B DQS P<0>		MEM B DQS P<0>											
		MEM A DM<0>		MEM A DM<0>		MEM B DM<0>		MEM B DM<0>											
		MEM A DQ<7>		MEM A DQ<7>		MEM B DQ<7>		MEM B DQ<7>											
		MEM A DQ<6>		MEM A DQ<6>		MEM B DQ<6>		MEM B DQ<6>											
		MEM A DQ<5>		MEM A DQ<5>		MEM B DQ<5>		MEM B DQ<5>											
		MEM A DQ<4>		MEM A DQ<4>		MEM B DQ<4>		MEM B DQ<4>											
		MEM A DQ<3>		MEM A DQ<3>		MEM B DQ<3>		MEM B DQ<3>											
		MEM A DQ<2>		MEM A DQ<2>		MEM B DQ<2>		MEM B DQ<2>											
		MEM A DQ<1>		MEM A DQ<1>		MEM B DQ<1>		MEM B DQ<1>											
		MEM A DQ<0>		MEM A DQ<0>		MEM B DQ<0>		MEM B DQ<0>											
		CPU CHANNEL A DQS 1 -> DIMM A DQS 1		CPU CHANNEL B DQS 1 -> DIMM B DQS 1															
		MEM A DQS N<1>		MEM A DQS N<1>		MEM B DQS N<1>		MEM B DQS N<1>											
		MEM A DQS P<1>		MEM A DQS P<1>		MEM B DQS P<1>		MEM B DQS P<1>											
		MEM A DM<1>		MEM A DM<1>		MEM B DM<1>		MEM B DM<1>											
		MEM A DQ<15>		MEM A DQ<15>		MEM B DQ<15>		MEM B DQ<15>											
		MEM A DQ<14>		MEM A DQ<14>		MEM B DQ<14>		MEM B DQ<14>											
		MEM A DQ<13>		MEM A DQ<13>		MEM B DQ<13>		MEM B DQ<13>											
		MEM A DQ<12>		MEM A DQ<12>		MEM B DQ<12>		MEM B DQ<12>											
		MEM A DQ<11>		MEM A DQ<11>		MEM B DQ<11>		MEM B DQ<11>											
		MEM A DQ<10>		MEM A DQ<10>		MEM B DQ<10>		MEM B DQ<10>											
		MEM A DQ<9>		MEM A DQ<9>		MEM B DQ<9>		MEM B DQ<9>											
		MEM A DQ<8>		MEM A DQ<8>		MEM B DQ<8>		MEM B DQ<8>											
		CPU CHANNEL A DQS 2 -> DIMM A DQS 2		CPU CHANNEL B DQS 2 -> DIMM B DQS 2															
		MEM A DQS N<2>		MEM A DQS N<2>		MEM B DQS N<2>		MEM B DQS N<2>											
		MEM A DQS P<2>		MEM A DQS P<2>		MEM B DQS P<2>		MEM B DQS P<2>											
		MEM A DM<2>		MEM A DM<2>		MEM B DM<2>		MEM B DM<2>											
		MEM A DQ<23>		MEM A DQ<23>		MEM B DQ<23>		MEM B DQ<23>											
		MEM A DQ<22>		MEM A DQ<22>		MEM B DQ<22>		MEM B DQ<22>											
		MEM A DQ<21>		MEM A DQ<21>		MEM B DQ<21>		MEM B DQ<21>											
		MEM A DQ<20>		MEM A DQ<20>		MEM B DQ<20>		MEM B DQ<20>											
		MEM A DQ<19>		MEM A DQ<19>		MEM B DQ<19>		MEM B DQ<19>											
		MEM A DQ<18>		MEM A DQ<18>		MEM B DQ<18>		MEM B DQ<18>											
		MEM A DQ<17>		MEM A DQ<17>		MEM B DQ<17>		MEM B DQ<17>											
		MEM A DQ<16>		MEM A DQ<16>		MEM B DQ<16>		MEM B DQ<16>											

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

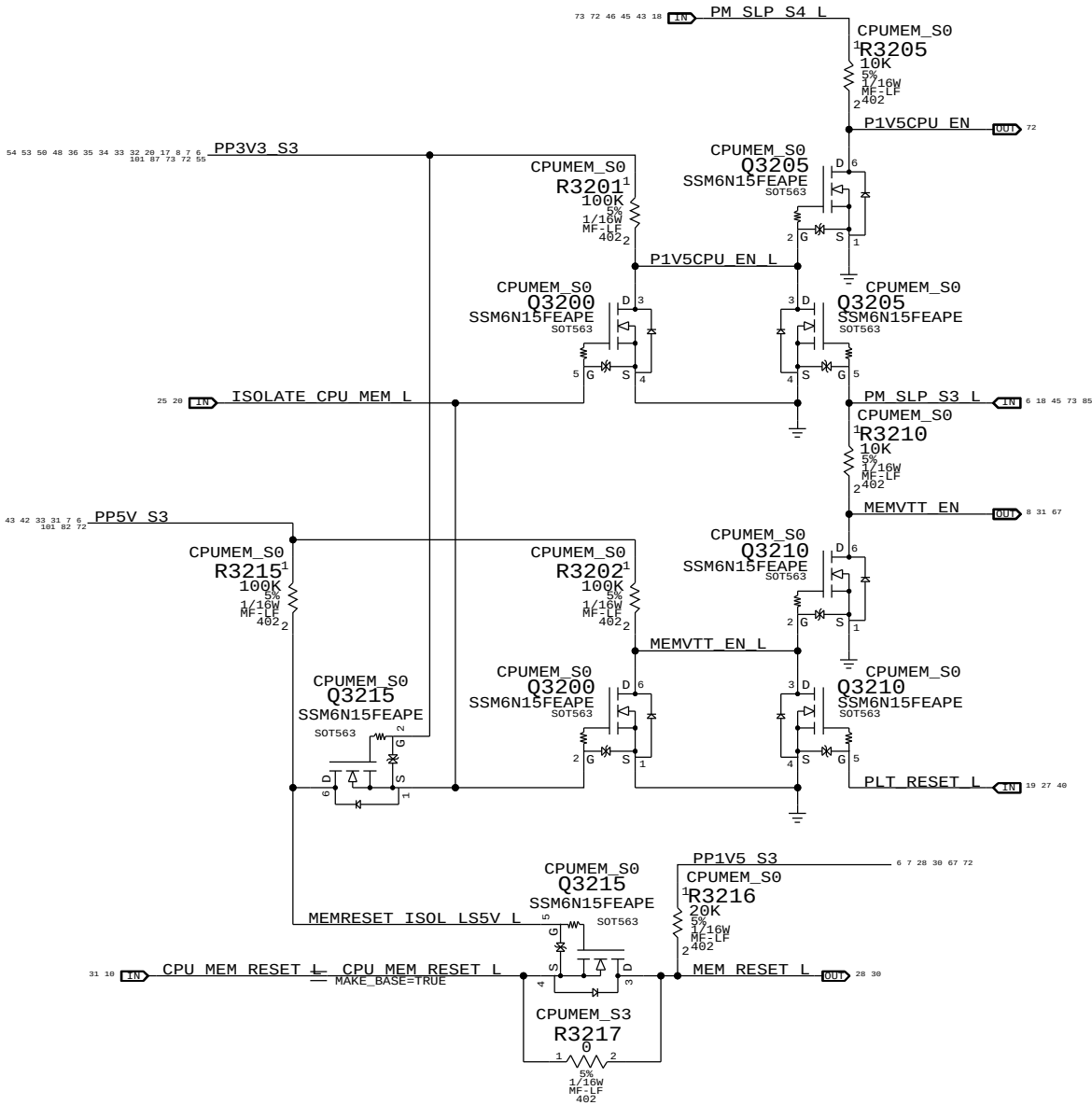
P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

D

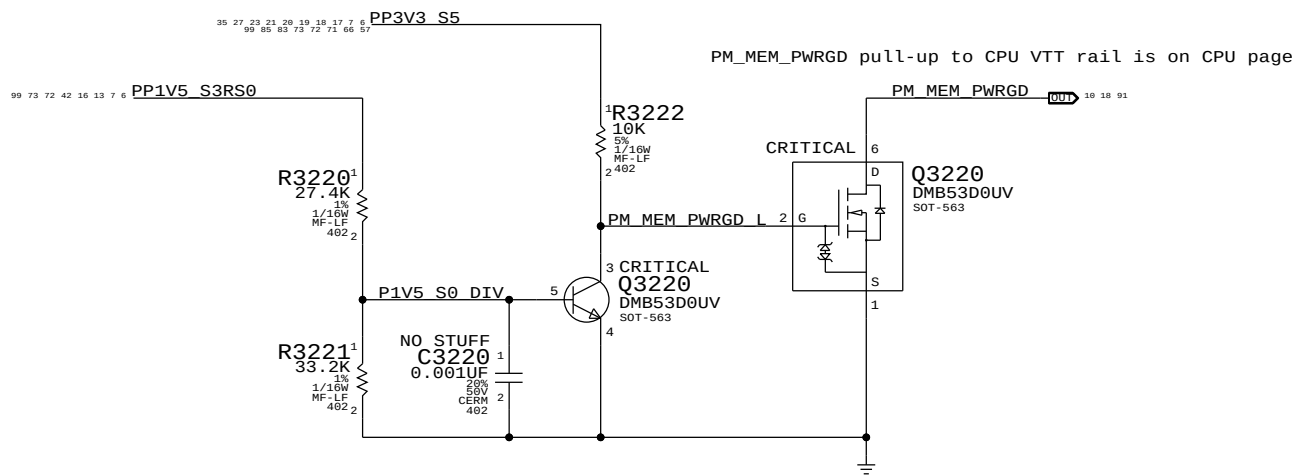
C

B

A

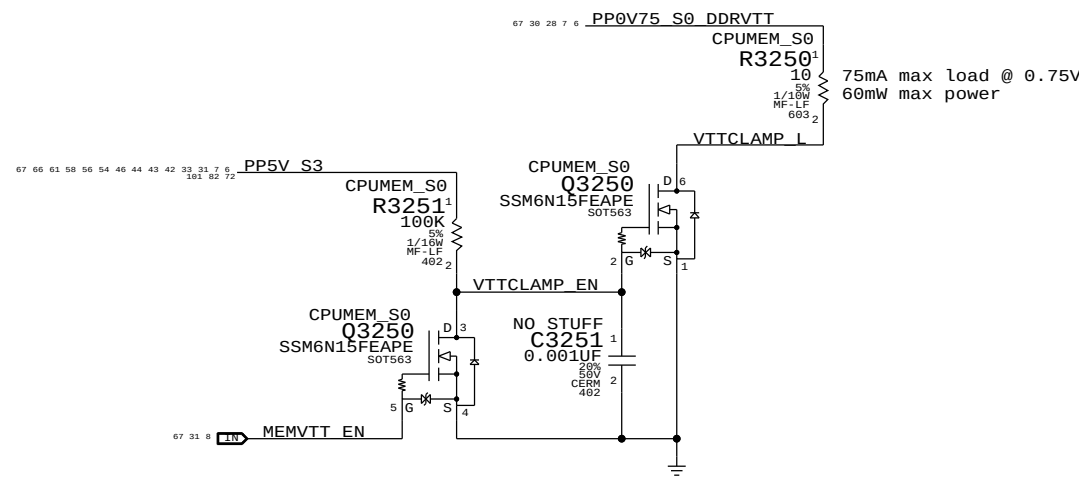


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RST_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=K17 REF

SYNC DATE=06/15/2009

CPU Memory S3 Support

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

<SCH_NUM>

REVISION

<E4LABEL>

BRANCH

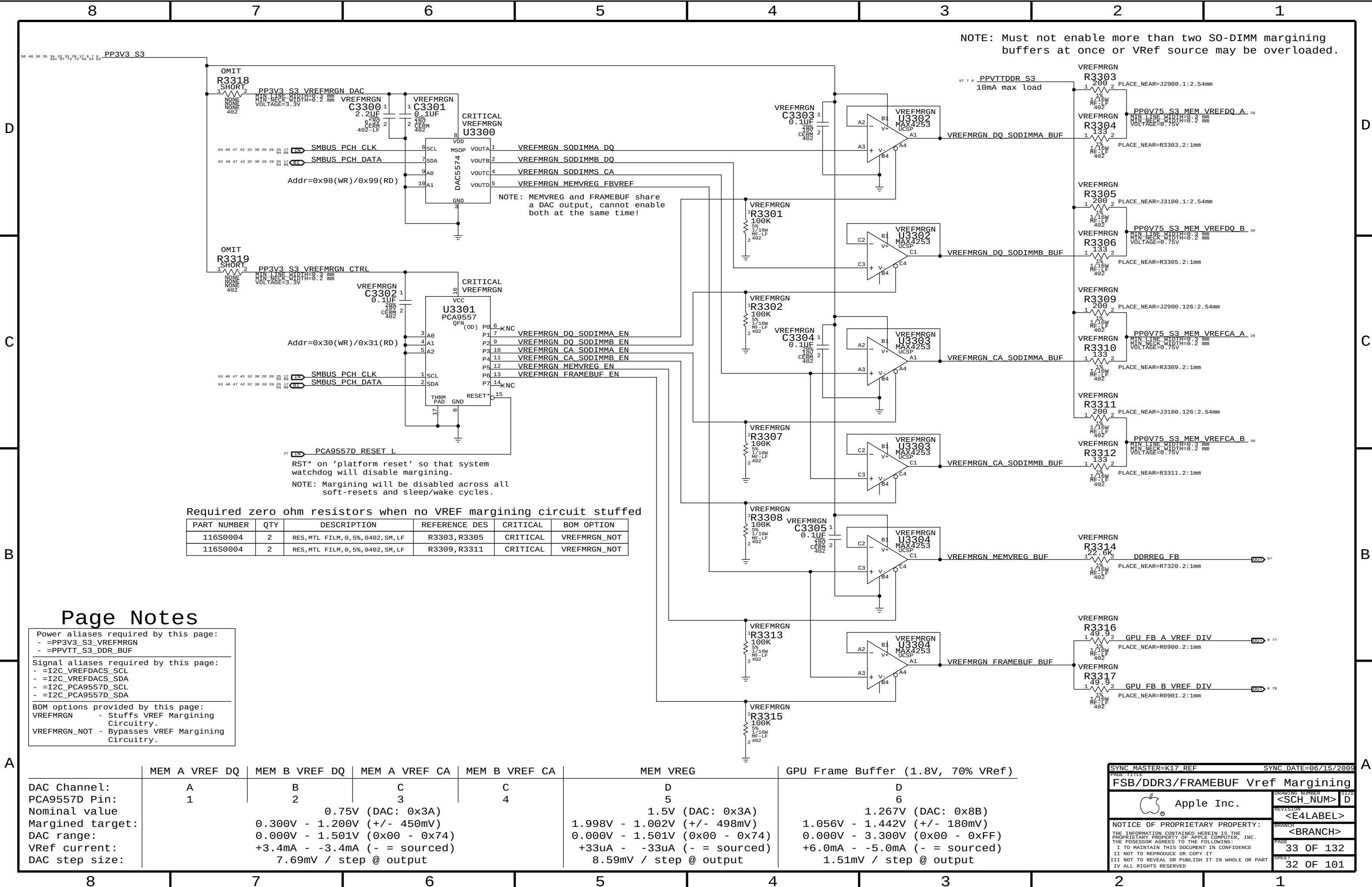
<BRANCH>

PAGE

32 OF 132

SHEET

31 OF 101



Page Notes

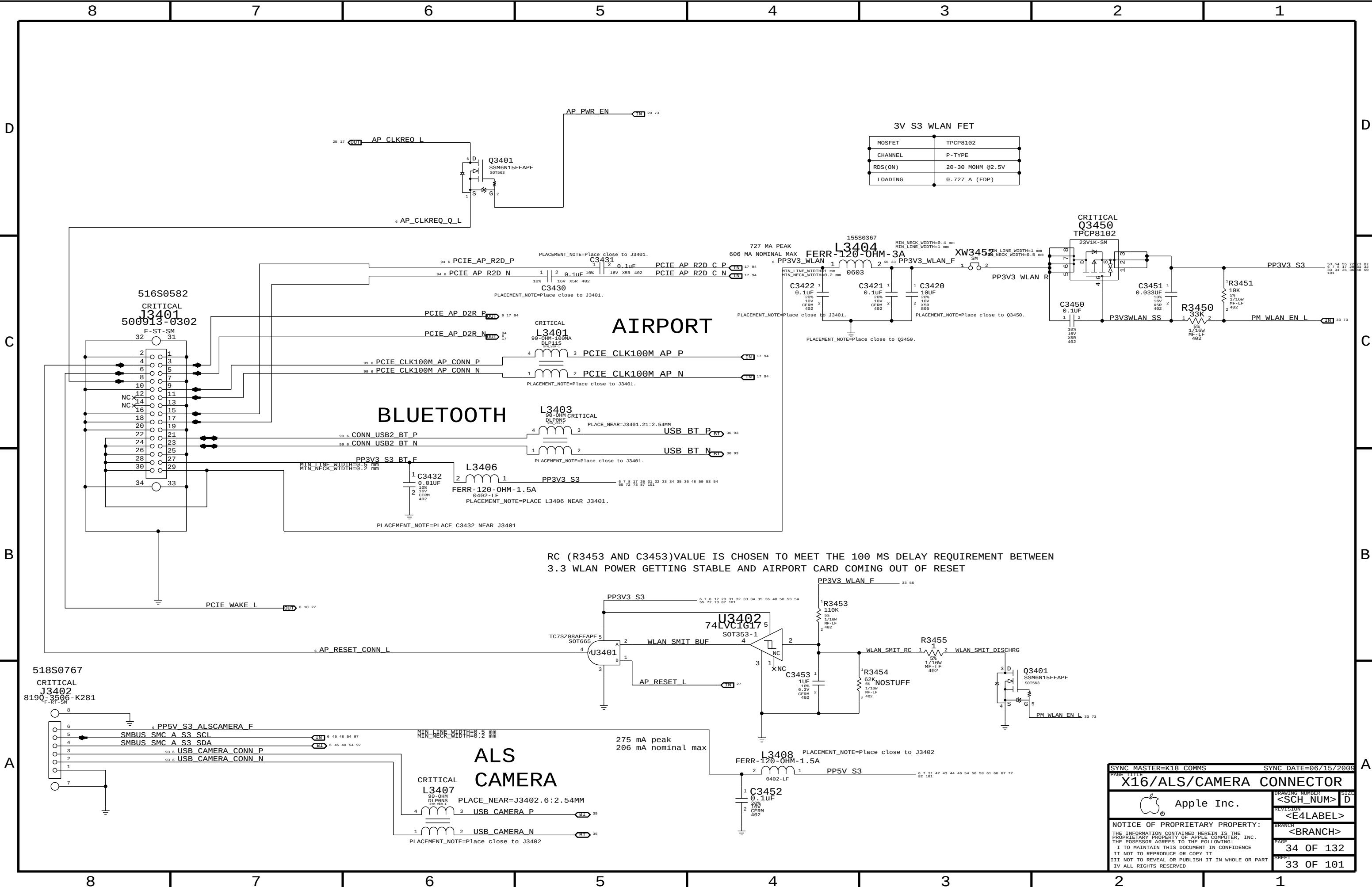
Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACs_SCL
- =I2C_VREFDACs_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN - Stuffs VREF Margining Circuitry.
VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

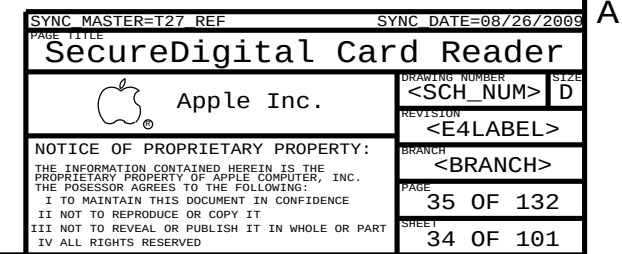
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
Vref current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
FSB/DDR3/FB Vref Margining			
		DRAWING NUMBER	SIZE
Apple Inc.		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE		<BRANCH>	
PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.		PAGE	
THE POSSESSOR AGREES TO THE FOLLOWING:		33 OF 132	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT		32 OF 101	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED.			



3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	0.727 A (EDP)

SYNC MASTER=K18 COMMS		SYNC DATE=06/15/2009	
PAGE TITLE		X16/ALS/CAMERA CONNECTOR	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	34 OF 132
		SHEET	33 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			

[illegible]

8 7 6 5 4 3 2 1

USB HUB-1

PP3V3 S3

FERR-120-0HM-1.5A

C3636

C3637

C3638

C3639

PPUSB_HUB1_VDDPLL3V3

C3618

C3642

C3643

C3644

C3645

C3646

C3647

PP3V3 S3

CRITICAL

Y3600

R3691

C3619

C3620

CRITICAL

USB_HUB1_TEST

USB_HUB1_RESET_L

USB_HUB1_XTAL1

USB_HUB1_XTAL2

USB_HUB1_LOCAL_PWR

USB_HUB1_SMBDATA

USB_HUB1_SMBCLK

USB_HUB1_CFG_SEL1

U3600

QFN

USX2061

OMIT

USBBDN1_DM/PRT_DIS_M1

USBBDN1_DP/PRT_DIS_P1

USBBDN2_DM/PRT_DIS_M2

USBBDN2_DP/PRT_DIS_P2

USBBDN3_DM/PRT_DIS_M3

USBBDN3_DP/PRT_DIS_P3

USBBDN4_DM/PRT_DIS_M4

USBBDN4_DP/PRT_DIS_P4

PRT_PWR1

PRT_PWR2

PRT_PWR3

PRT_PWR4

IPU_OCS1*

IPU_OCS2*

IPU_OCS3*

IPU_OCS4*

RBIAS

VBUS_DET

USBUP_DM

USBUP_DP

THRML_PAD

Camera

IR Receiver

External B

External C

PP3V3 S3

R3604

C3634

U3614

AT24C02B

WP_HUB1

R3692

R3694

R3665

R3666

R3667

R3668

R3682

R3640

R3641

C3641

Q3640

Q3640

D3600

BAT54XV2T1

SEL1

SEL0

DESCRIPTION

K17/K18 configuration:

0

0

Internal Default with Self powered Operation

0

1

SMBUS Slave Config

1

0

Internal Default with Bus powered Operation

1

1

EEPROM Supported

NON_REM1

NON_REM0

DESCRIPTION

0

0

All ports are removable

0

1

Port 1 is non removable

1

0

Port 1 and 2 are non removable

1

1

Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	SMSC USB2514	U3600, U3700	CRITICAL	USBHUB_2514
338S0824	2	SMSC USB2514B	U3600, U3700	CRITICAL	USBHUB_2514B
338S0721	2	SMSC USX2061	U3600, U3700	CRITICAL	USBHUB_2061

BOM GROUP	BOM OPTIONS
HUB1_ALLREM	HUB1_NONREM1_0, HUB1_NONREM0_0
HUB1_1NONREM	HUB1_NONREM1_0, HUB1_NONREM0_1
HUB1_2NONREM	HUB1_NONREM1_1, HUB1_NONREM0_0
HUB1_3NONREM	HUB1_NONREM1_1, HUB1_NONREM0_1

SYNC MASTER=K18 MLB SYNC DATE=10/07/2009

PAGE TITLE

USB HUB 1

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

<SCH_NUM>

REVISION

<E4LABEL>

BRANCH

<BRANCH>

PAGE

36 OF 132

SHEET

35 OF 101

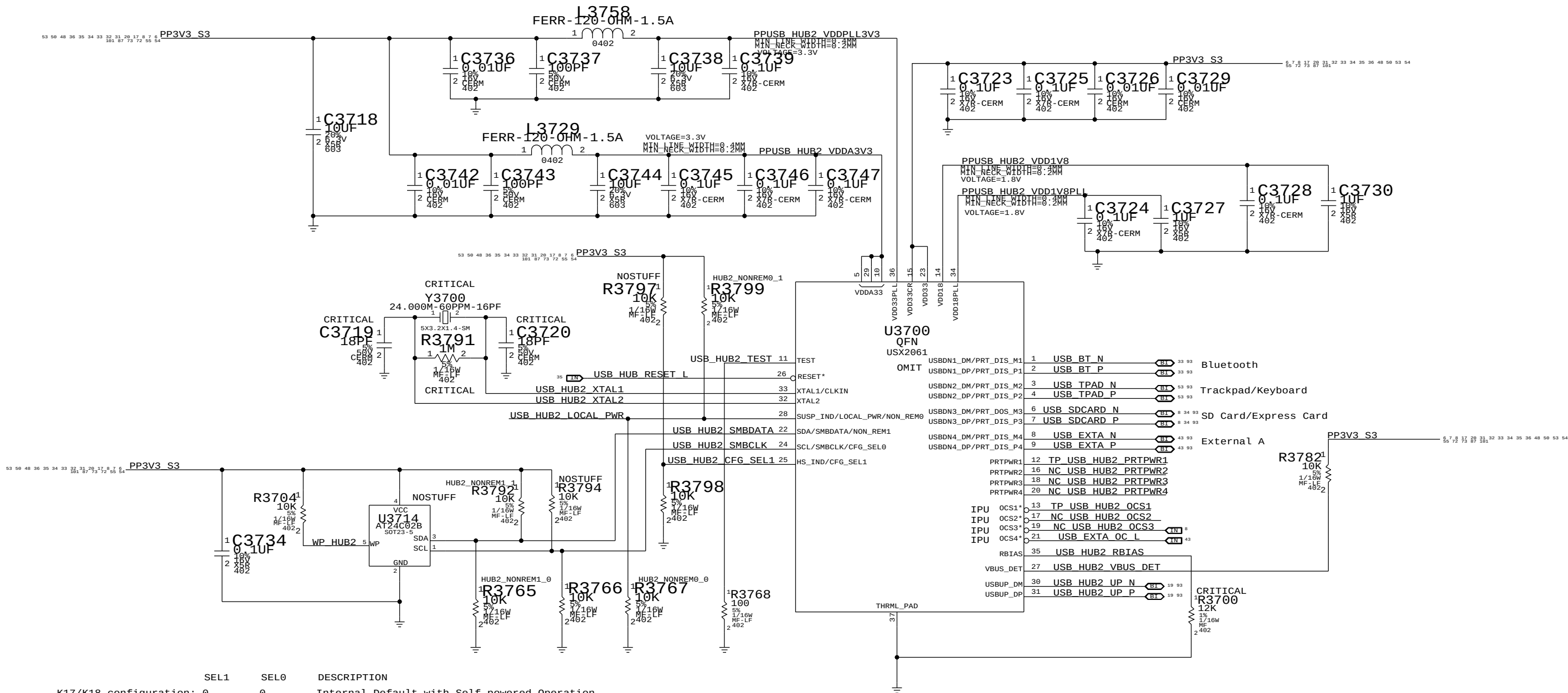
8 7 6 5 4 3 2 1

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	SMSC USB2514	U3600, U3700	CRITICAL	USBHUB_2514
338S0824	2	SMSC USB2514B	U3600, U3700	CRITICAL	USBHUB_2514B
338S0721	2	SMSC USX2061	U3600, U3700	CRITICAL	USBHUB_2061

BOM GROUP	BOM OPTIONS
HUB1_ALLREM	HUB1_NONREM1_0, HUB1_NONREM0_0
HUB1_1NONREM	HUB1_NONREM1_0, HUB1_NONREM0_1
HUB1_2NONREM	HUB1_NONREM1_1, HUB1_NONREM0_0
HUB1_3NONREM	HUB1_NONREM1_1, HUB1_NONREM0_1

SYNC MASTER=K18 MLB SYNC DATE=10/07/2009			
PAGE TITLE			
USB HUB 1			
	Apple Inc.	DRAWING NUMBER	SIZE
		<SCH_NUM>	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
		<E4LABEL>	
THE INFORMATION CONTAINED HEREIN IS THE PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT IIT NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		<BRANCH>	
		PAGE	36 OF 132
		SHEET	35 OF 101

USB HUB-2



	SEL1	SEL0	DESCRIPTION
K17/K18 configuration:	0	0	Internal Default with Self powered Operation
	0	1	SMBUS Slave Config
	1	0	Internal Default with Bus powered Operation
	1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM GROUP	BOM OPTIONS
HUB2_ALLREM	HUB2_NONREM1_0, HUB2_NONREM0_0
HUB2_1NONREM	HUB2_NONREM1_0, HUB2_NONREM0_1
HUB2_2NONREM	HUB2_NONREM1_1, HUB2_NONREM0_0
HUB2_3NONREM	HUB2_NONREM1_1, HUB2_NONREM0_1

SYNC MASTER=K23F

SYNC DATE=10/06/2009

PAGE TITLE

USB HUB 2

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
III NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

<SCH_NUM> D

REVISION

<E4LABEL>

BRANCH

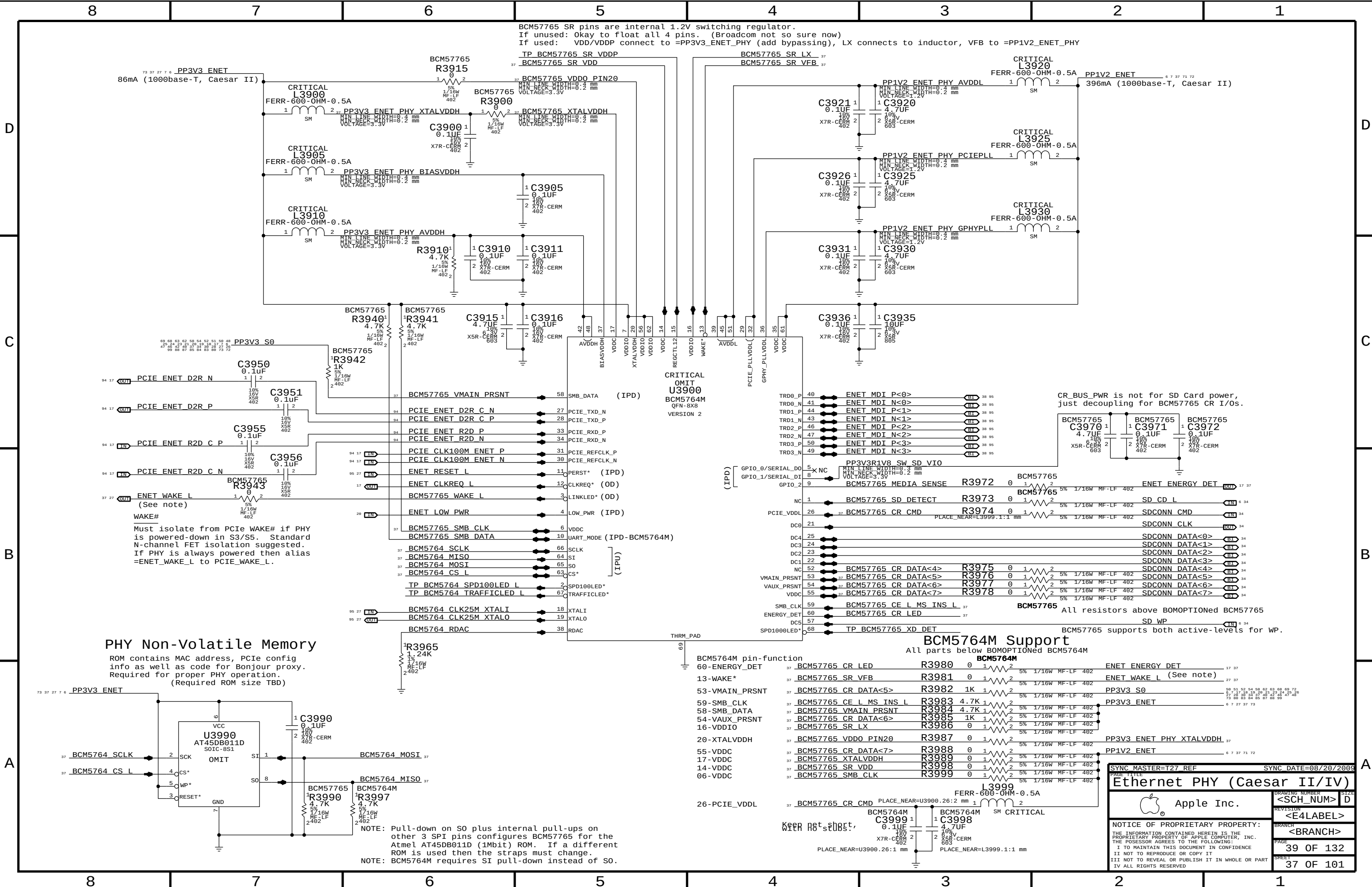
<BRANCH>

PAGE

37 OF 132

SHEET

36 OF 101



PHY Non-Volatile Memory

ROM contains MAC address, PCIE config info as well as code for Bonjour proxy. Required for proper PHY operation. (Required ROM size TBD)

BCM5764M pin-function

- 60-ENERGY_DET
- 13-WAKE*
- 53-VMAIN_PRSENT
- 59-SMB_CLK
- 58-SMB_DATA
- 54-VAUX_PRSENT
- 16-VDDIO
- 20-XTALVDDH
- 55-VDDC
- 17-VDDC
- 14-VDDC
- 06-VDDC
- 26-PCIE_VDDL

BCM5764M Support

All parts below BOMOPTIONED BCM5764M

- BCM57765 CR LED R3980
- BCM57765 SR VFB R3981
- BCM57765 CR DATA<5> R3982
- BCM57765 CE L MS INS L R3983
- BCM57765 VMAIN_PRSENT R3984
- BCM57765 CR DATA<6> R3985
- BCM57765 SR LX R3986
- BCM57765 VDD0 PIN20 R3987
- BCM57765 CR DATA<7> R3988
- BCM57765 XTALVDDH R3989
- BCM57765 SR VDD R3990
- BCM57765 SMB_CLK R3999

SYNC MASTER=T27 REF

SYNC DATE=08/20/2009

Ethernet PHY (Caesar II/IV)

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

<SCH_NUM>

<E4LABEL>

<BRANCH>

39 OF 132

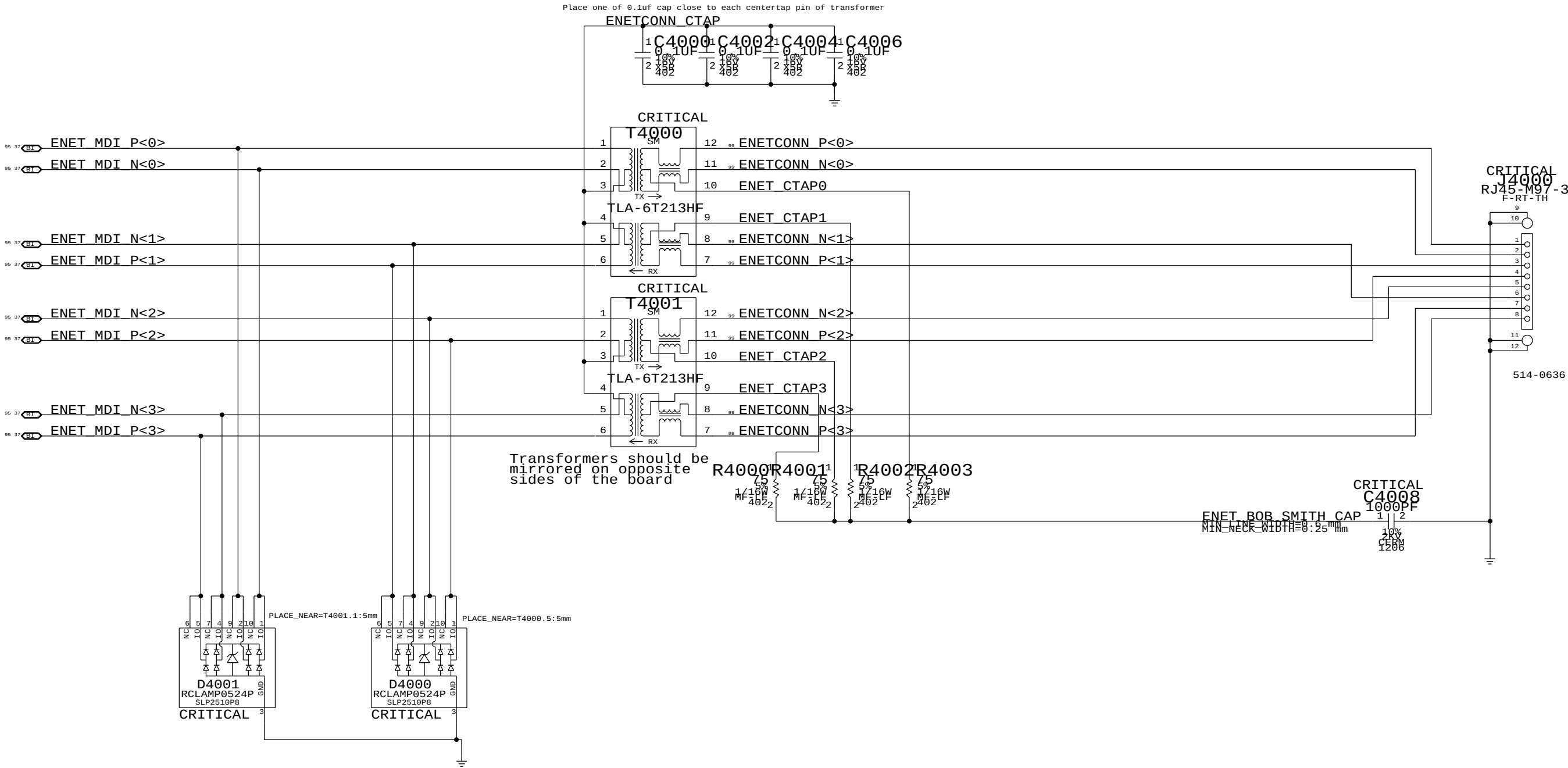
37 OF 101

Page Notes

Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
Ethernet Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	40 OF 132
		SHEET	38 OF 101

Page Notes

Power aliases required by this page:
- =PPBUS_S5_FWPWRSM (system supply for bus power)
- =PP3V3_FW_LATEVG_ACTIVE
- =PPVP_FW_SUMNODE (power passthru summation node)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

3.3V FW FET

I(max) = 1.7A (85C)

1.05V FW FET

FireWire Port Power Switch

Late-VG Protection

SYNC MASTER=K19 MLB		SYNC DATE=05/29/2009	
PAGE TITLE			
FireWire Port Power			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	<BRANCH>
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	42 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	40 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

D

C

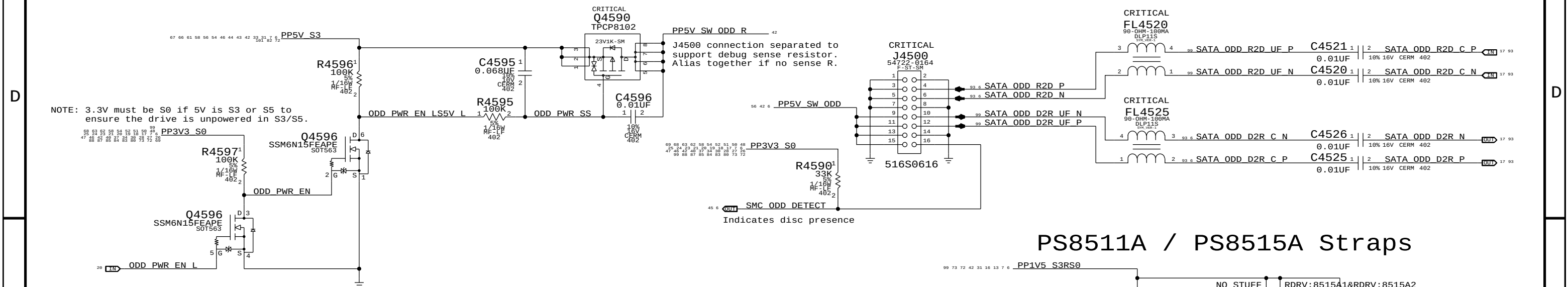
B

A

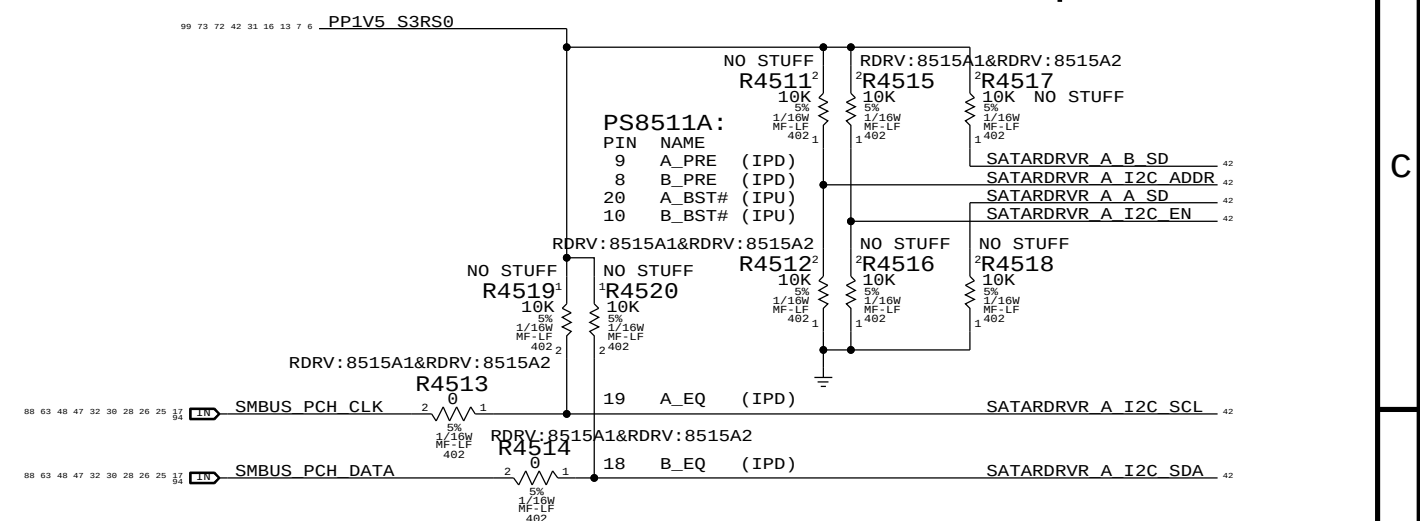
DRAWING NUMBER	SIZE
<SCH_NUM>	D
REVISION	
<E4LABEL>	
BRANCH	
<BRANCH>	
PAGE	
43 OF 132	
SHEET	
41 OF 101	

ODD Power Control

SATA ODD Connector



PS8511A / PS8515A Straps

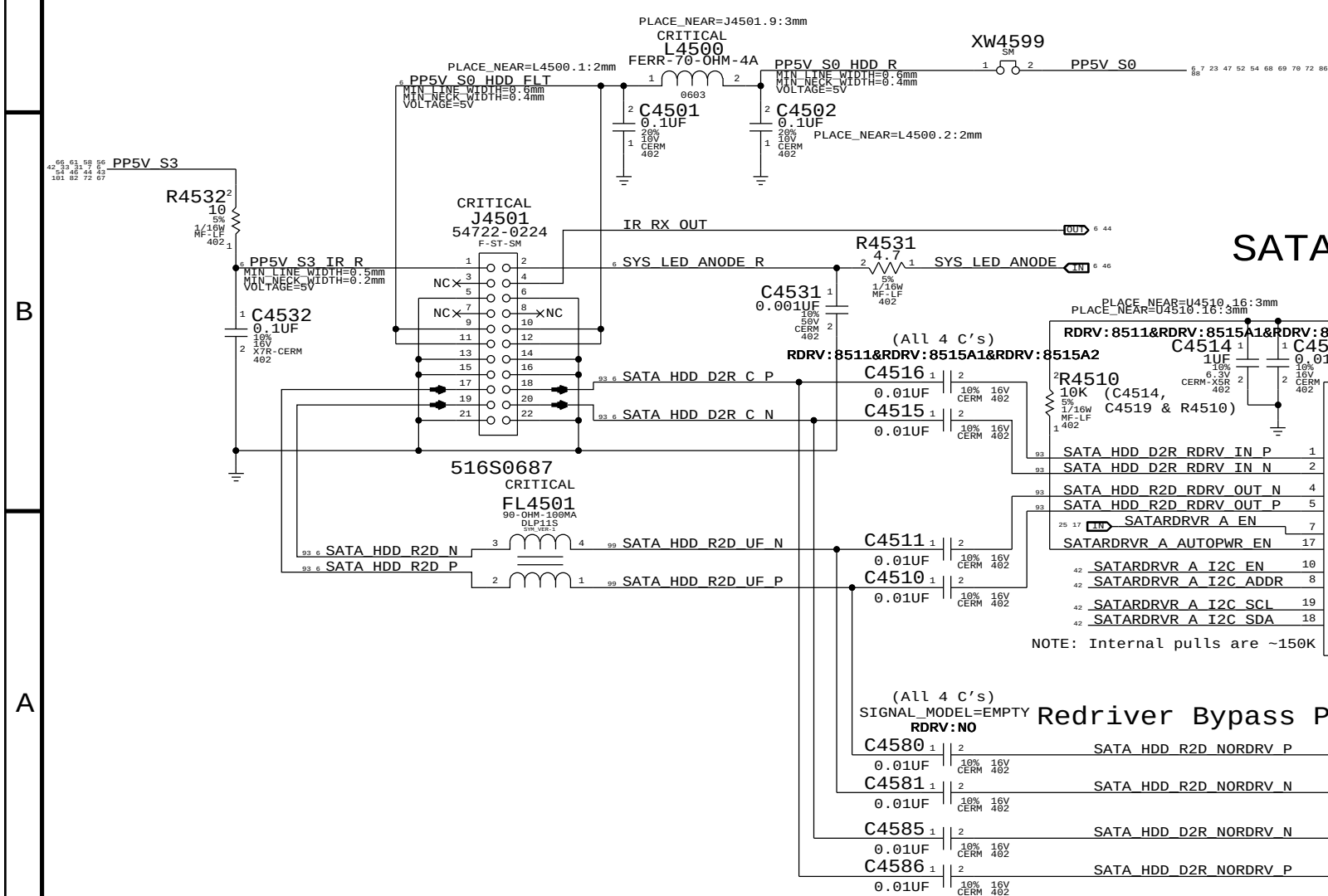


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S0769	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV:8511
338S0778	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV:8515A1

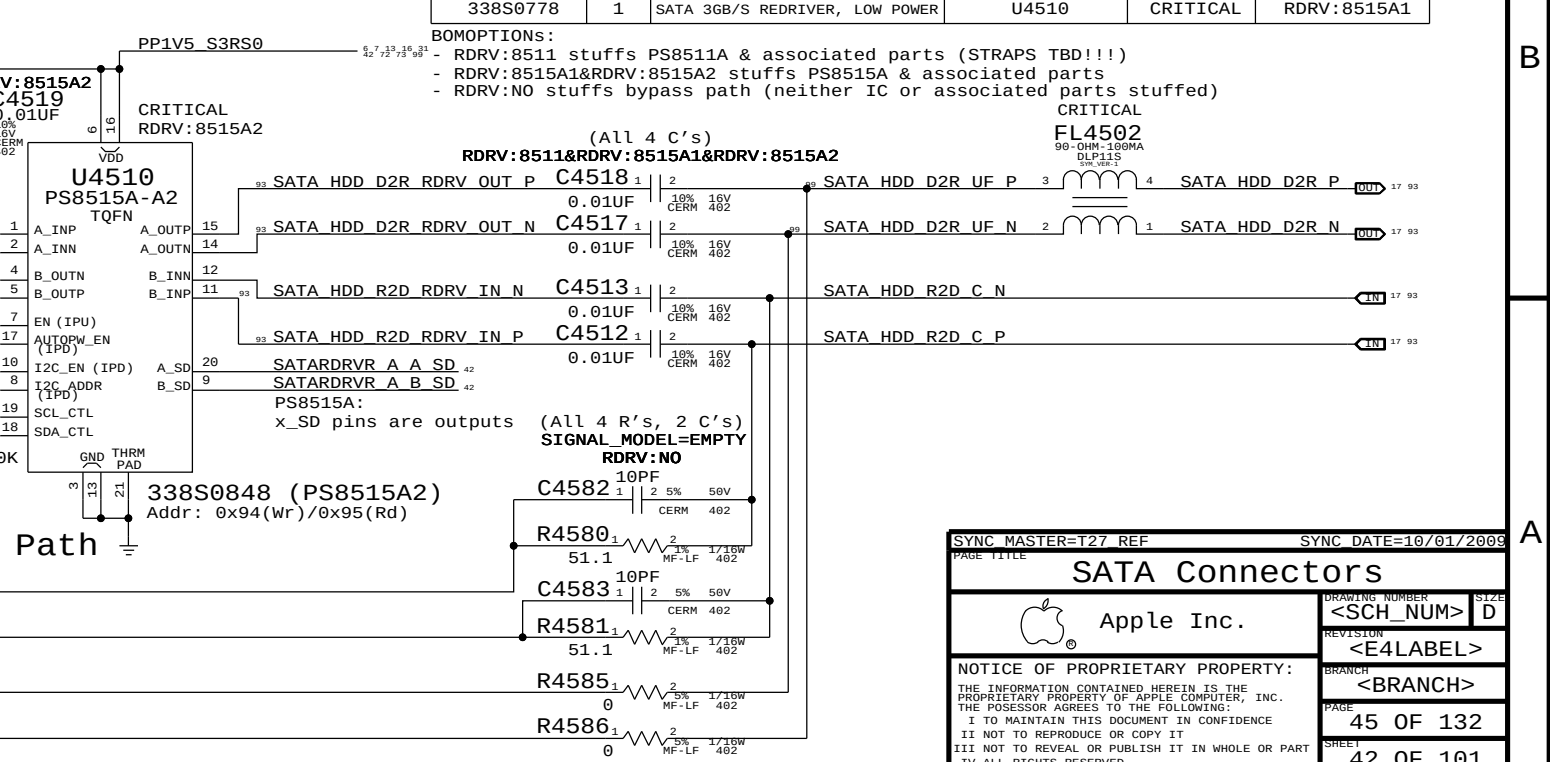
BOMOPTIONS:

- RDRV:8511 stuffs PS8511A & associated parts (STRAPS TBD!!!)
- RDRV:8515A1&RDRV:8515A2 stuffs PS8515A & associated parts
- RDRV:NO stuffs bypass path (neither IC or associated parts stuffed)

SATA HDD/IR/SIL Connector



SATA Redriver



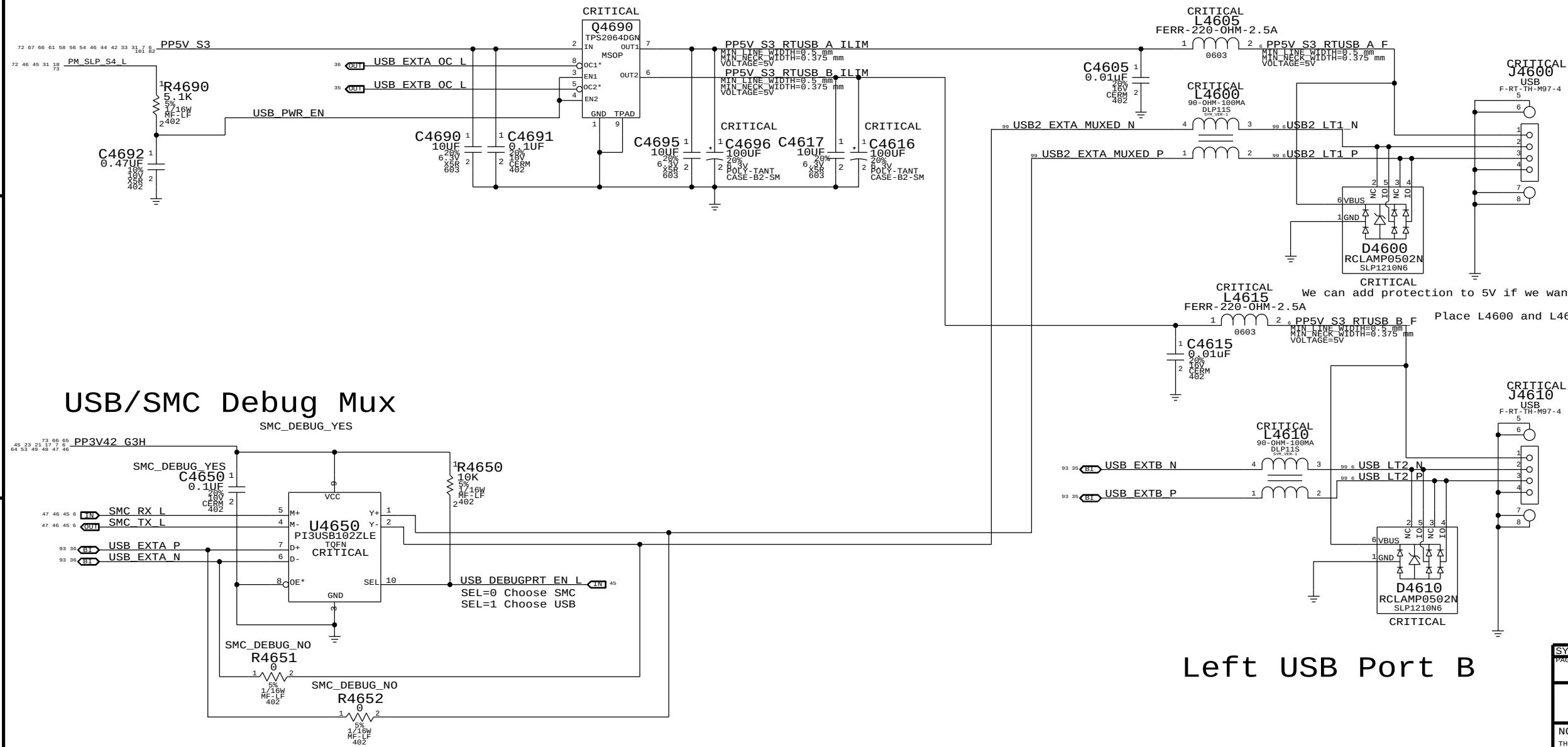
SYNC MASTER=T27 REF		SYNC DATE=10/01/2009	
PAGE TITLE			
SATA Connectors			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
		REVISION <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH <BRANCH>	
		PAGE 45 OF 132	
		SHEET 42 OF 101	

Port Power Switch

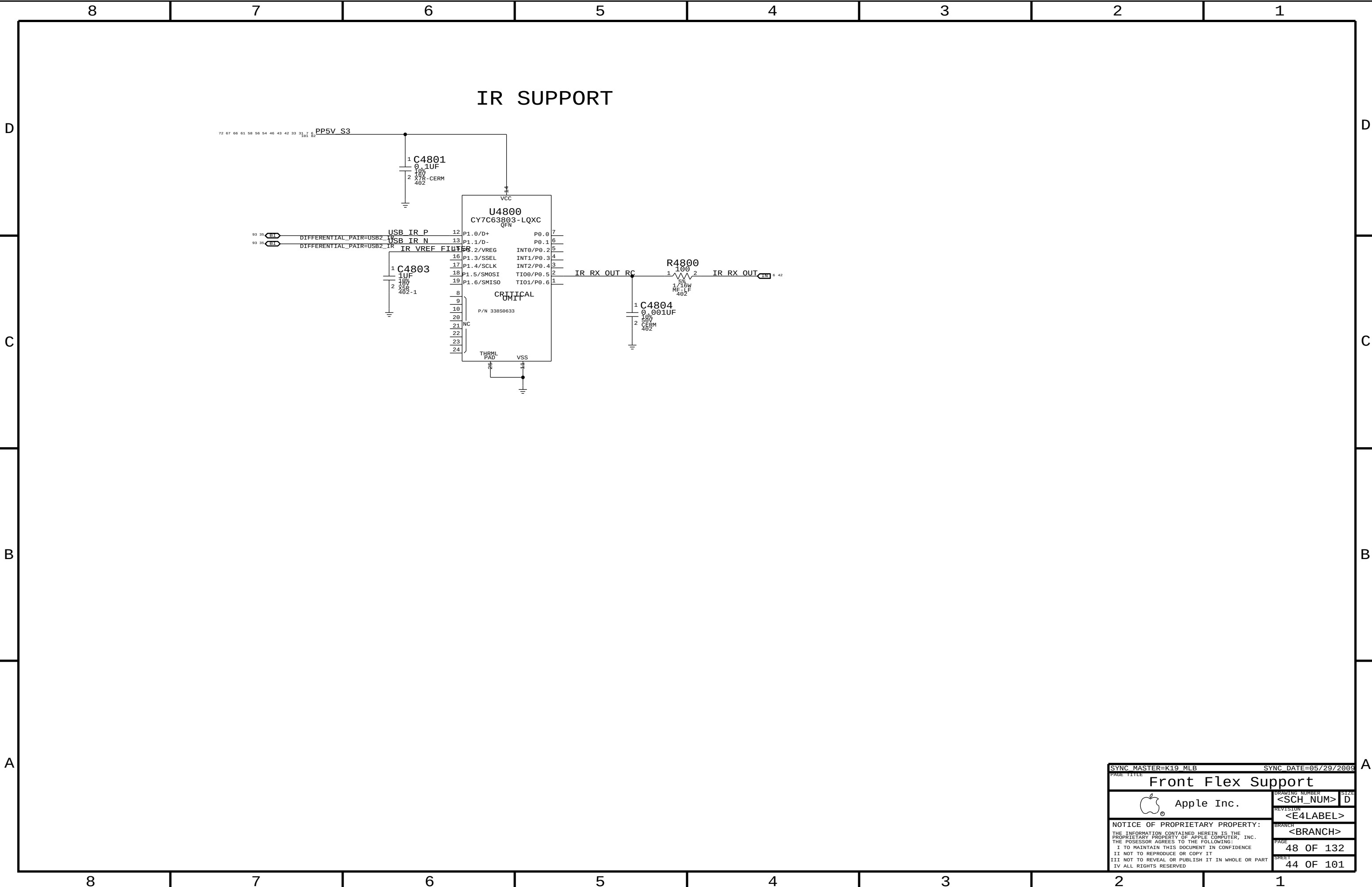
Left USB Port A

USB/SMC Debug Mux

Left USB Port B



SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
External USB Connectors		DRAWING NUMBER	
Apple Inc.		<SCH_NUM>	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		<E4LABEL>	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		<BRANCH>	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		PAGE	
IV ALL RIGHTS RESERVED		46 OF 132	
		SHEET	
		43 OF 101	

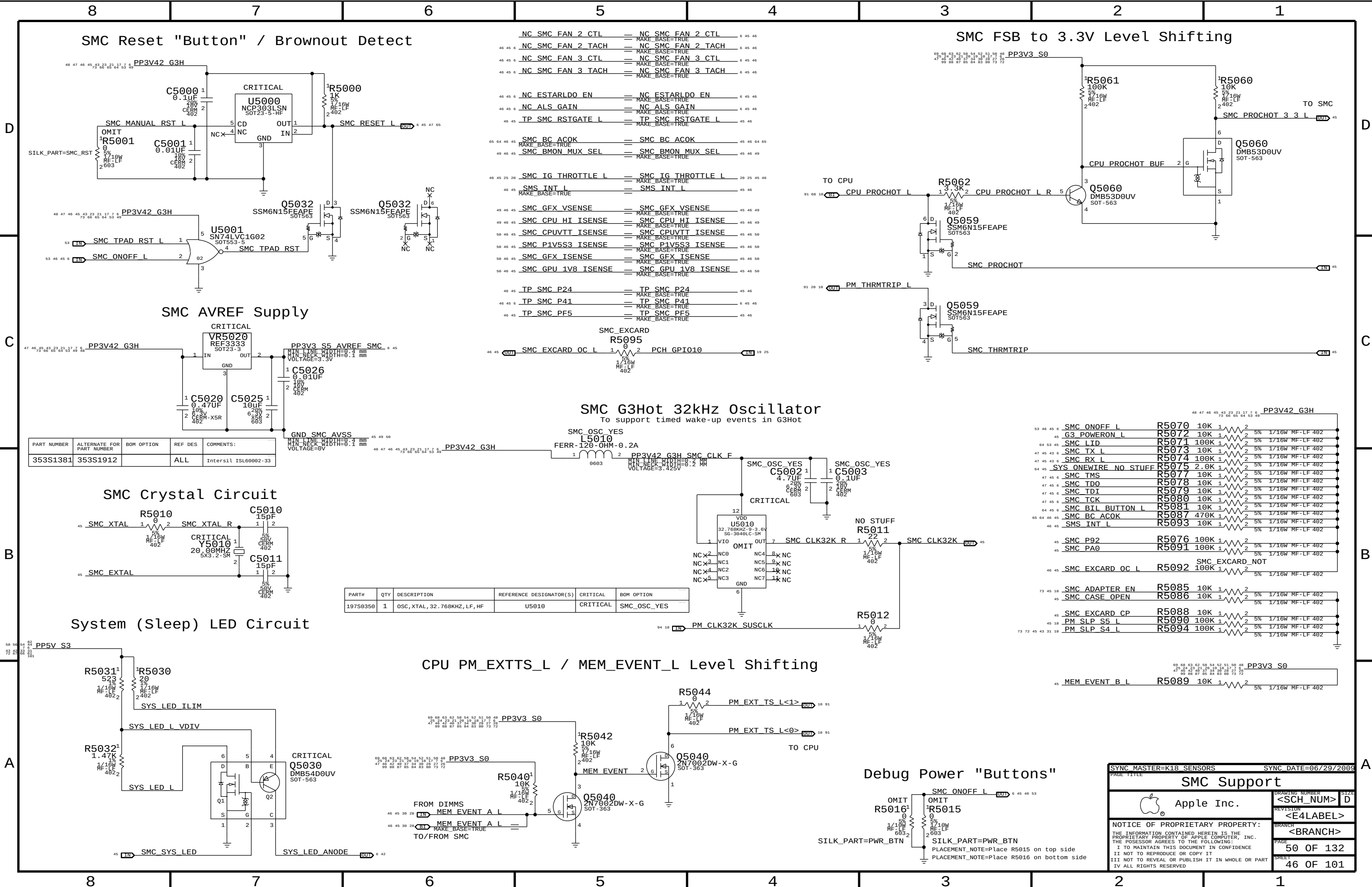


A

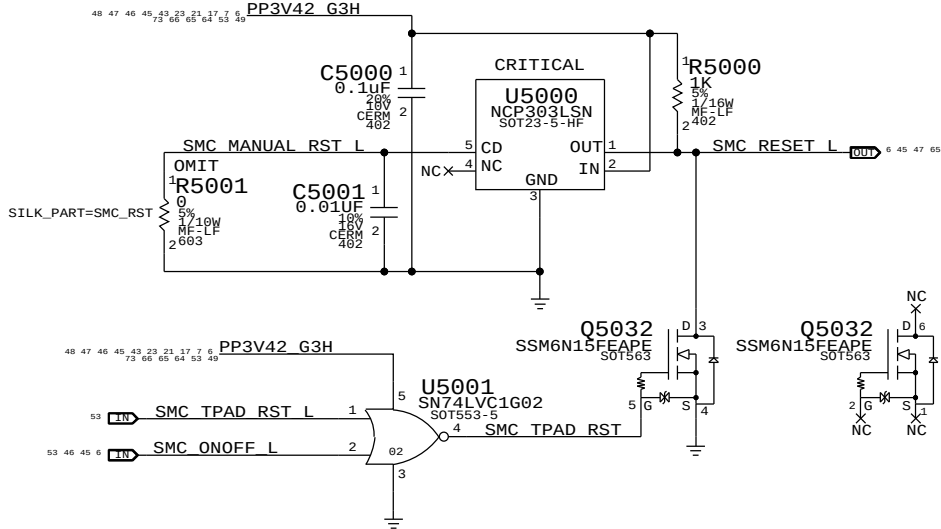


NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

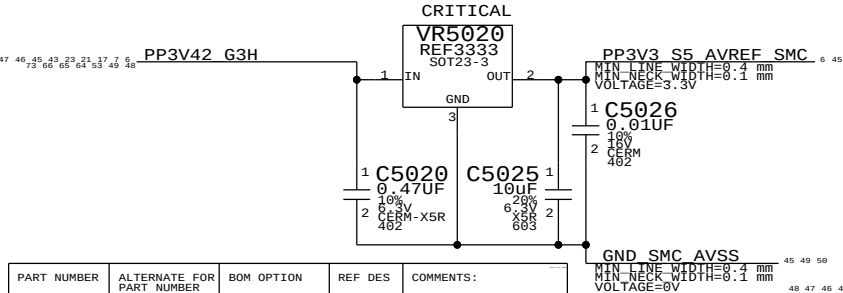




SMC Reset "Button" / Brownout Detect

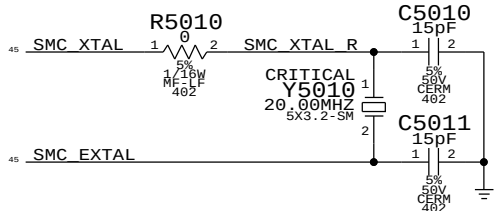


SMC AVREF Supply



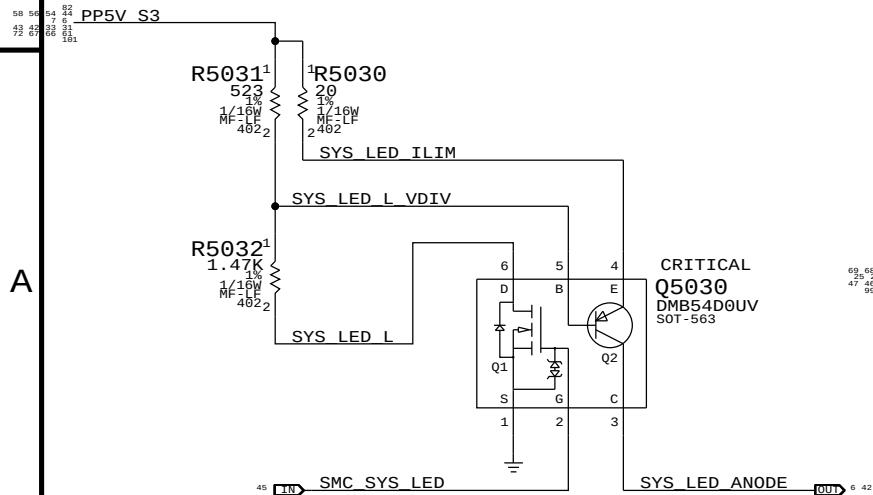
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	Intersil ISL60002-33

SMC Crystal Circuit

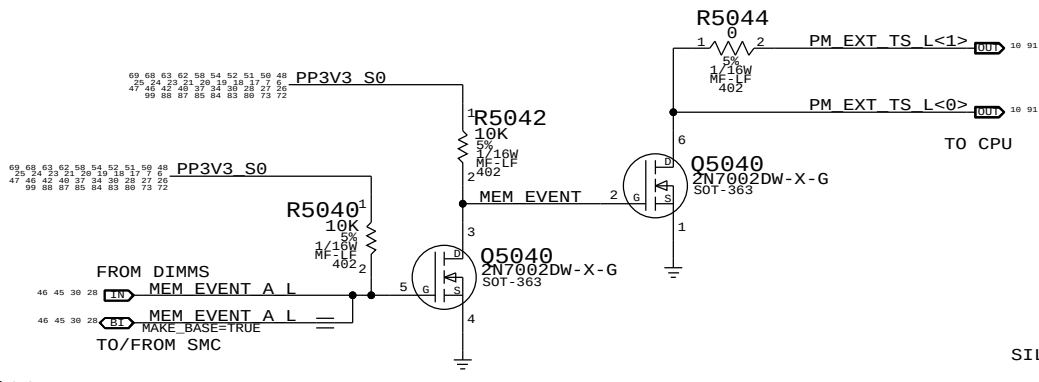


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
197S0350	1	OSC, XTAL, 32.768KHZ, LF, HF	U5010	CRITICAL	SMC_OSC_YES

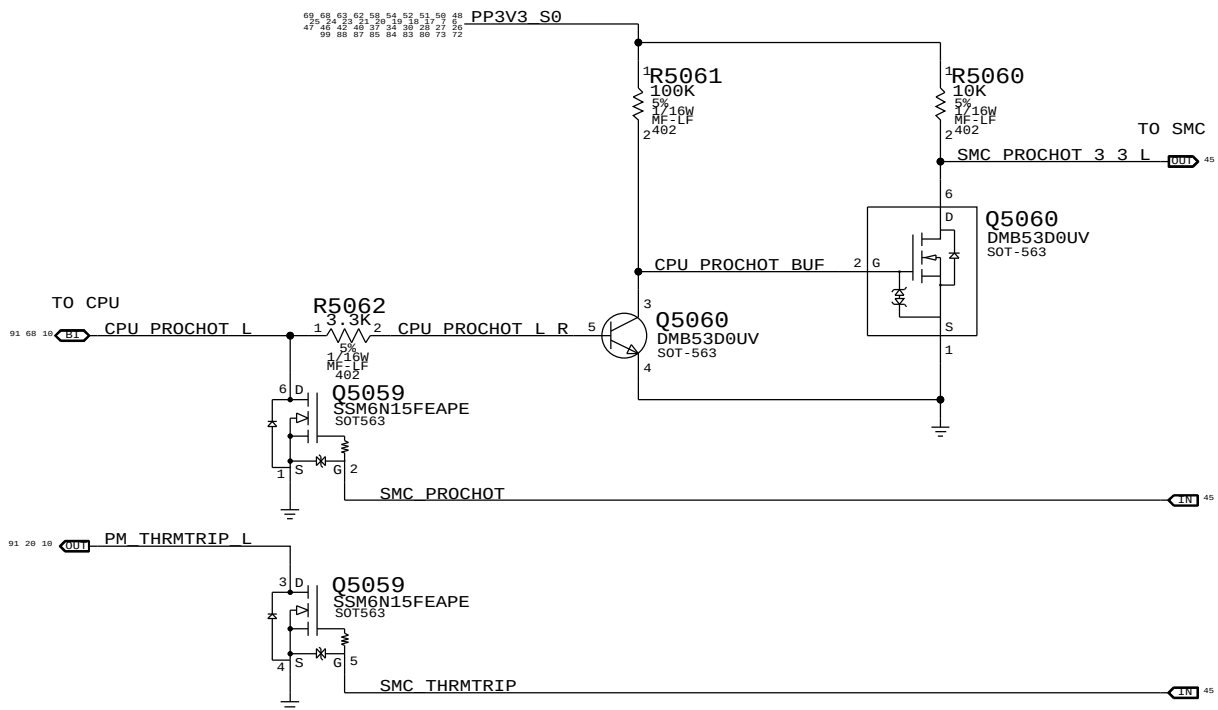
System (Sleep) LED Circuit



CPU PM_EXTTS_L / MEM_EVENT_L Level Shifting

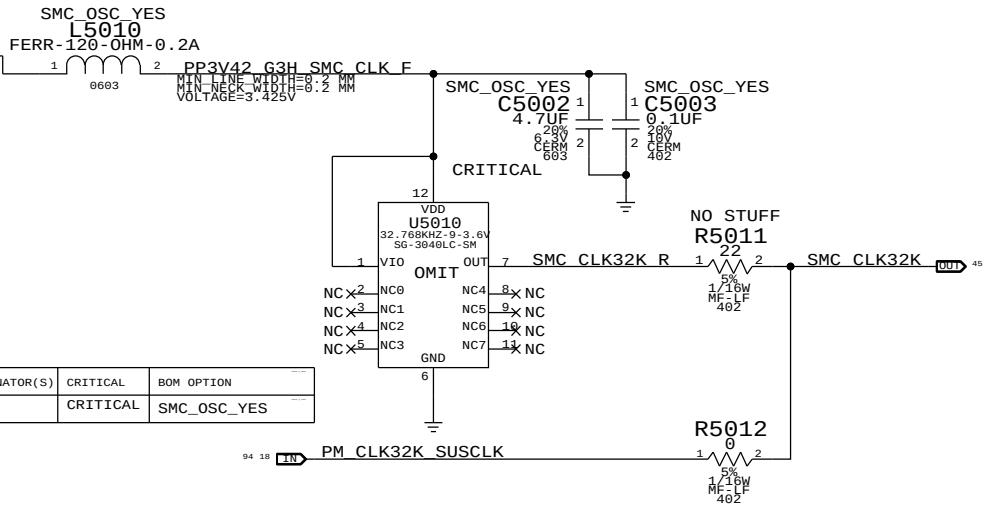


SMC FSB to 3.3V Level Shifting



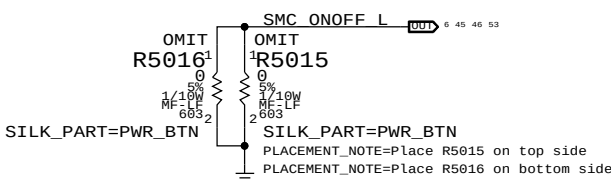
SMC G3Hot 32kHz Oscillator

To support timed wake-up events in G3Hot



53 46 45 6	SMC_ONOFF_L	R5070	10K	1	2	5%	1/16W	MF-LF	402
45	G3_POWERON_L	R5072	10K	1	2	5%	1/16W	MF-LF	402
64 53 45	SMC_LID	R5071	100K	1	2	5%	1/16W	MF-LF	402
47 45 43 6	SMC_TX_L	R5073	10K	1	2	5%	1/16W	MF-LF	402
64 45 6	SMC_RX_L	R5074	100K	1	2	5%	1/16W	MF-LF	402
47 45 6	SYS_ONEWIRE_NO_STUFF	R5075	2.0K	1	2	5%	1/16W	MF-LF	402
47 45 6	SMC_TMS	R5077	10K	1	2	5%	1/16W	MF-LF	402
47 45 6	SMC_TDO	R5078	10K	1	2	5%	1/16W	MF-LF	402
47 45 6	SMC_TDI	R5079	10K	1	2	5%	1/16W	MF-LF	402
47 45 6	SMC_TCK	R5080	10K	1	2	5%	1/16W	MF-LF	402
64 45 6	SMC_BIL_BUTTON_L	R5081	10K	1	2	5%	1/16W	MF-LF	402
65 64 45 6	SMC_BC_ACOK	R5087	470K	1	2	5%	1/16W	MF-LF	402
46 45	SMS_INT_L	R5093	10K	1	2	5%	1/16W	MF-LF	402
45	SMC_P92	R5076	100K	1	2	5%	1/16W	MF-LF	402
45	SMC_PA0	R5091	100K	1	2	5%	1/16W	MF-LF	402
46 45	SMC_EXCARD_OC_L	R5092	100K	1	2	5%	1/16W	MF-LF	402
73 45 18	SMC_ADAPTER_EN	R5085	10K	1	2	5%	1/16W	MF-LF	402
45	SMC_CASE_OPEN	R5086	10K	1	2	5%	1/16W	MF-LF	402
45	SMC_EXCARD_CP	R5088	10K	1	2	5%	1/16W	MF-LF	402
45 18	PM_SLP_S5_L	R5090	100K	1	2	5%	1/16W	MF-LF	402
73 72 45 43 31 18	PM_SLP_S4_L	R5094	100K	1	2	5%	1/16W	MF-LF	402

Debug Power "Buttons"



SYNC MASTER=K18_SENSORS

SYNC DATE=06/29/2009

Apple Inc.

Apple logo

SMC Support

<SCH_NUM> D

<E4LABEL>

<BRANCH>

50 OF 132

46 OF 101

NOTICE OF PROPRIETARY PROPERTY:

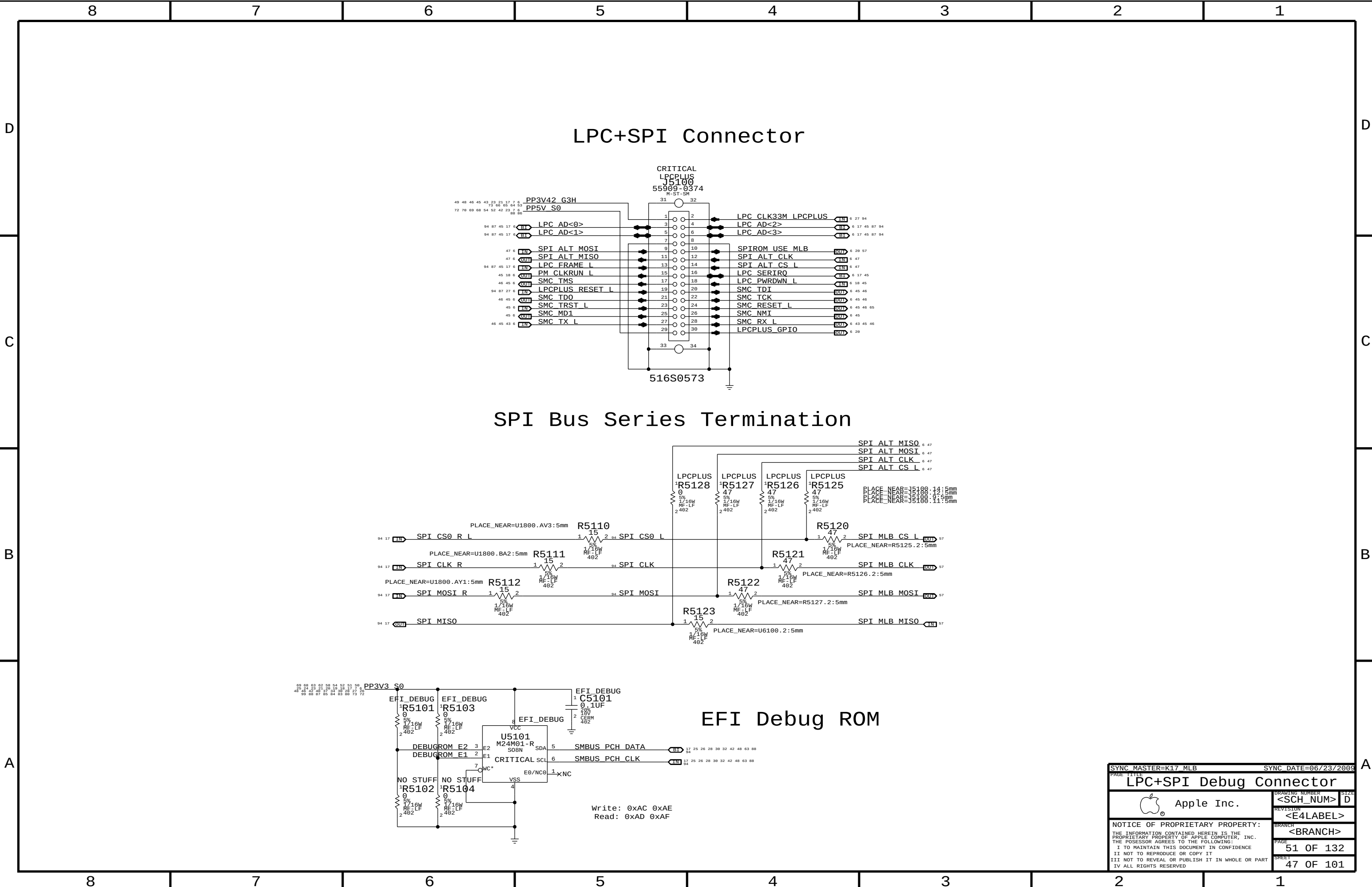
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

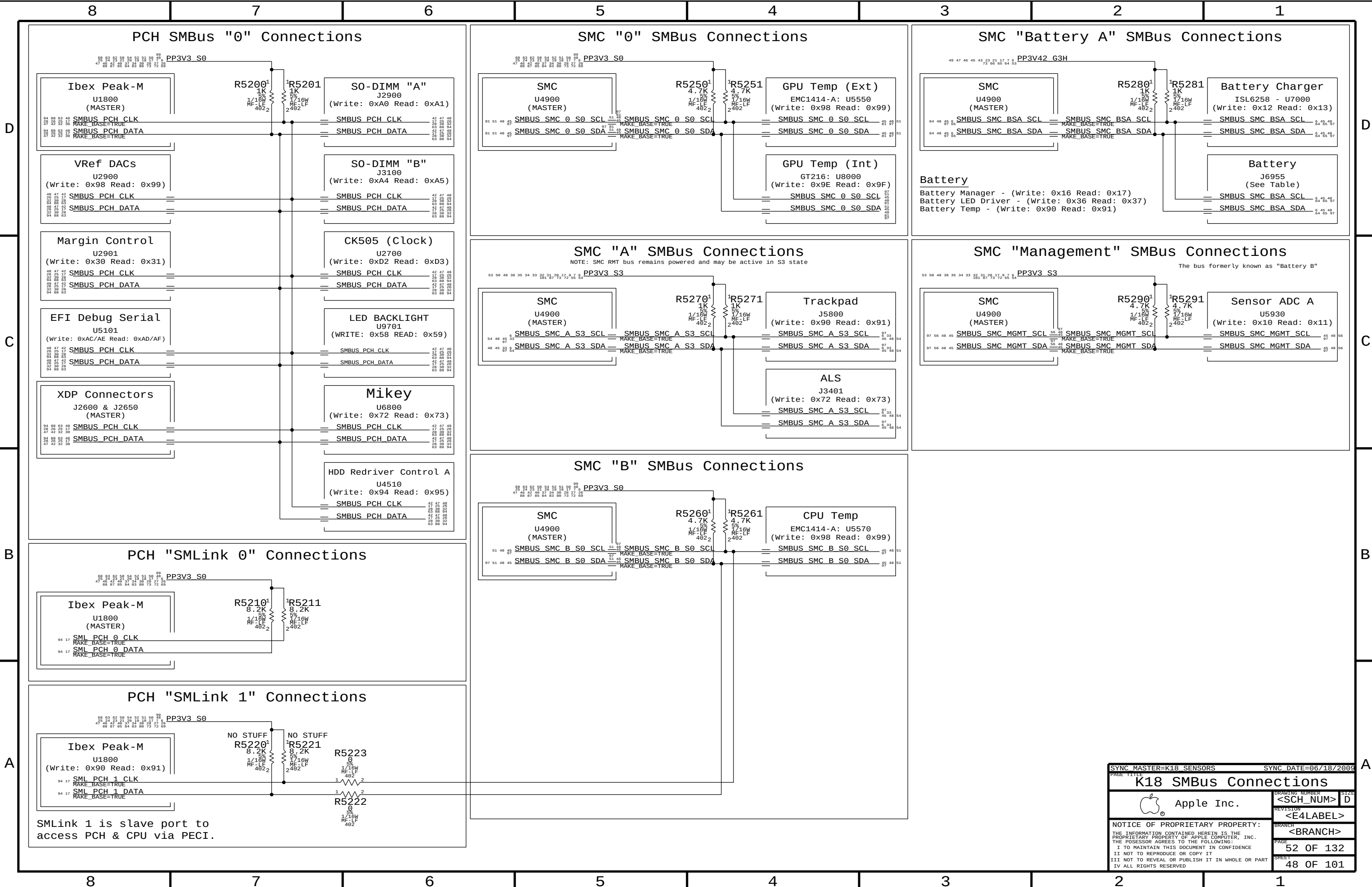
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

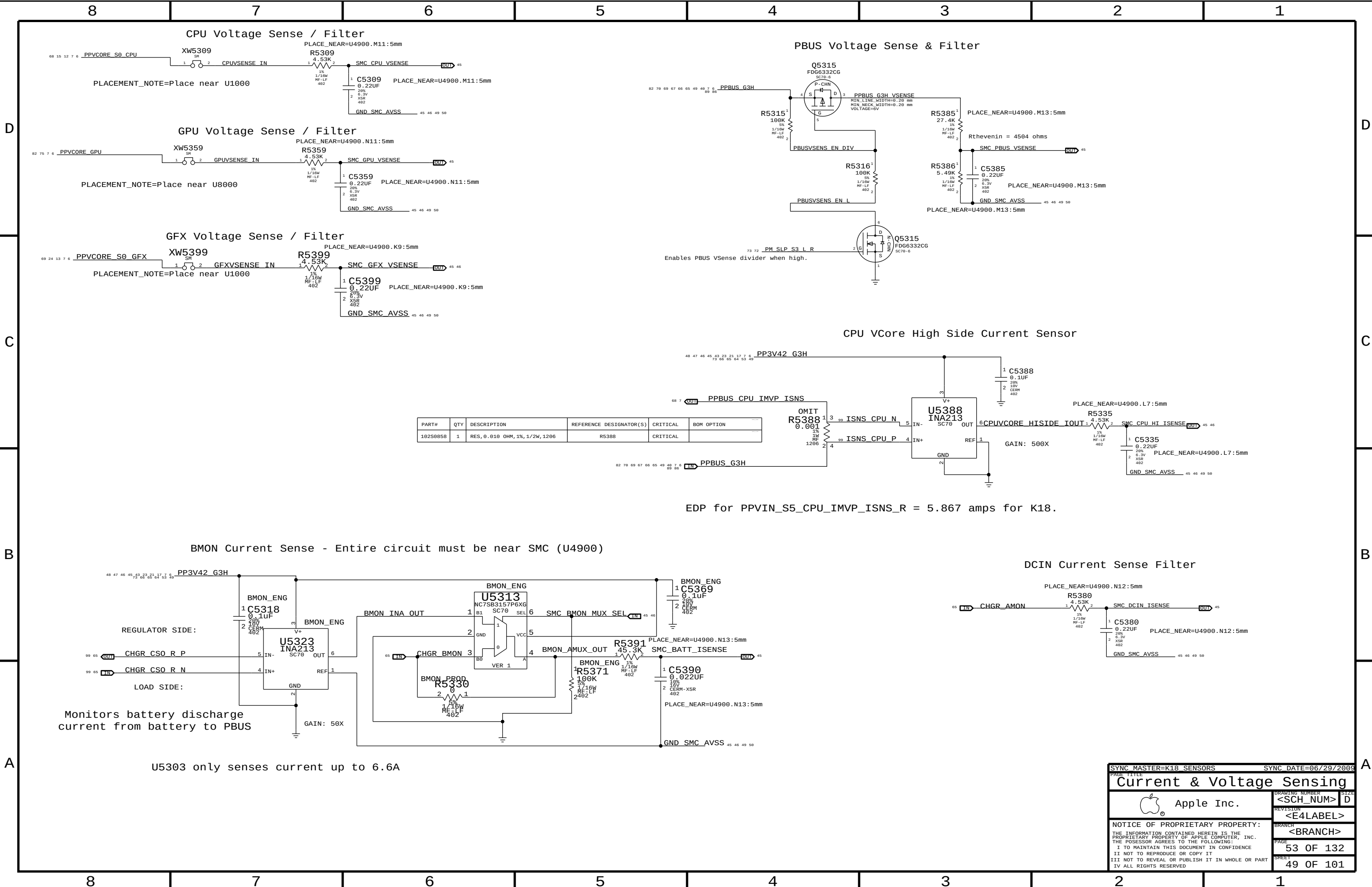
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

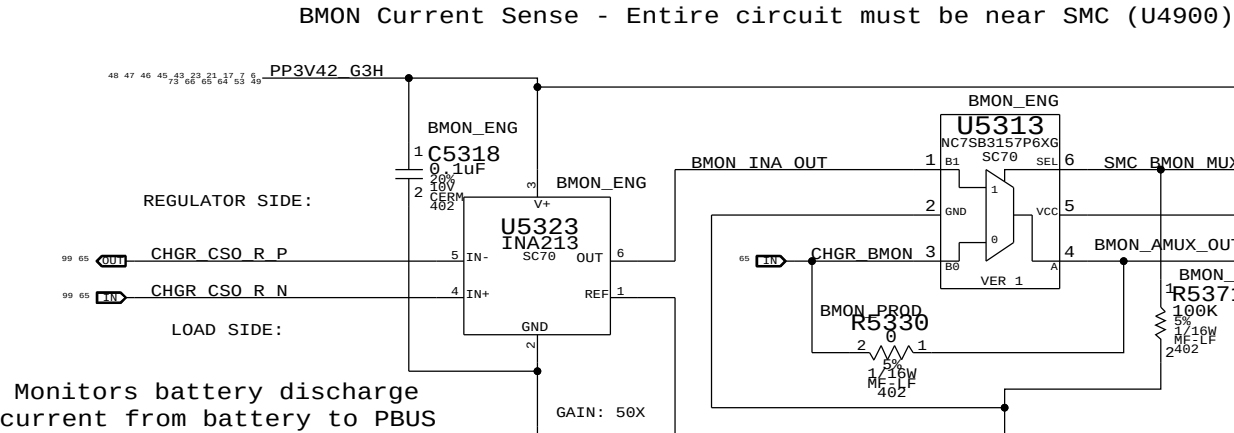
IV ALL RIGHTS RESERVED





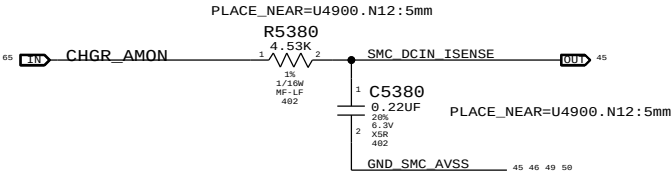


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
10250858	1	RES, 0.010 OHM, 1%, 1/2W, 1206	R5388	CRITICAL	



U5303 only senses current up to 6.6A

DCIN Current Sense Filter



SYNC MASTER=K18 SENSORS

SYNC DATE=06/29/2009

Current & Voltage Sensing

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

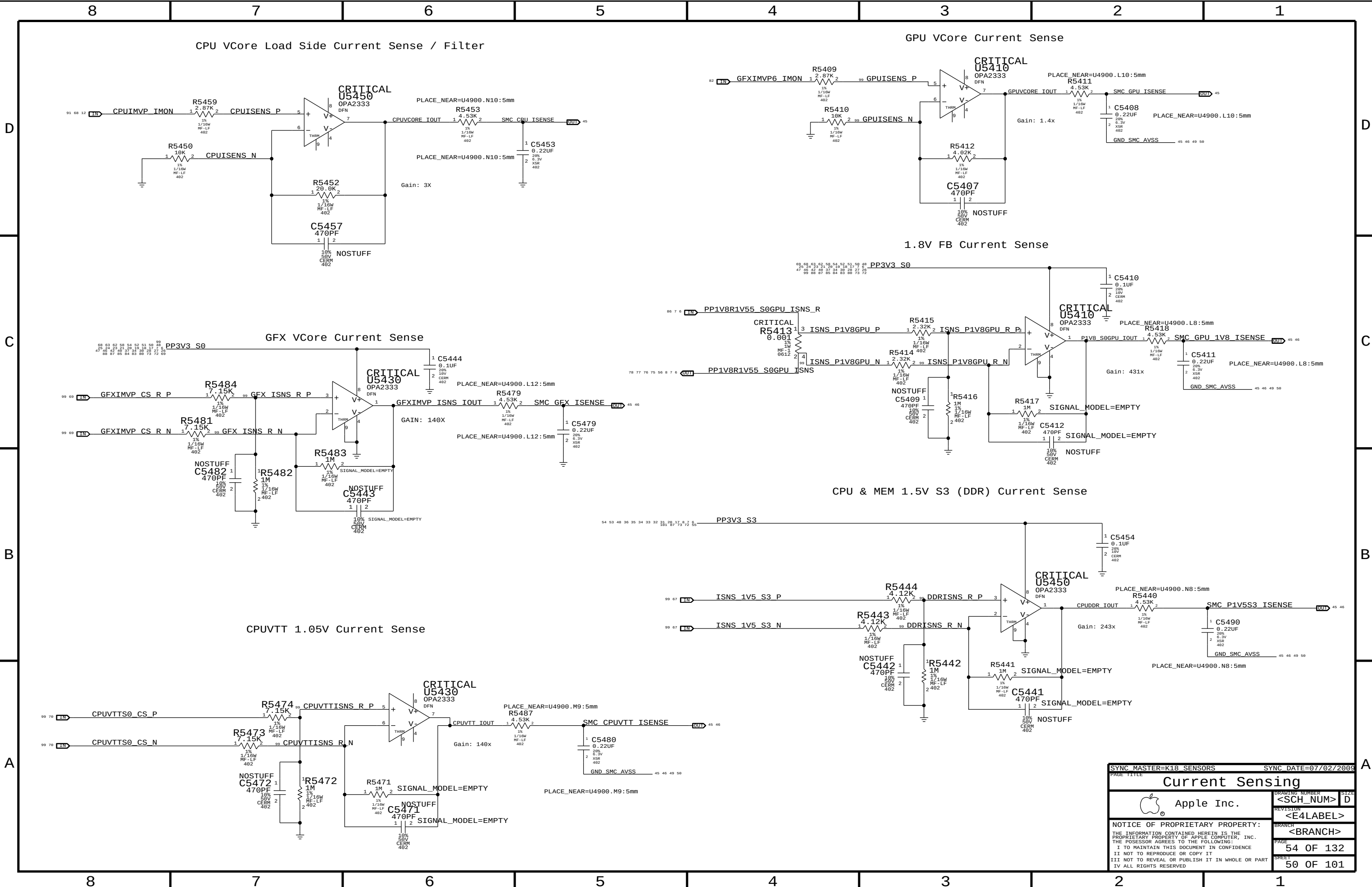
DRAWING NUMBER
<SCH_NUM> D

REVISION
<E4LABEL>

BRANCH
<BRANCH>

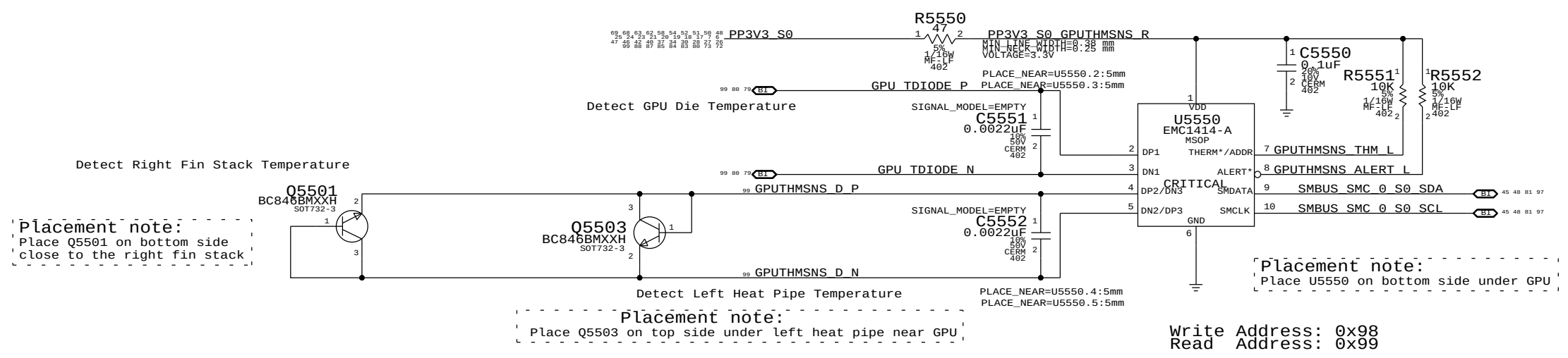
PAGE
53 OF 132

SHEET
49 OF 101

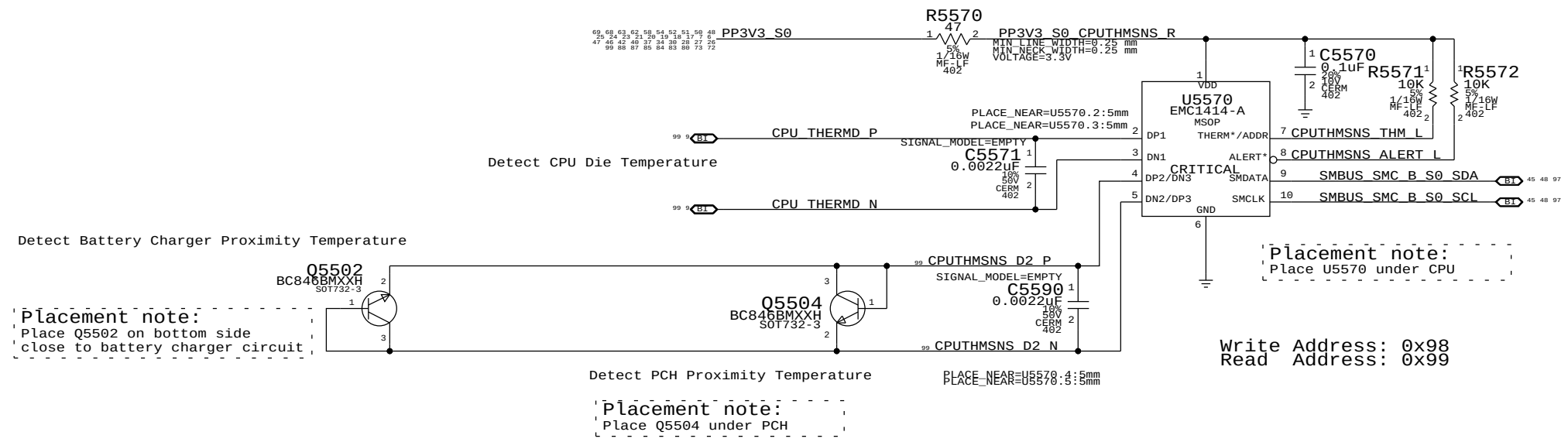


SYNC MASTER=K18 SENSORS		SYNC DATE=07/02/2009	
PAGE TITLE		Current Sensing	
Apple Inc.		DRAWING NUMBER	<SCH_NUM> D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	<E4LABEL>
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	<BRANCH>
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	54 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	50 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

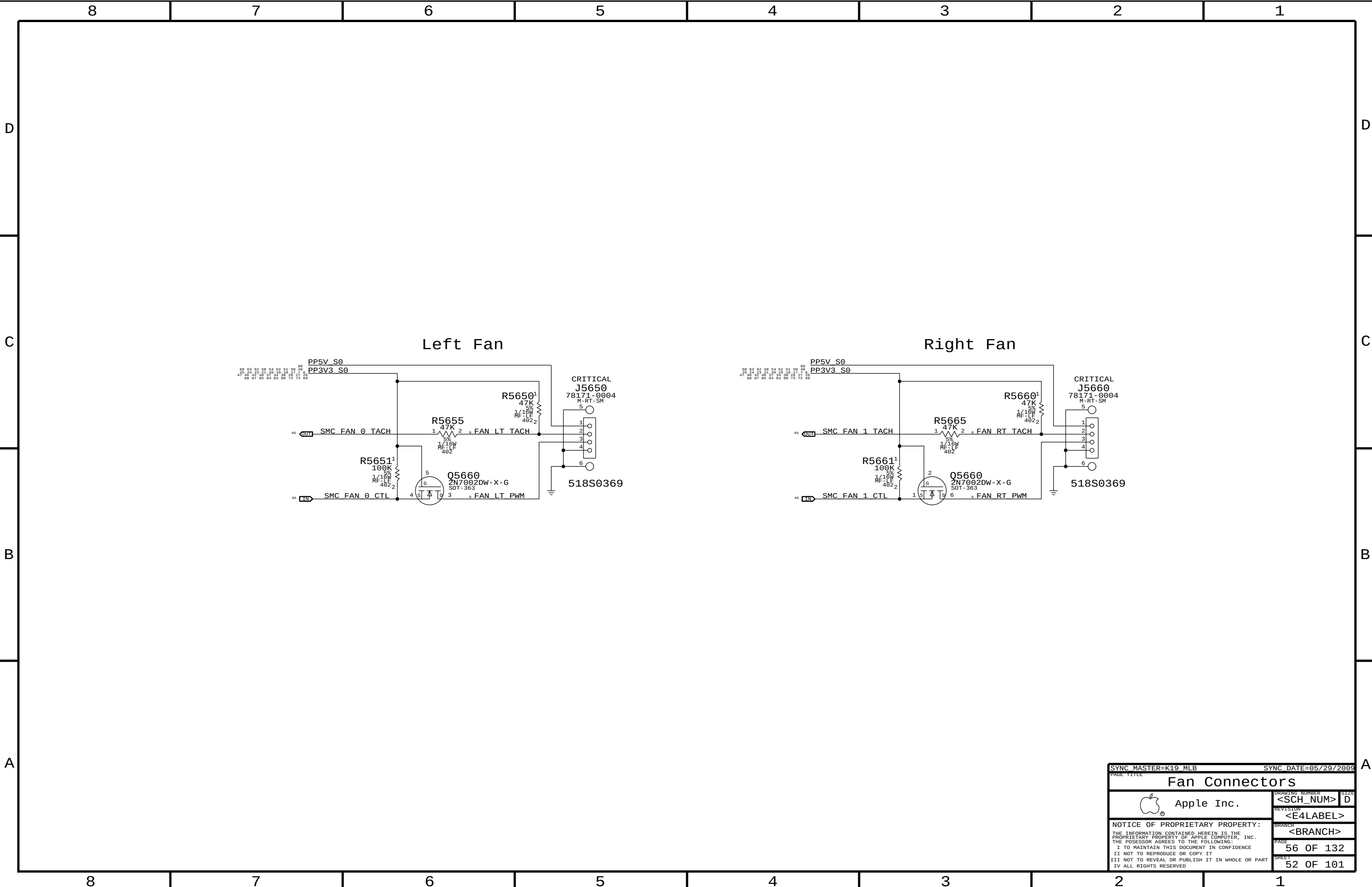
GPU Proximity/GPU Die/Left Heat Pipe/Right Fin Stack

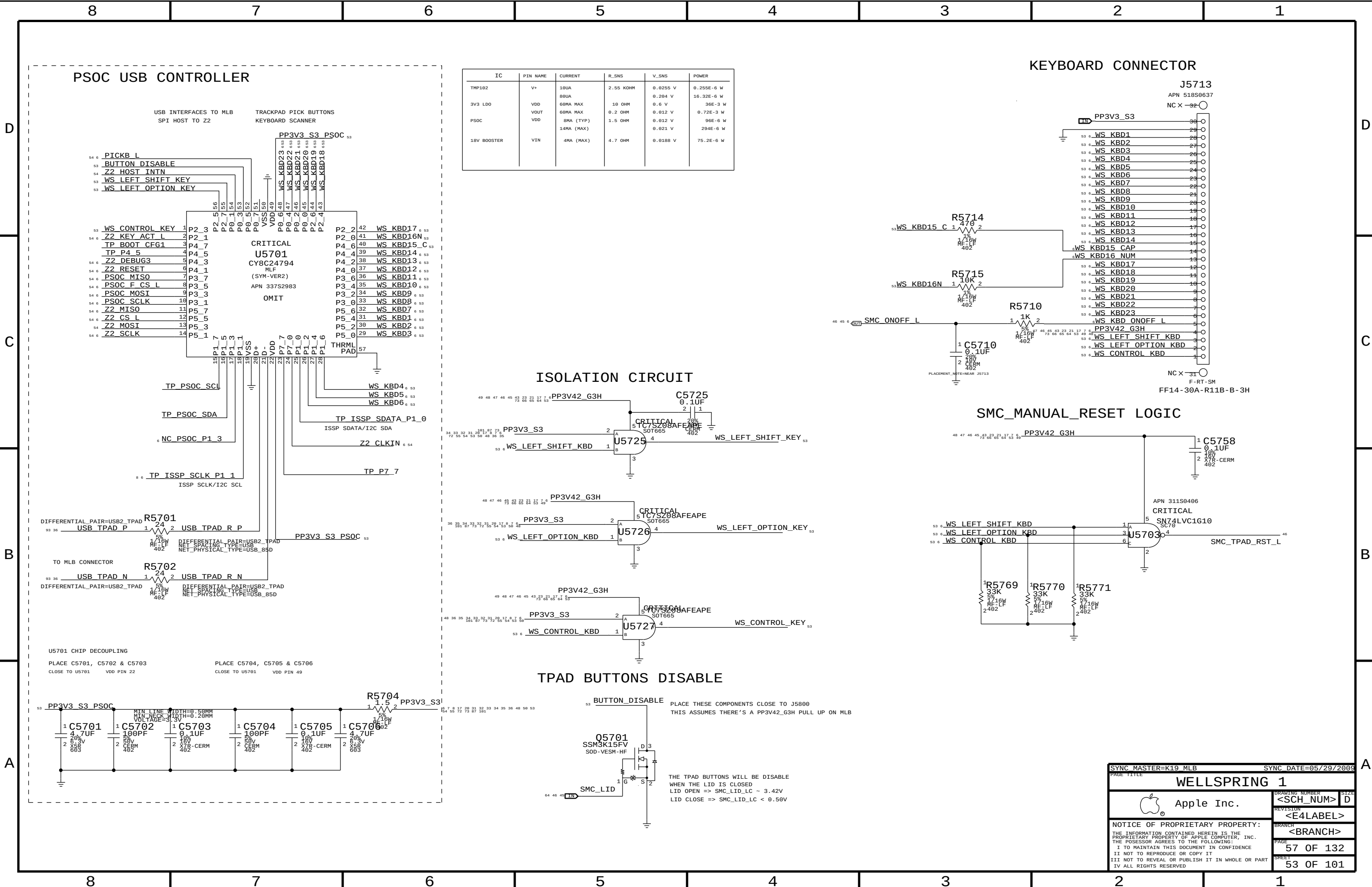


CPU Proximity/CPU Die/PCH Proximity/Battery Charger Proximity



Note: EMC1414 can perform Beta Compensation for External Diode 1 only





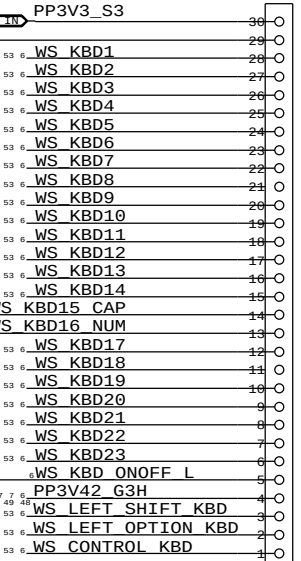
IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA	2.55 KOHM	0.0255 V	0.255E-6 W
3V3 LDO	VDD	98UA	10 OHM	0.204 V	16.32E-6 W
	VOUT	68MA MAX	0.2 OHM	0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

KEYBOARD CONNECTOR

J5713

APN 518S0637

NC X -32

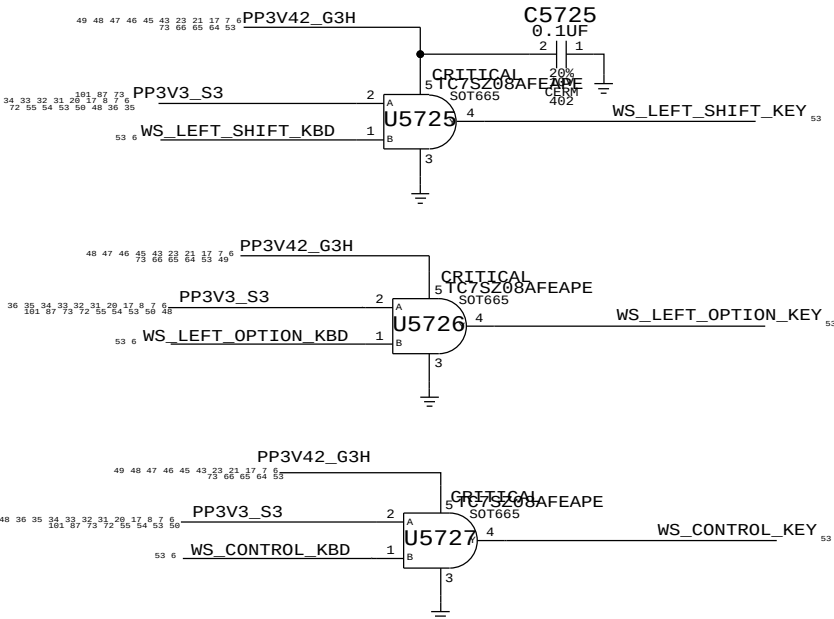


NC X -31

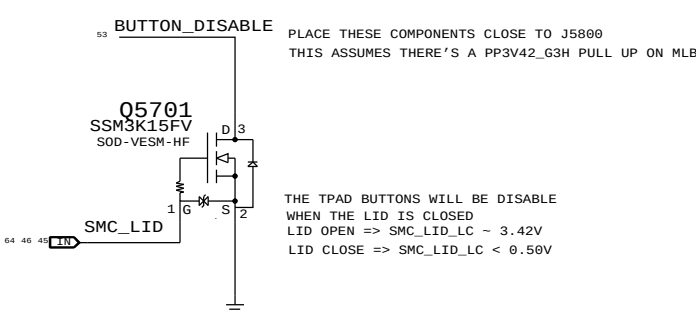
F-RT-SM

FF14-30A-R11B-B-3H

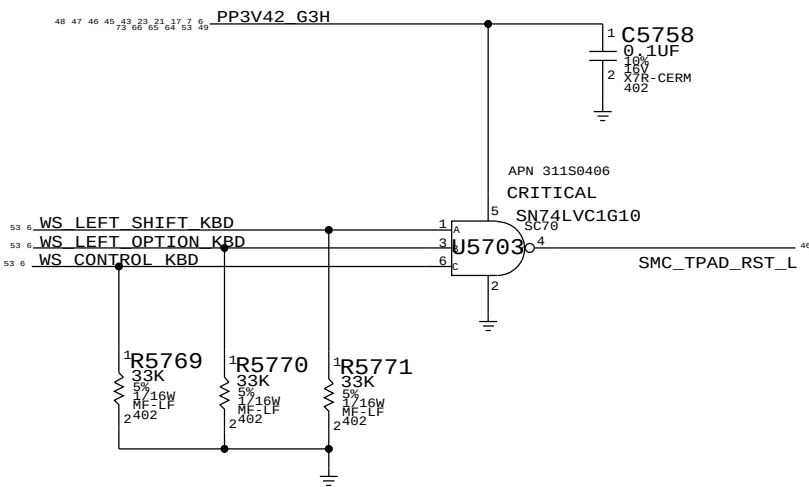
ISOLATION CIRCUIT



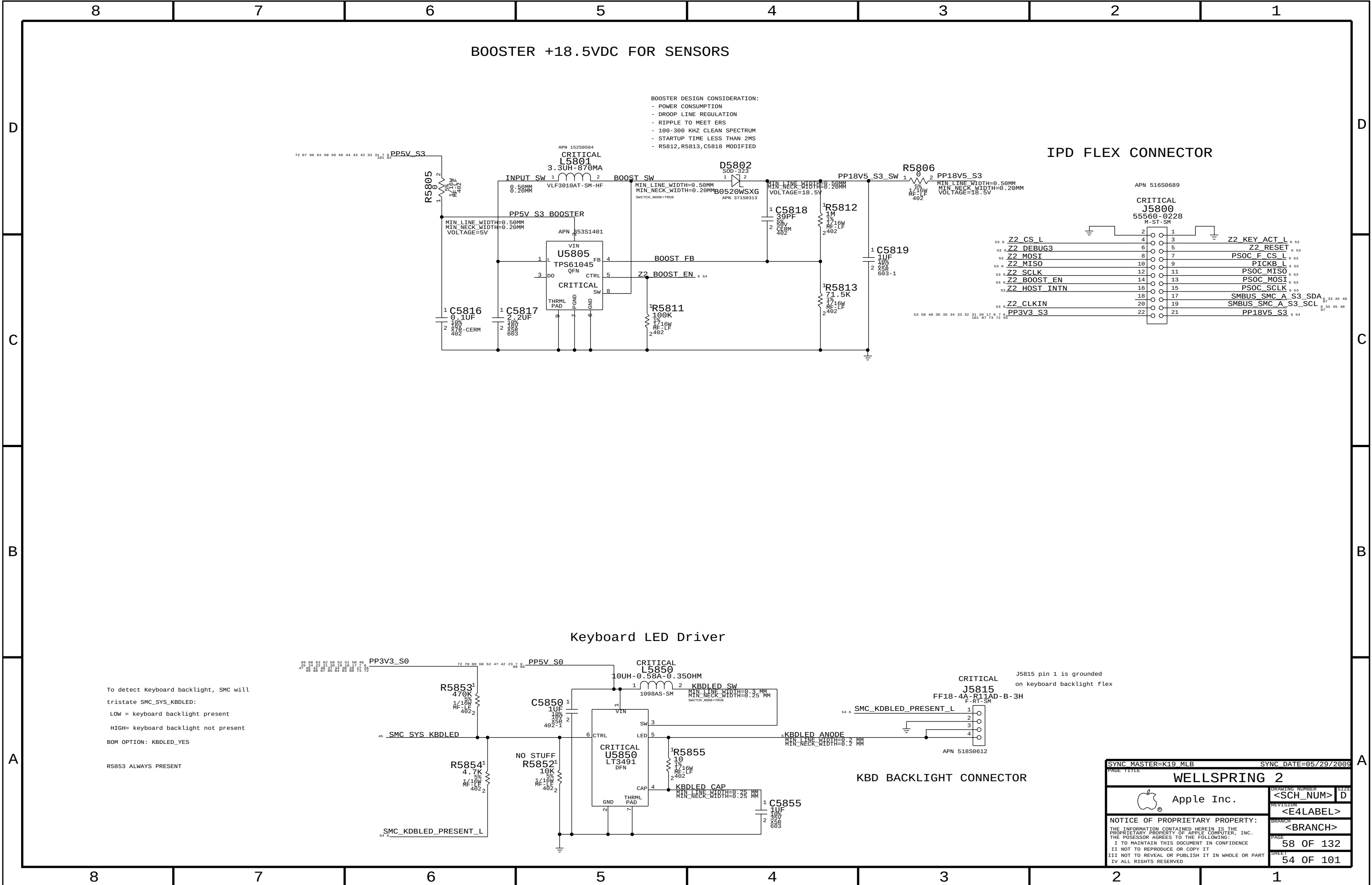
TPAD BUTTONS DISABLE

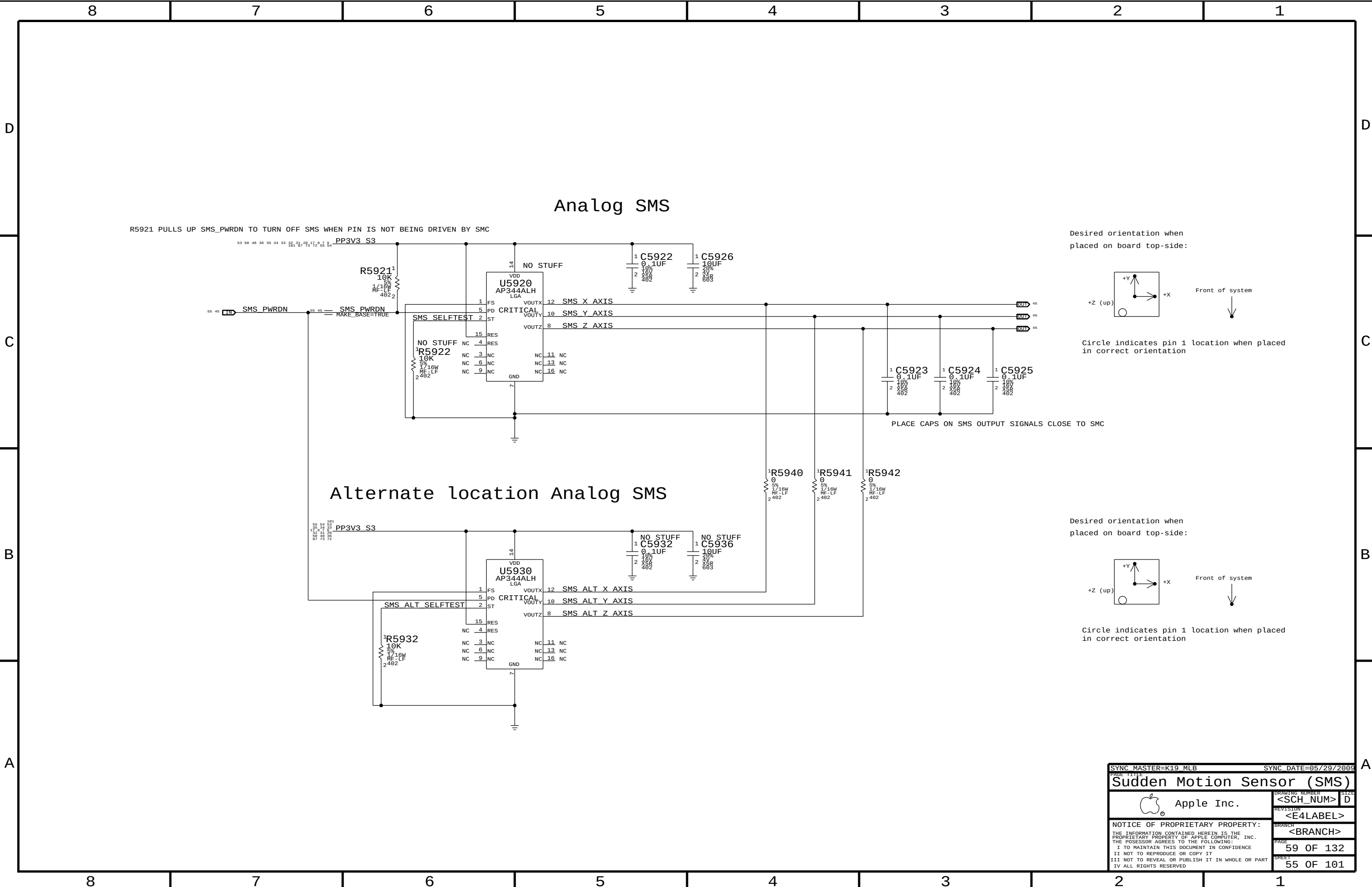


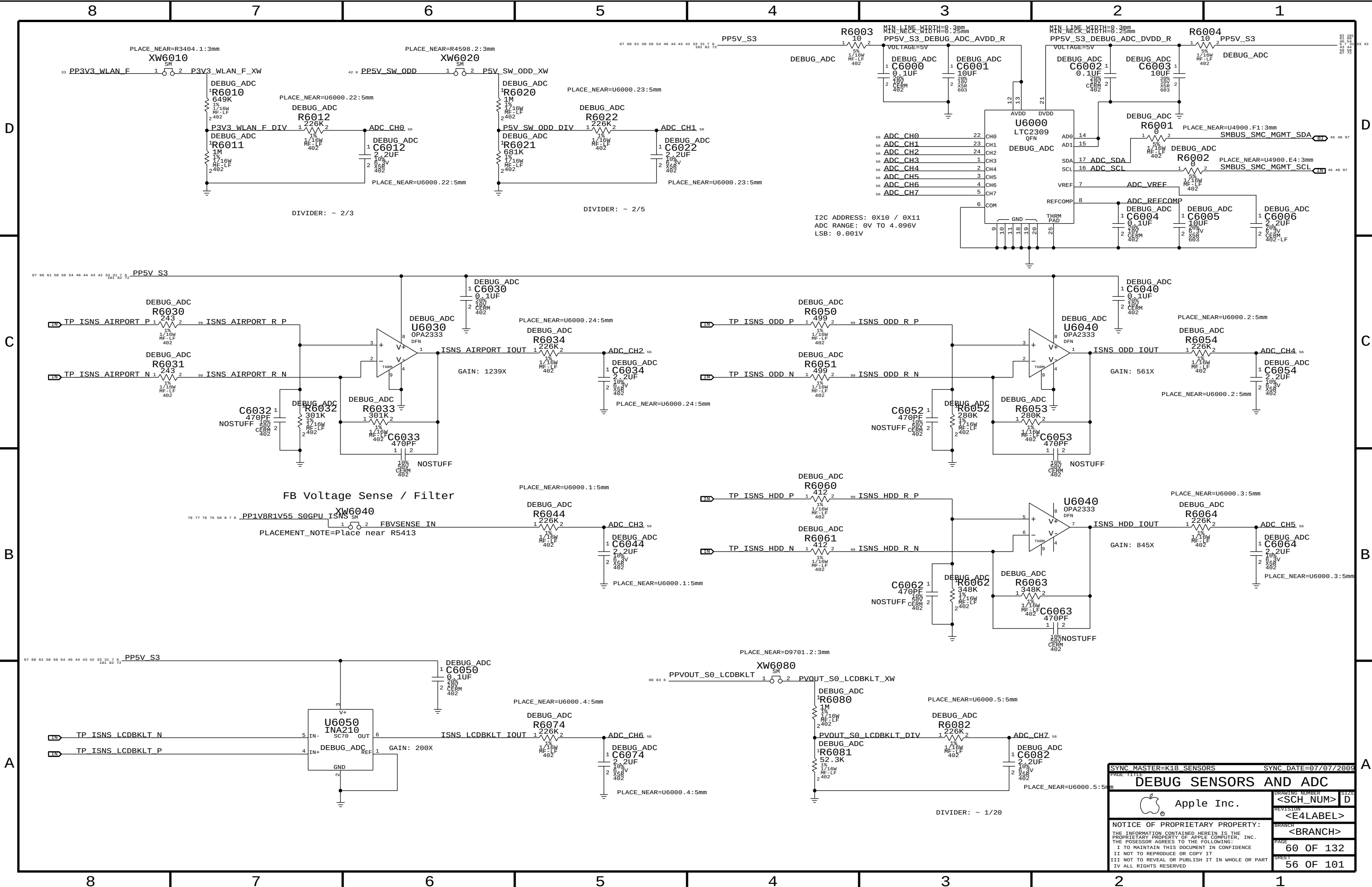
SMC_MANUAL_RESET LOGIC



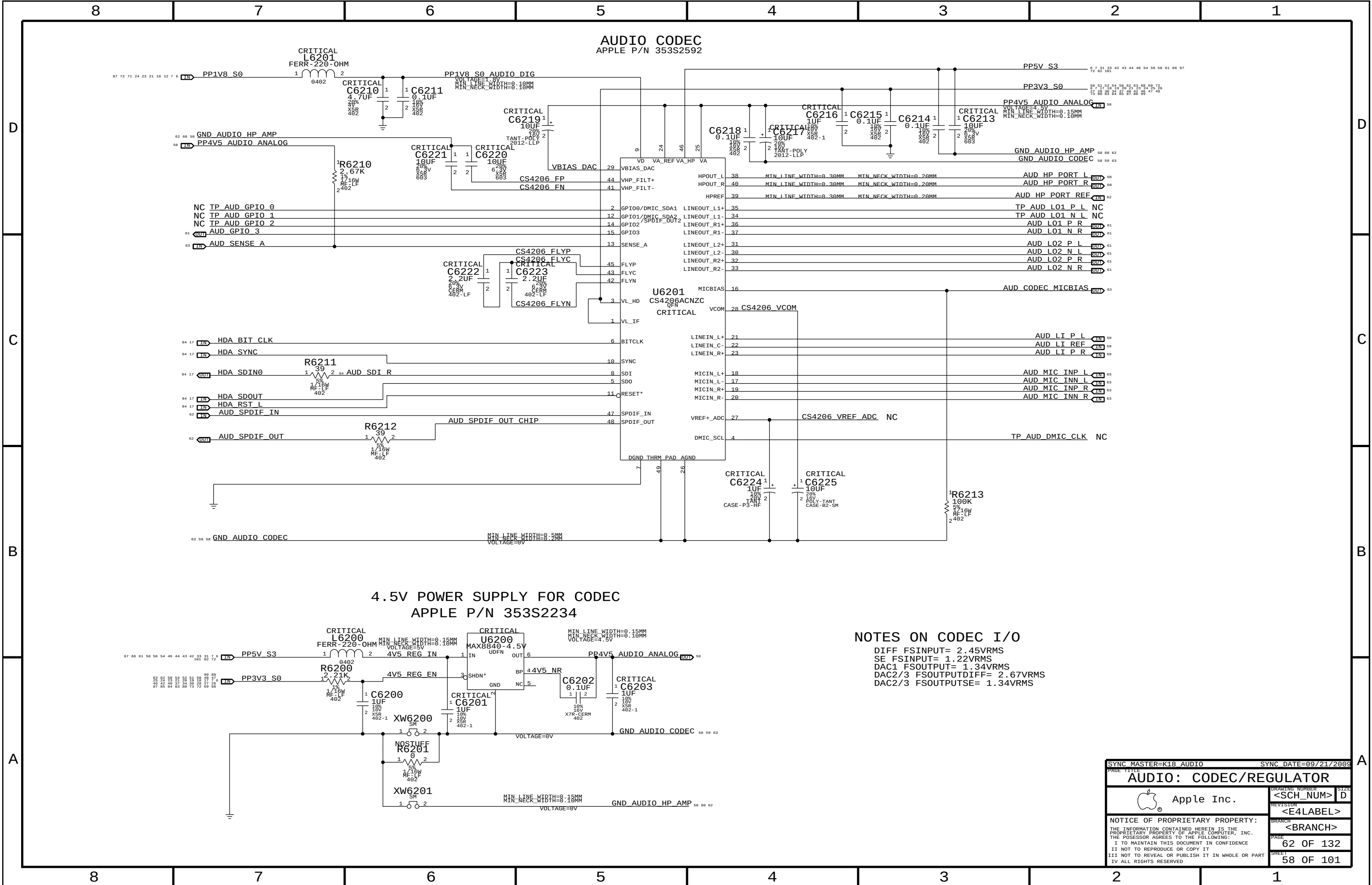
SYNC MASTER=K19 MLB		SYNC DATE=05/29/2009	
PAGE TITLE		WELLSPRING 1	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	57 OF 132
		SHEET	53 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			







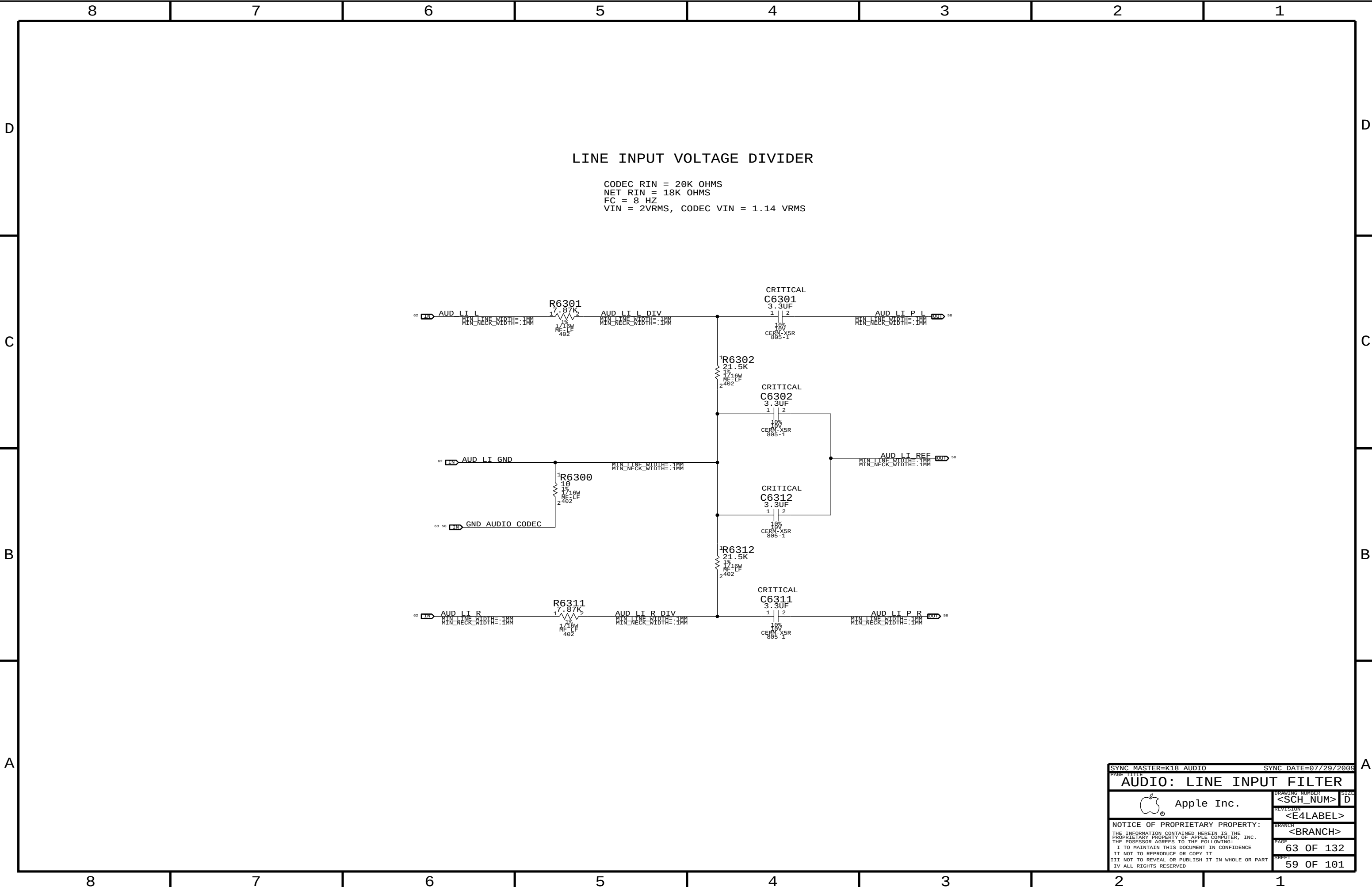
SYNC MASTER=K18 SENSORS		SYNC DATE=07/07/2009	
PAGE TITLE		DRAWING NUMBER	
DEBUG SENSORS AND ADC		<SCH_NUM> D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	
		60 OF 132	
		SHEET	
		56 OF 101	

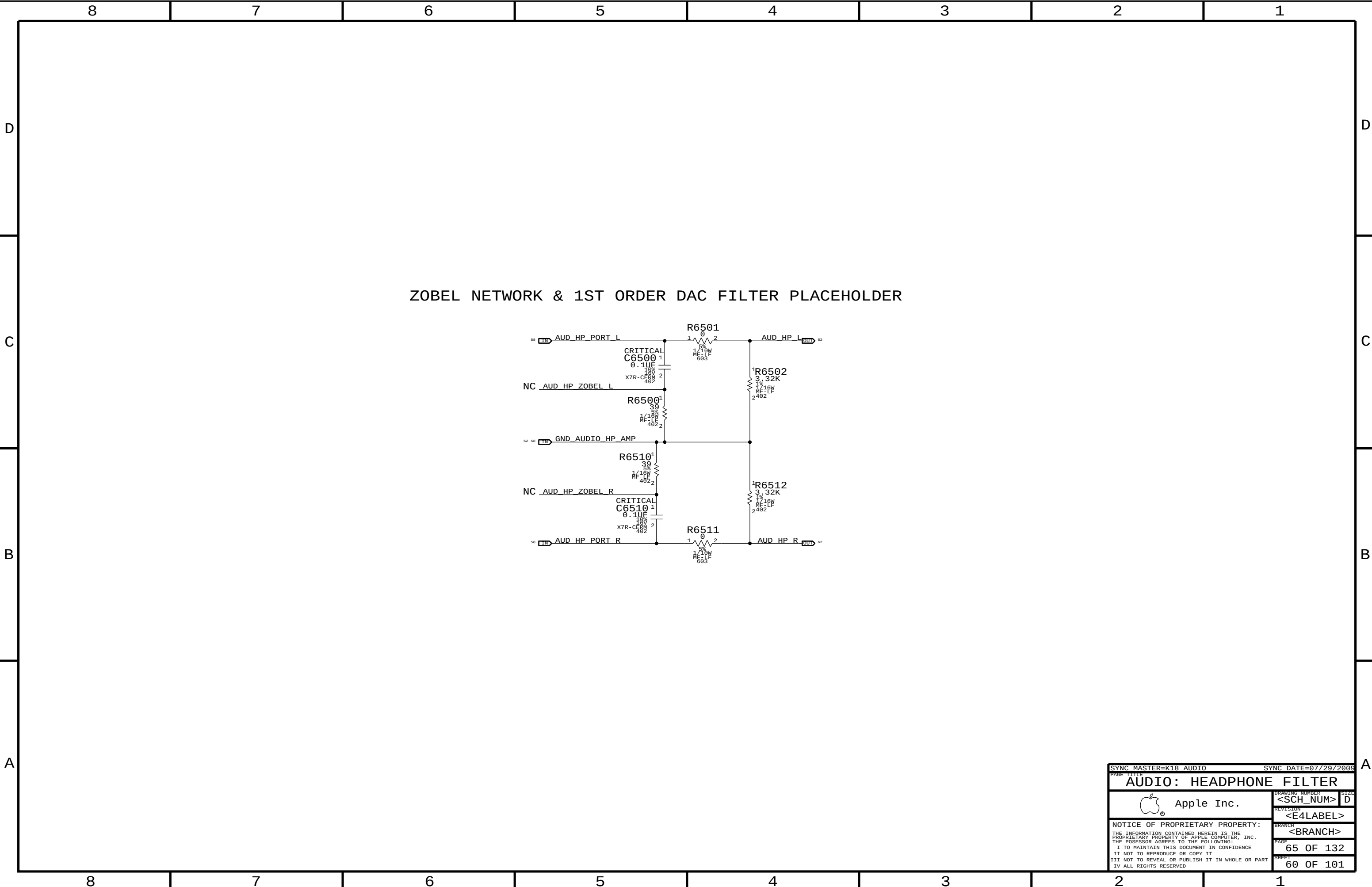


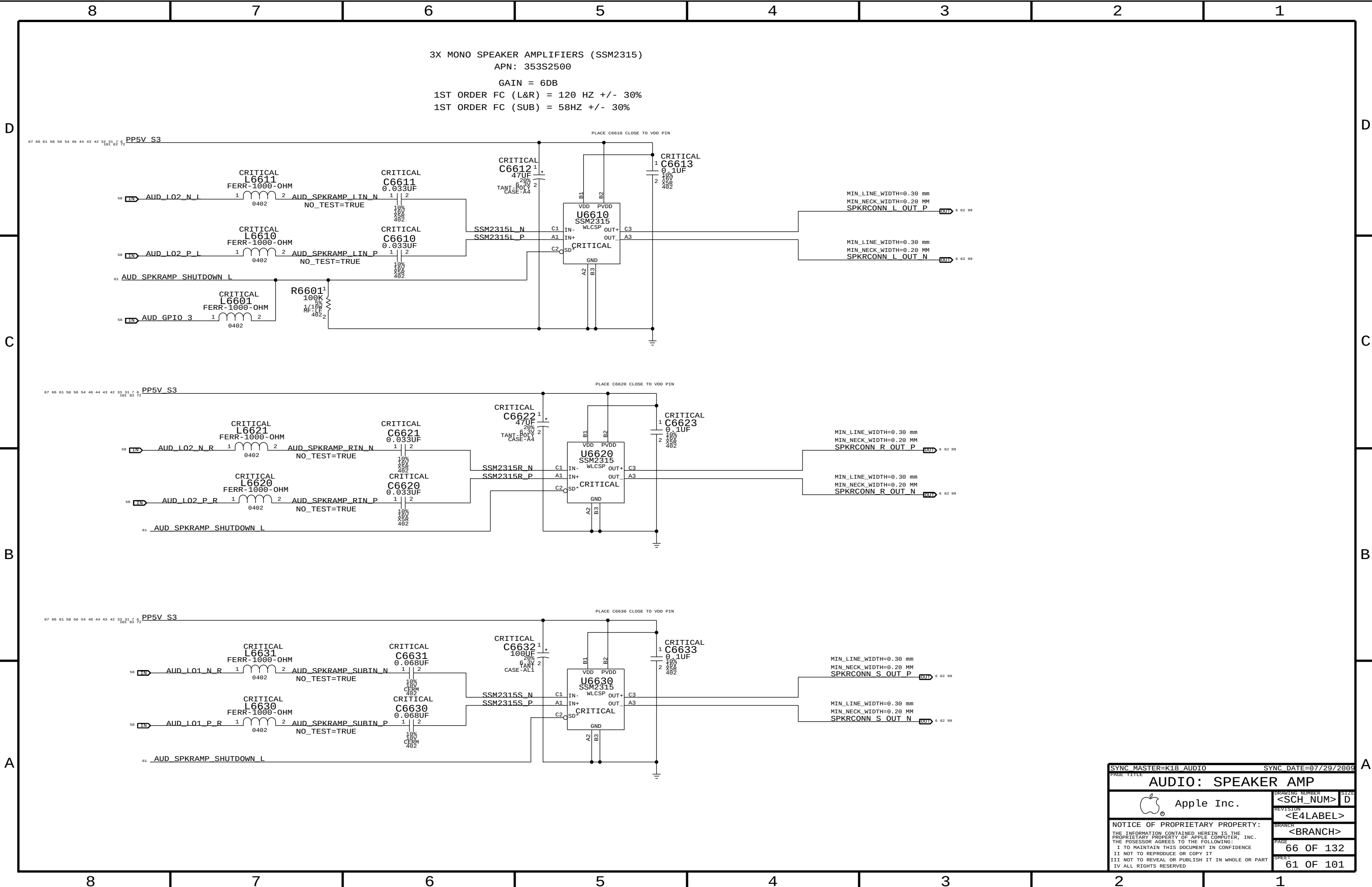
NOTES ON CODEC I/O

DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=K18 AUDIO		SYNC DATE=09/21/2009	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
DRAWING NUMBER		SIZE	
<SCH_NUM>		D	
REVISION		BRANCH	
<E4LABEL>		<BRANCH>	
NOTICE OF PROPRIETARY PROPERTY:		PAGE	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		62 OF 132	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	
II NOT TO REPRODUCE OR COPY IT		58 OF 101	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			







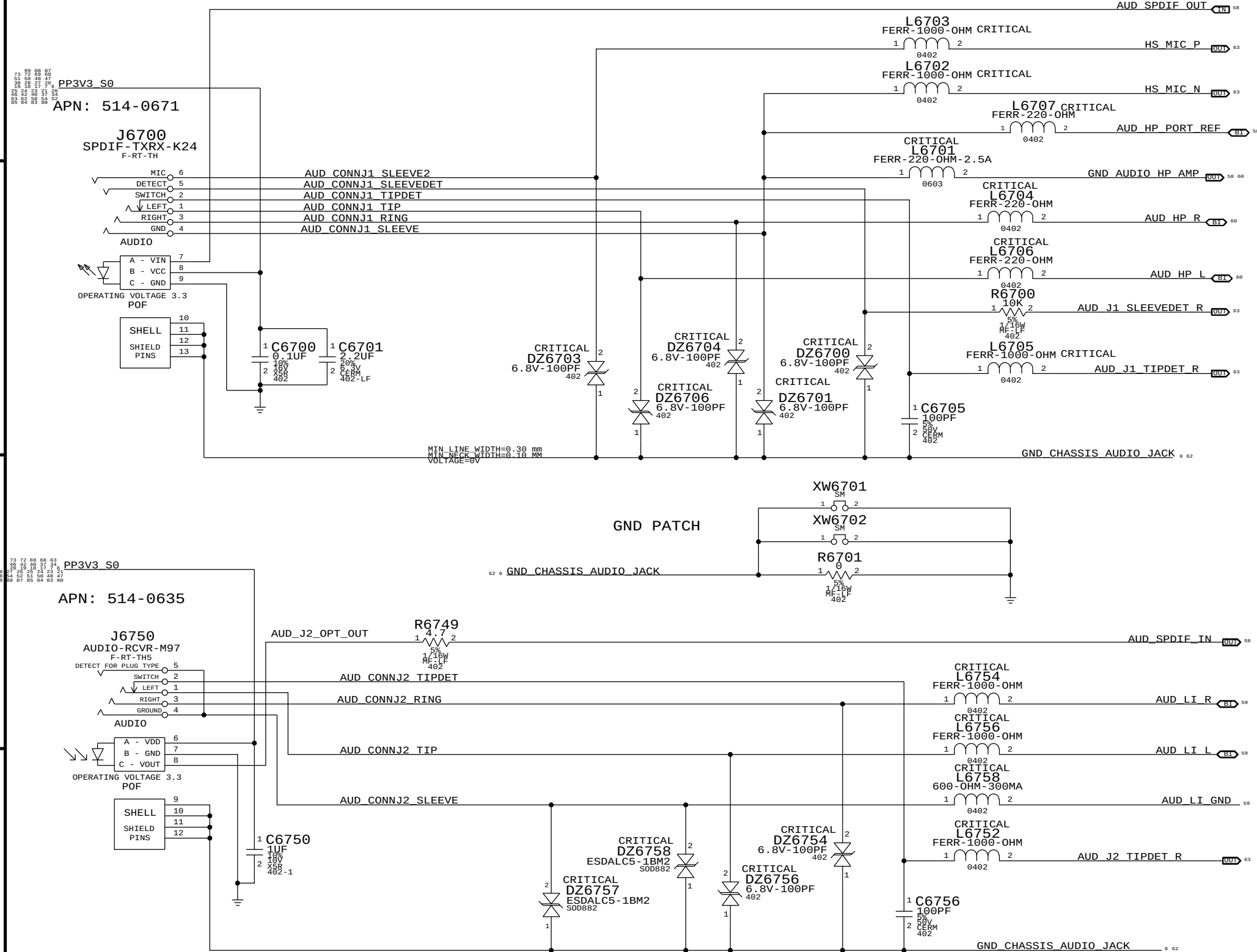
AUDIO JACK 1 LO/HP JACK, SPDIF TX

D

C

B

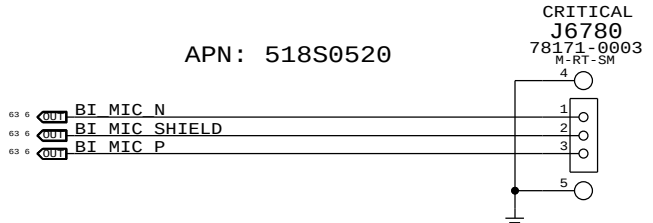
A



AUDIO JACK 2 LINE IN JACK, SPDIF RX

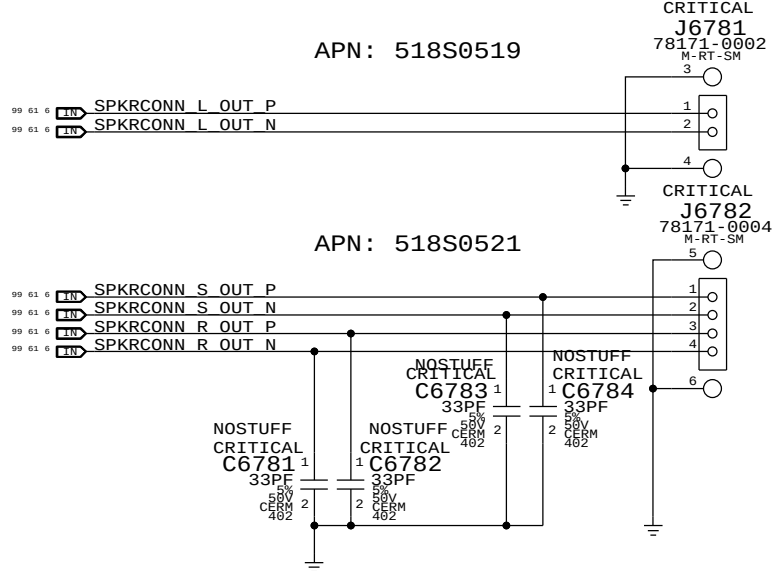
MIC CONNECTOR

APN: 518S0520

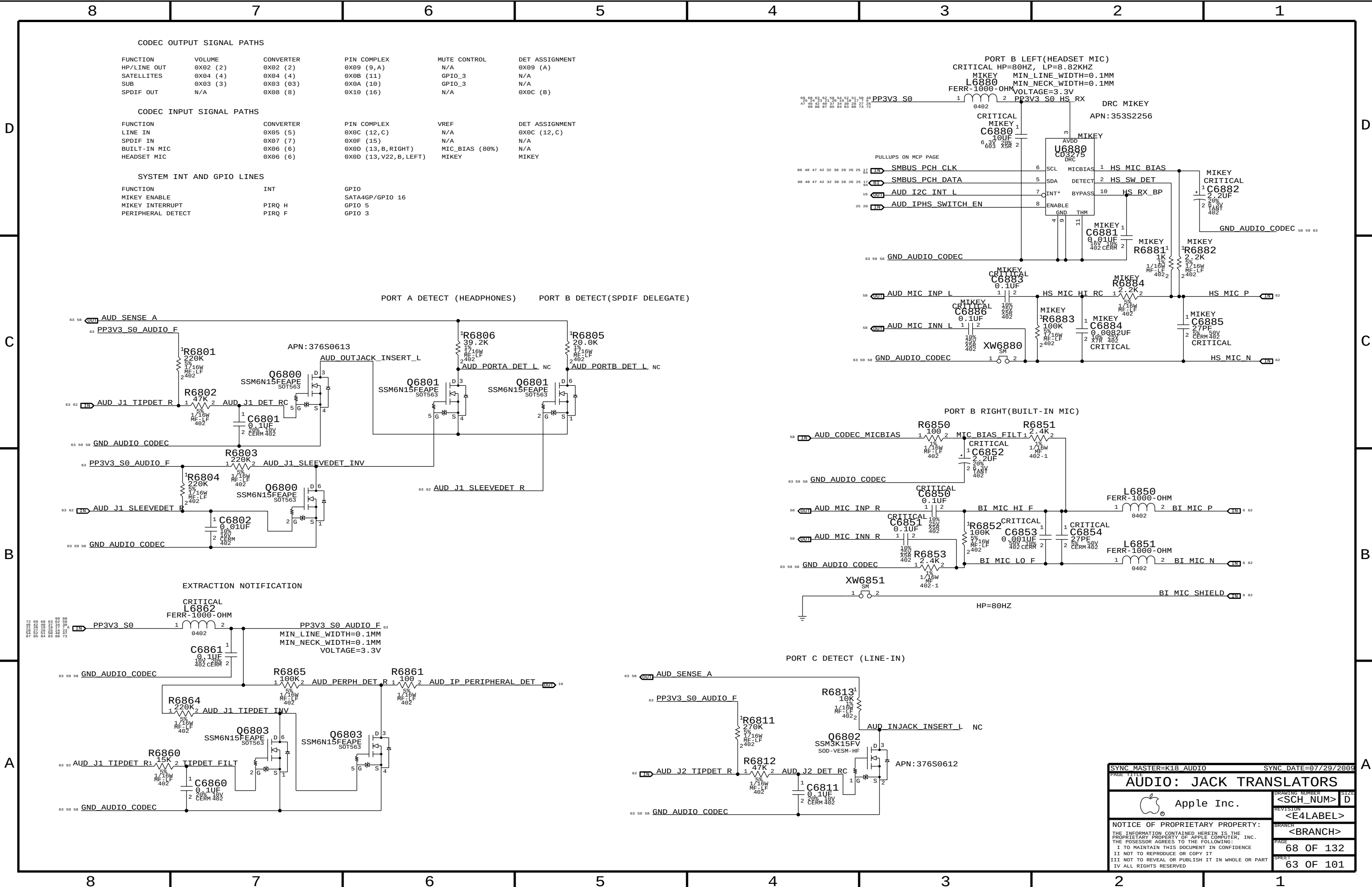


SPEAKER CONNECTOR

APN: 518S0519



SYNC MASTER=K18 AUDIO		SYNC DATE=07/29/2009	
PAGE TITLE		AUDIO: JACKS	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	67 OF 132
		SHEET	62 OF 101



CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	N/A	0X09 (A)
SATELLITES	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	0X0C (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X0C (12,C)	N/A	0X0C (12,C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
BUILT-IN MIC	0X06 (6)	0X0D (13,B,RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY

SYSTEM INT AND GPIO LINES

FUNCTION	INT	GPIO
MIKEY ENABLE		SATA4GP/GPIO 16
MIKEY INTERRUPT	PIRQ H	GPIO 5
PERIPHERAL DETECT	PIRQ F	GPIO 3

PORT A DETECT (HEADPHONES) PORT B DETECT (SPDIF DELEGATE)

PORT B LEFT (HEADSET MIC)
CRITICAL HP=80HZ, LP=8.82KHZ
MIKEY MIN_LINE_WIDTH=0.1MM
L6880 MIN_NECK_WIDTH=0.1MM
FERR-1000-OHM VOLTAGE=3.3V

PP3V3_S0 1 2 PP3V3_S0 HS_RX
0402
CRITICAL MIKEY
C6880 10UF
6 3V 28V 2
683 XSR

AVDD
U6880 CD3275
MIKEY
DRC MIKEY
APN:353S2256

PULLUPS ON MCP PAGE
SMBUS_PCH_CLK 6 SCL MICBIAS
SMBUS_PCH_DATA 5 SDA DETECT
AUD_I2C_INT_L 7 INT* BYPASS
AUD_IPHS_SWITCH_EN 8 ENABLE GND THM

MIKEY 1 HS_MIC_BIAS
C6881 0.01UF
482 CERN 2
MIKEY 1 R6881 1K
482 CERN 2

MIKEY 1 HS_SW_DET
C6882 2.2UF
482 CERN 2
MIKEY 1 HS_RX_BP
C6883 0.1UF
482 CERN 2

MIKEY 1 HS_MIC_P
C6884 0.01UF
482 CERN 2
MIKEY 1 HS_MIC_N
C6885 27PE
482 CERN 2

MIKEY 1 HS_MIC_HI_RC
R6884 2.2K
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6851 2.4K
482-1

MIKEY 1 HS_MIC_LO_F
R6852 100K
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6853 2.4K
482-1

MIKEY 1 HS_MIC_LO_F
R6854 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6855 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6856 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6857 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6858 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6859 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6860 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6861 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6862 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6863 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6864 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6865 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6866 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6867 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6868 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6869 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6870 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6871 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6872 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6873 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6874 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6875 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6876 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6877 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6878 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6879 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6880 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6881 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6882 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6883 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6884 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6885 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6886 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6887 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6888 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6889 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6890 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6891 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6892 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6893 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6894 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6895 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6896 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6897 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6898 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6899 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6900 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6901 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6902 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6903 27PE
482 CERN 2

MIKEY 1 HS_MIC_LO_F
R6904 27PE
482 CERN 2
MIKEY 1 HS_MIC_LO_F
R6905 27PE
482 CERN 2

SYNC MASTER=K18 AUDIO		SYNC DATE=07/29/2009	
PAGE TITLE		AUDIO: JACK TRANSLATORS	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	68 OF 132
		SHEET	63 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			

D



C

C



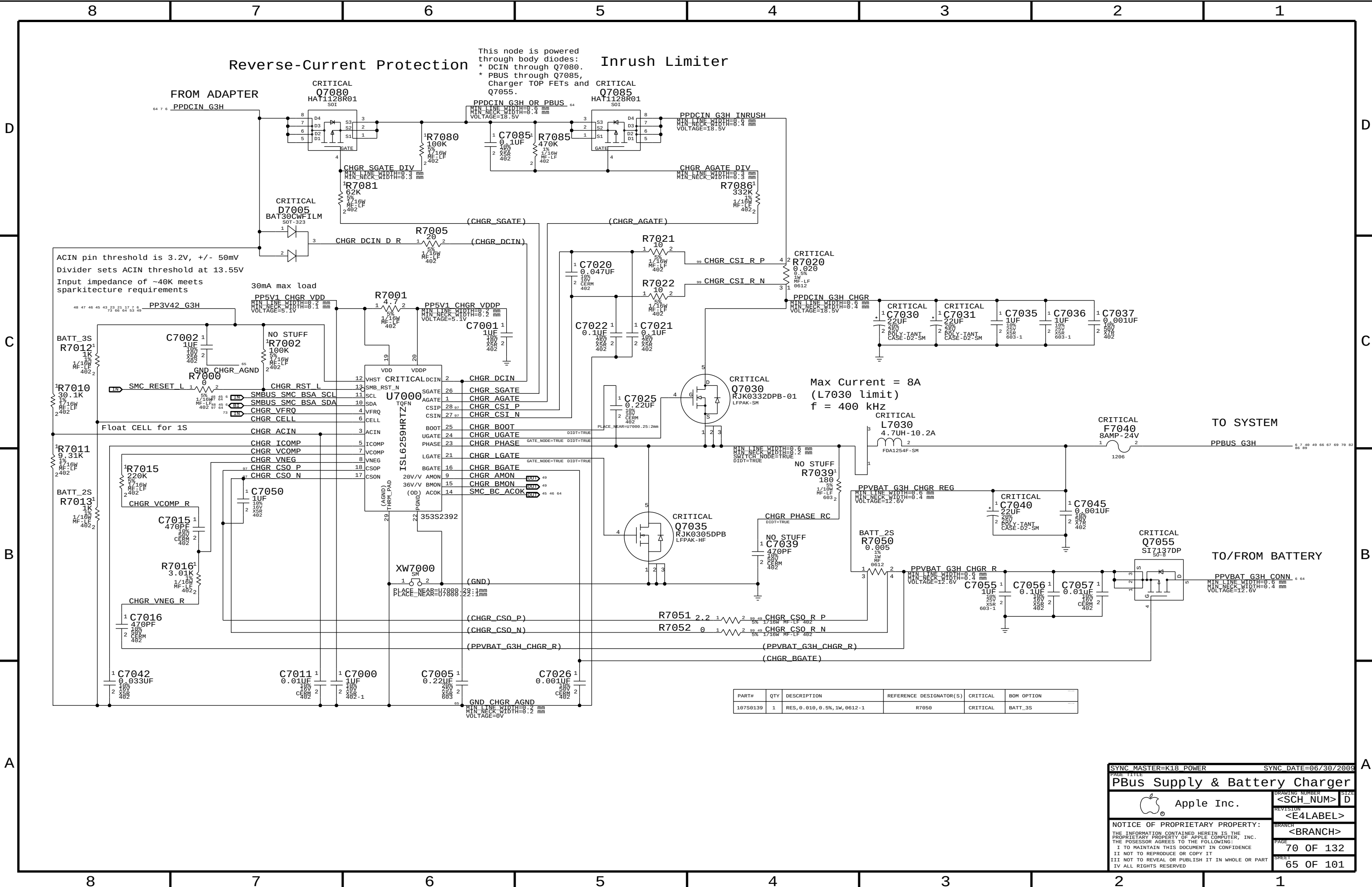
B

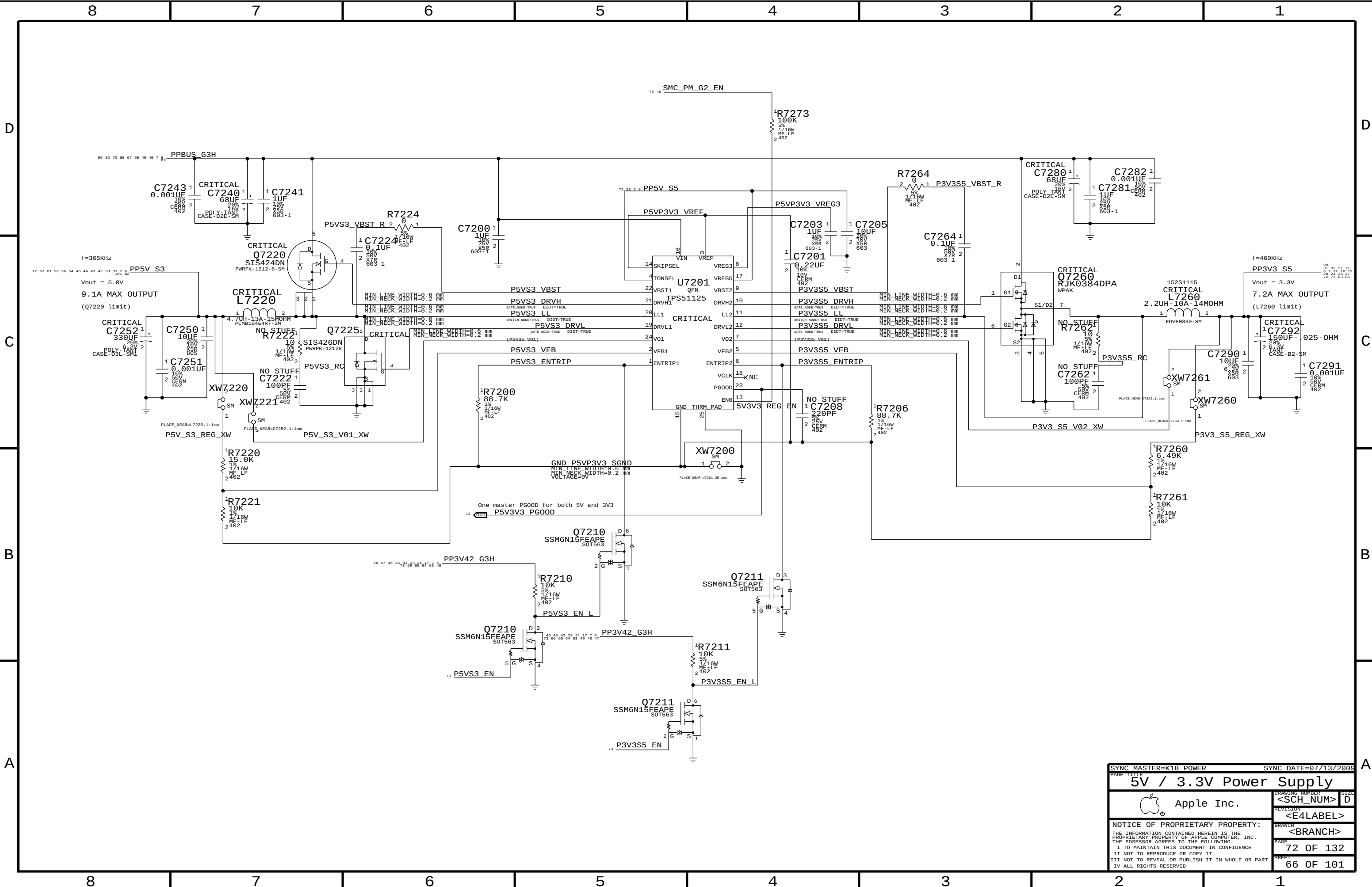
A



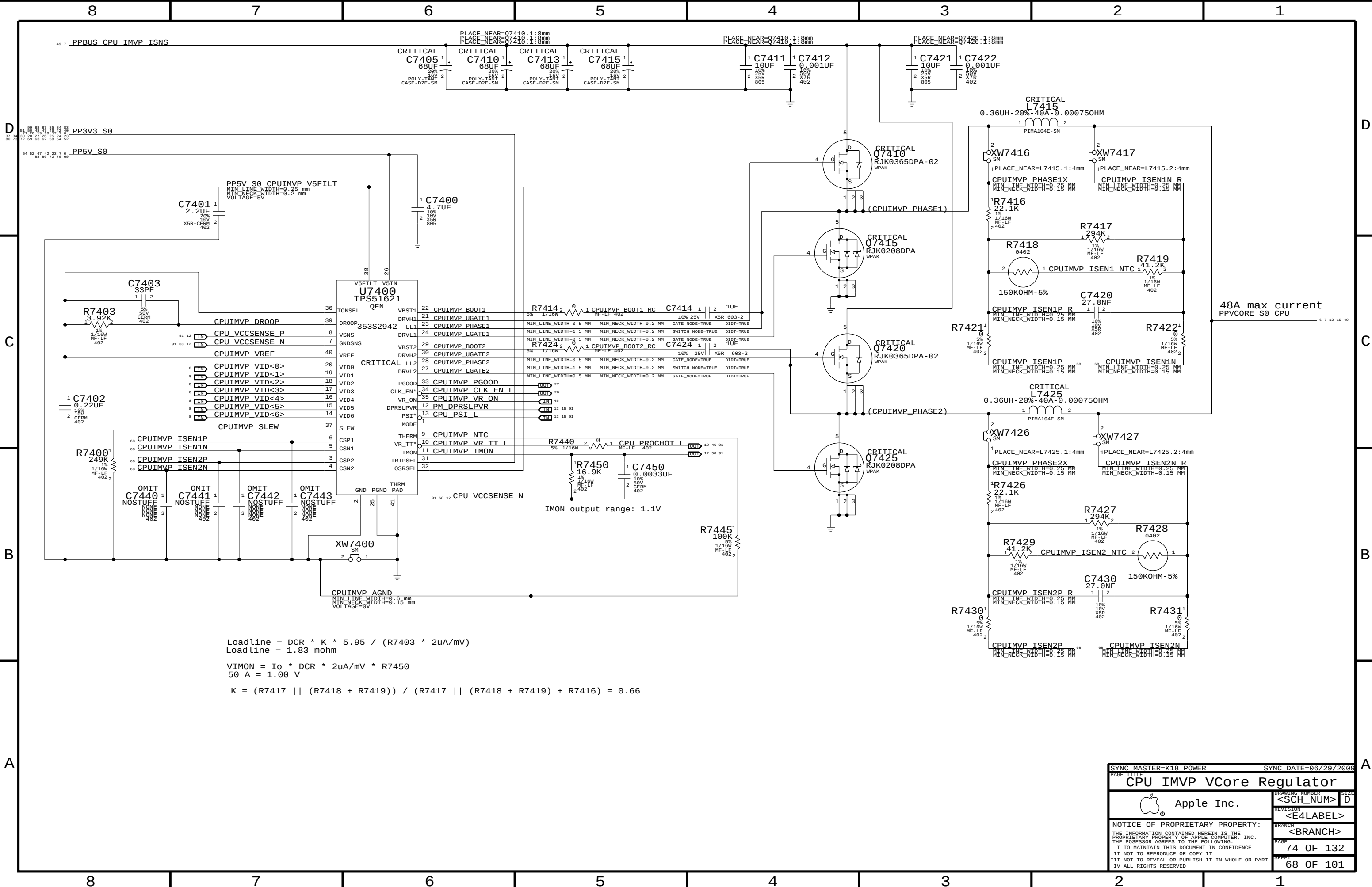
A







SYNC MASTER=K18 POWER		SYNC DATE=07/13/2009	
PAGE TITLE		5V / 3.3V Power Supply	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		<BRANCH>	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	72 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	66 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



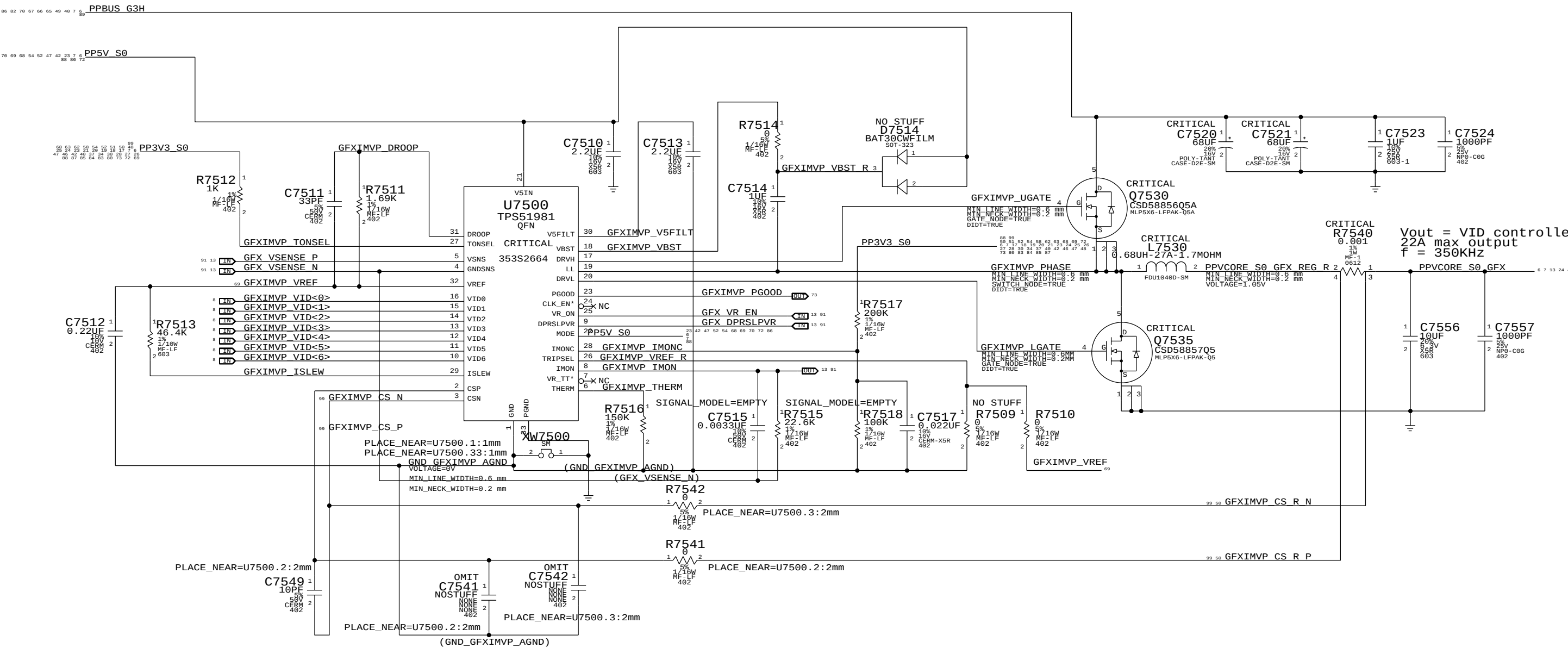
Loadline = $DCR * K * 5.95 / (R7403 * 2\mu A/mV)$
Loadline = 1.83 mohm

$VIMON = I_o * DCR * 2\mu A/mV * R7450$
50 A = 1.00 V

$K = (R7417 || (R7418 + R7419)) / (R7417 || (R7418 + R7419) + R7416) = 0.66$

SYNC MASTER=K18 POWER		SYNC DATE=06/29/2009	
PAGE TITLE			
CPU IMVP VCore Regulator			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	74 OF 132
		SHEET	68 OF 101

GFX IMVP VCore



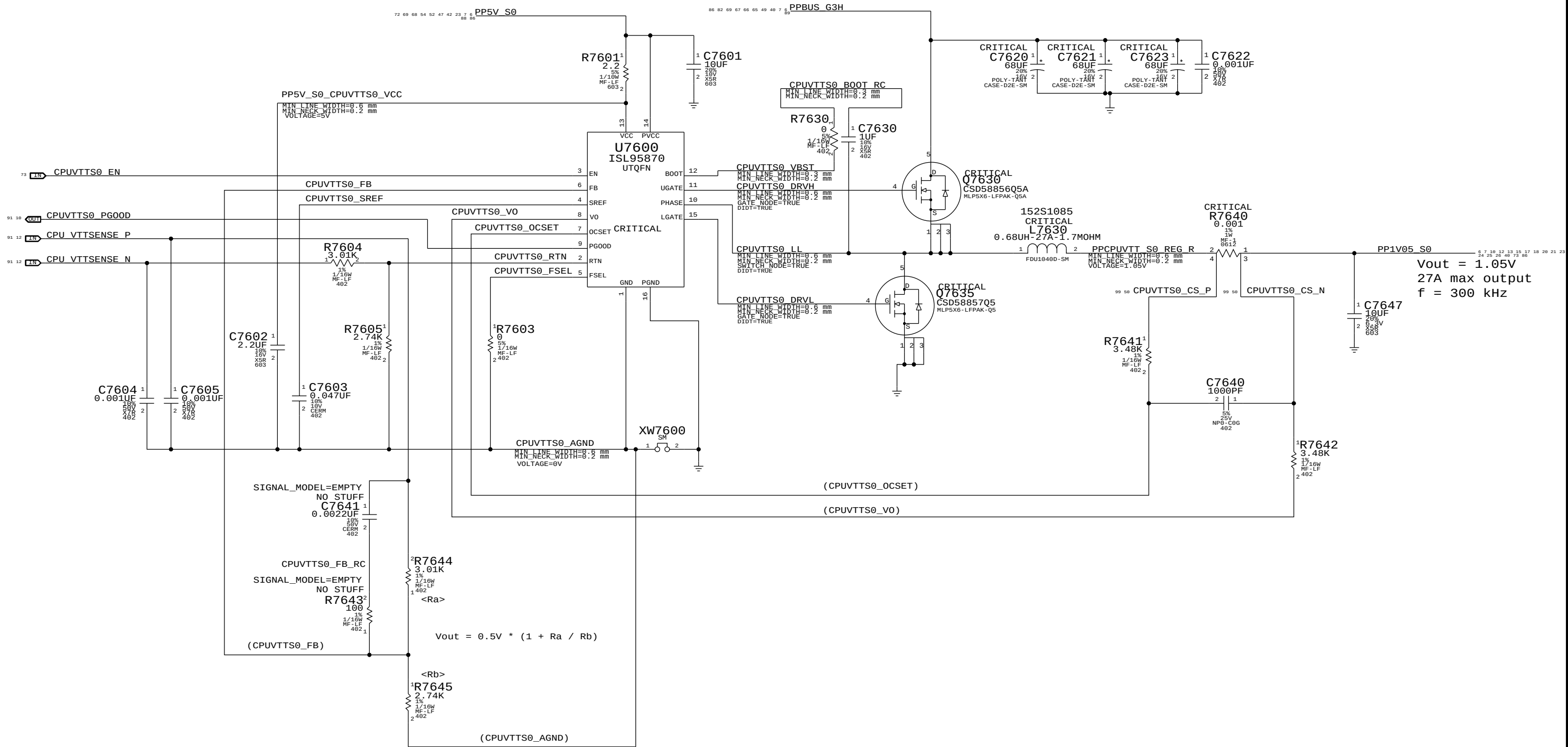
$I_{mon} = I_o \times R7540 \times 2\mu A/mV \times R7515$

$I_{mon} = I_o \times 45.2mV/A$

$22A \Rightarrow 1V$

SYNC MASTER=K18 POWER		SYNC DATE=07/08/2009	
PAGE TITLE			
GFX IMVP VCore Regulator			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	75 OF 132
		SHEET	69 OF 101

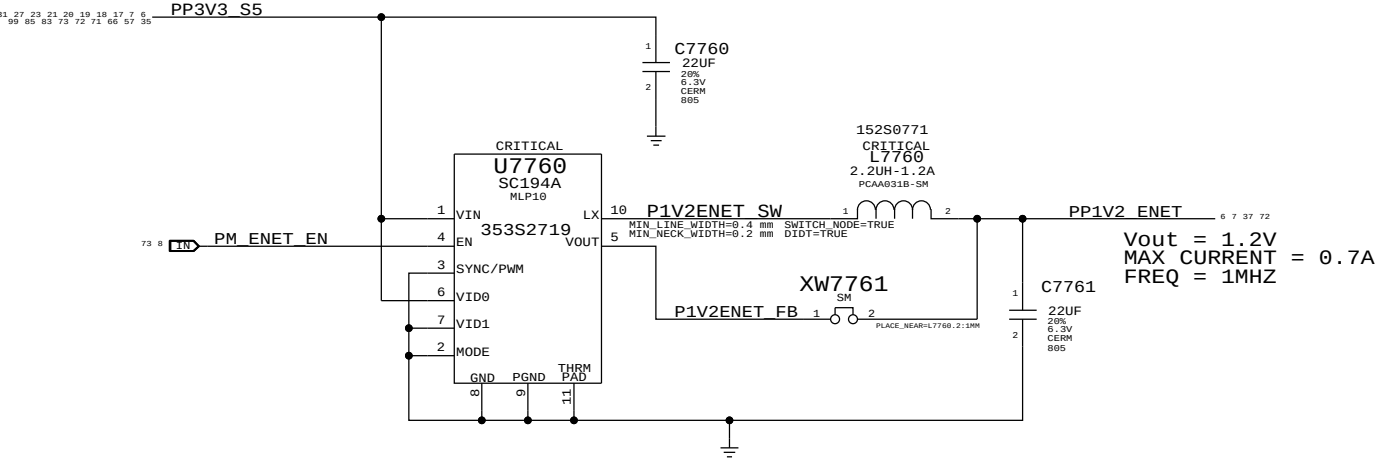
CPU VTT (1.05V S0) Regulator



SYNC MASTER=K18 POWER		SYNC DATE=07/14/2009	
PAGE TITLE			
CPUVTT (1.05V) Power Supply			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	76 OF 132
		SHEET	70 OF 101

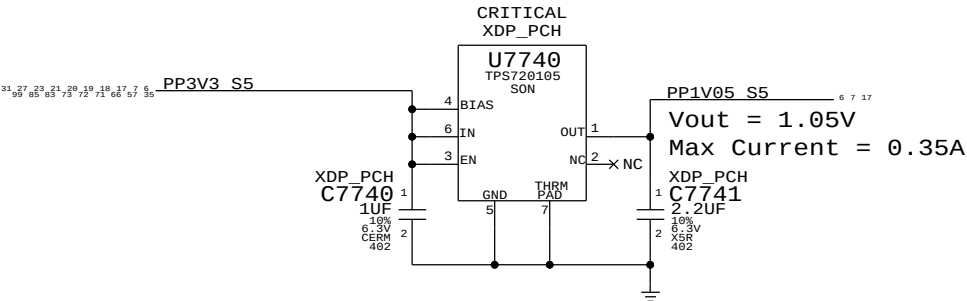
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

1.2V S3 Regulator

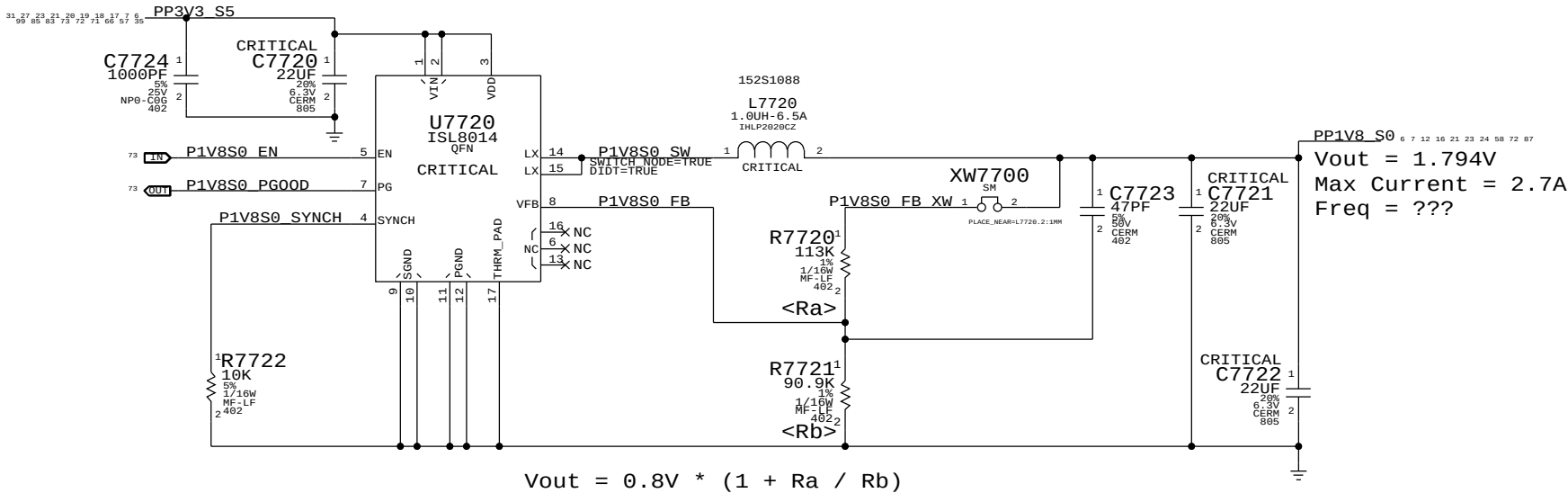


1.05V S5 LDO

Ibex Peak-M requires JTAG pull-ups to be powered at 1.05V in S5. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.

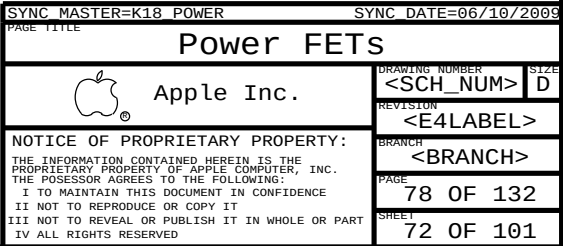


1.8V S0 Regulator

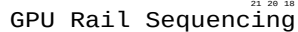
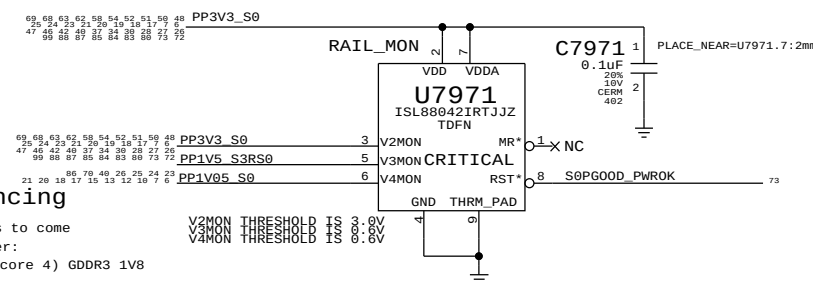
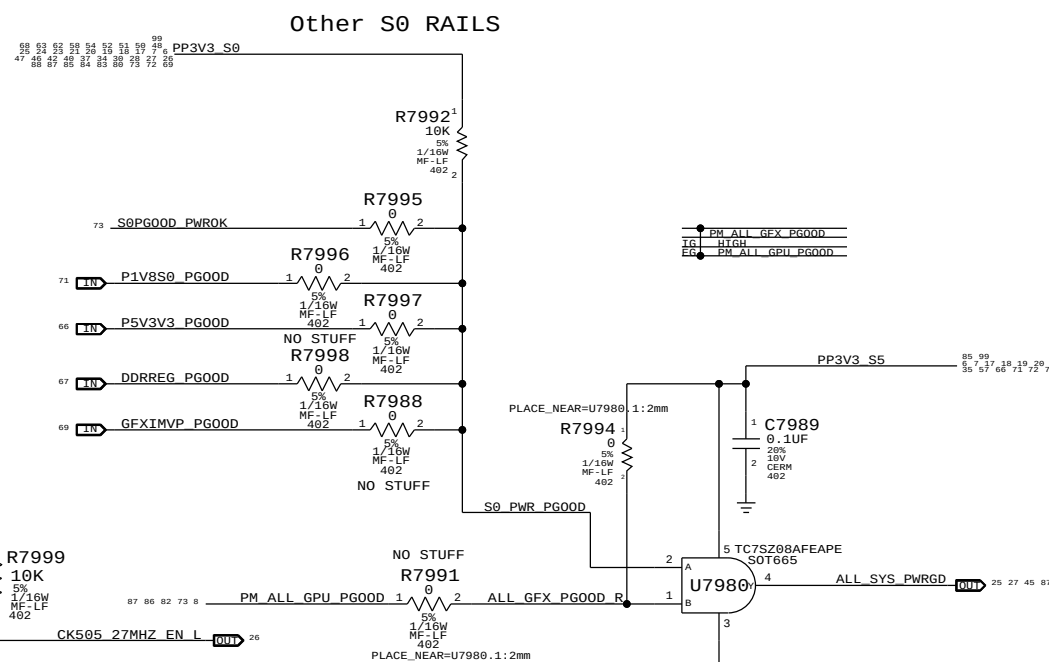
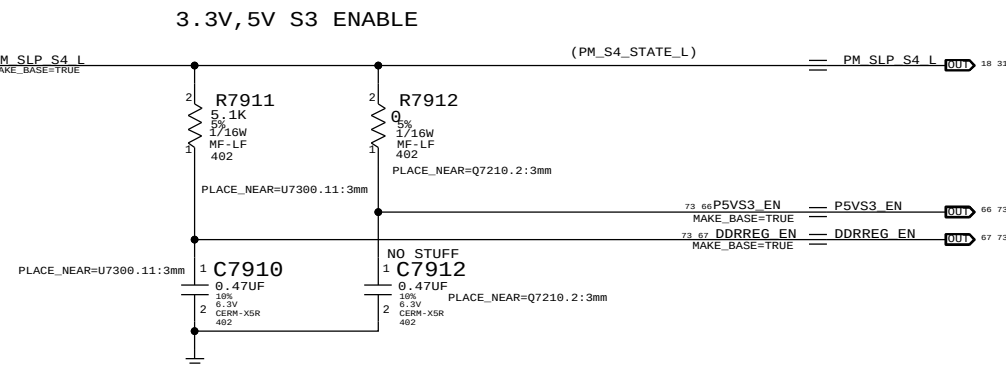


$$V_{out} = 0.8V * (1 + R_a / R_b)$$

SYNC MASTER=K18 POWER		SYNC DATE=06/29/2009	
PAGE TITLE			
Misc Power Supplies			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	77 OF 132
		SHEET	71 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			



State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0

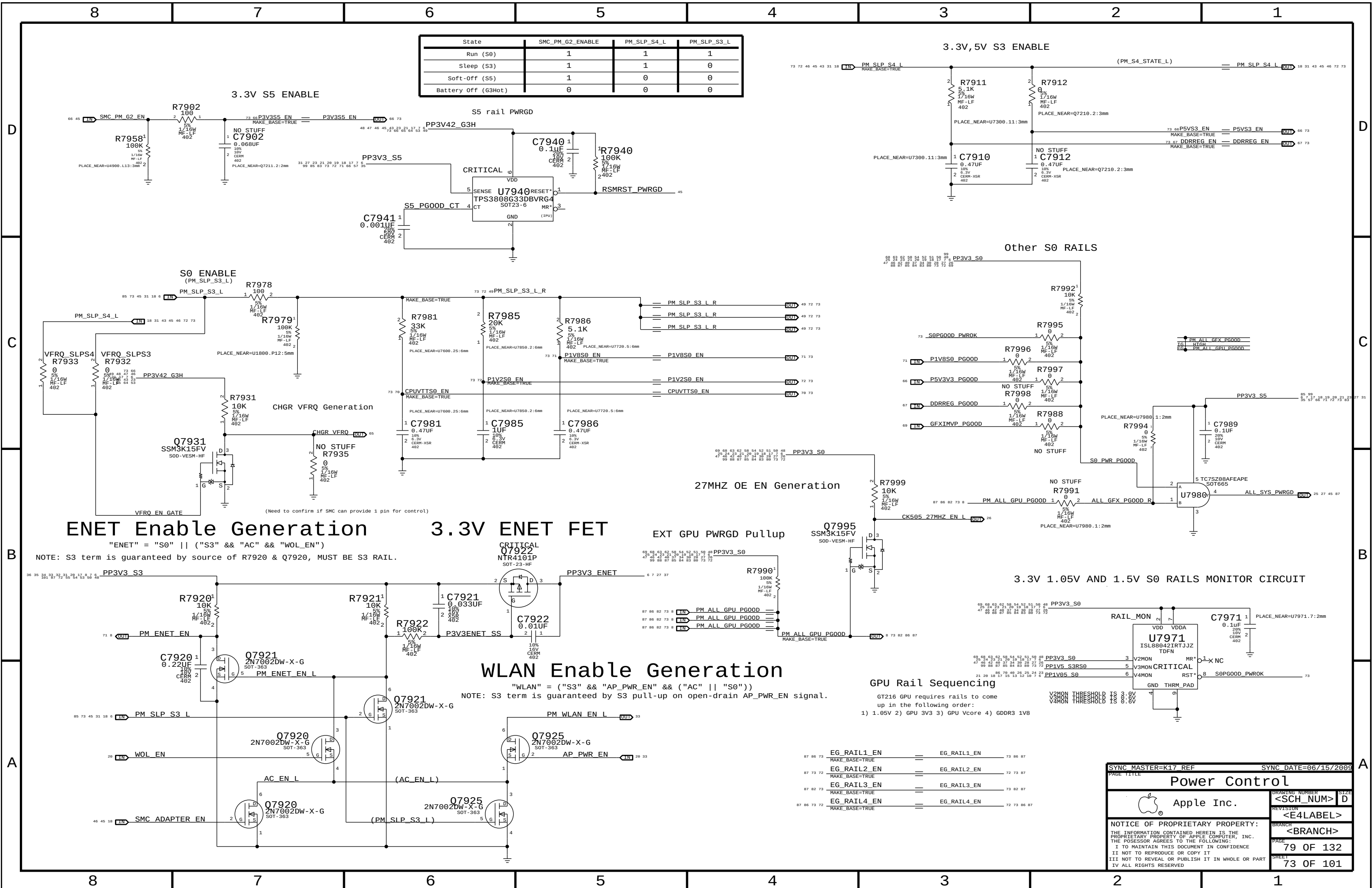


GT216 GPU requires rails to come up in the following order:

- 1) 1.05V
- 2) GPU 3V3
- 3) GPU Vcore
- 4) GDDR3 1V8

87	86	73	EG_RAI1_EN	==	EG_RAI1_EN	73	86	87
			MAKE_BASE_TRUE					
87	73	72	EG_RAI2_EN	==	EG_RAI2_EN	72	73	87
			MAKE_BASE_TRUE					
87	82	73	EG_RAI3_EN	==	EG_RAI3_EN	73	82	87
			MAKE_BASE_TRUE					
87	86	73	EG_RAI4_EN	==	EG_RAI4_EN	72	73	86
			MAKE_BASE_TRUE					

SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
Power Control			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
		SIZE D	
		REVISION <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		<BRANCH> PAGE 79 OF 132 SHEET 73 OF 101	



ENET Enable Generation

```
"ENET" = "S0" || ("S3" && "AC" && "WOL_EN")
```

NOTE: S3 term is guaranteed by source of R7920 & Q7920, MUST BE S3 RAIL.

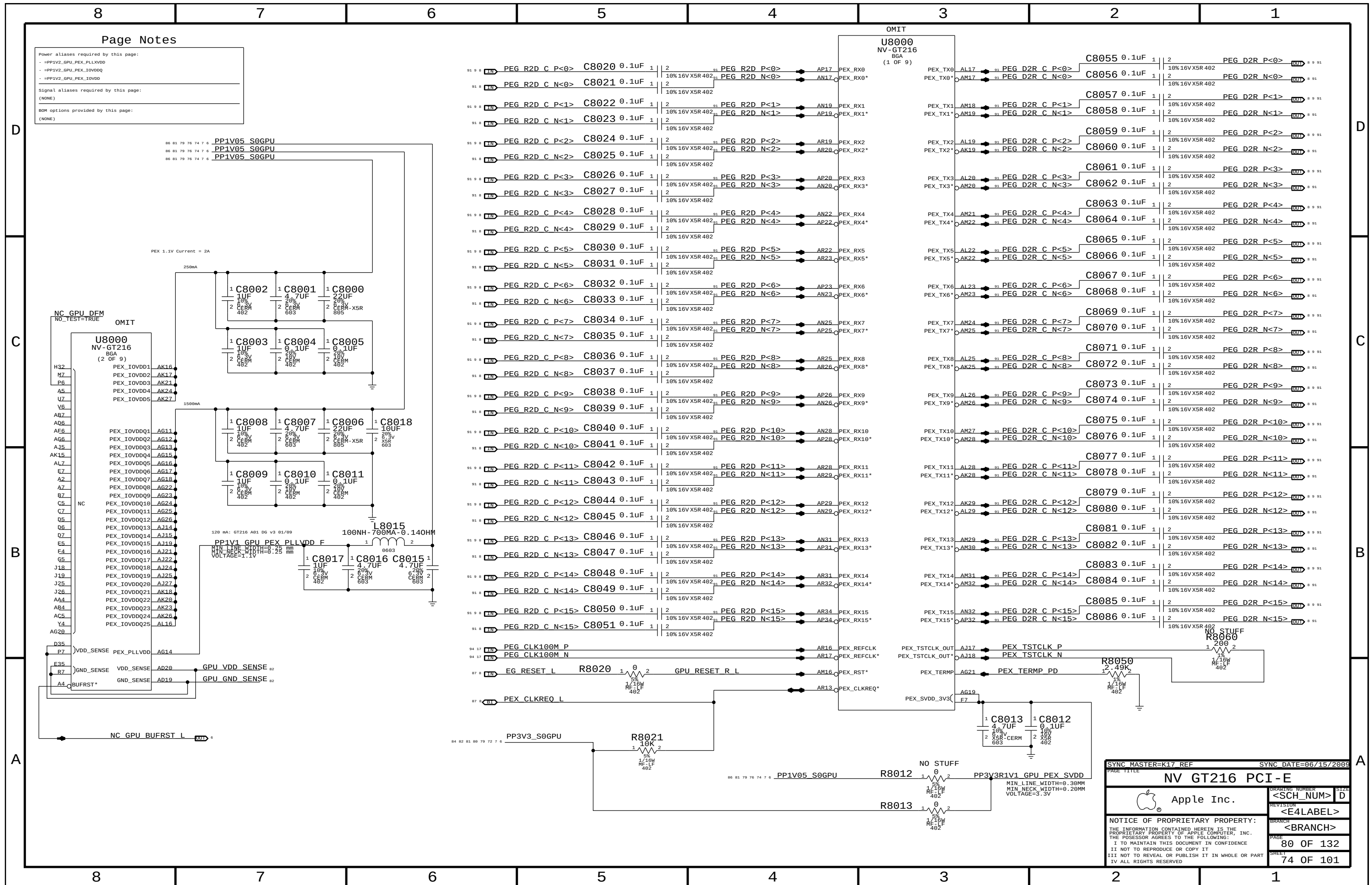
3.3V ENET FET

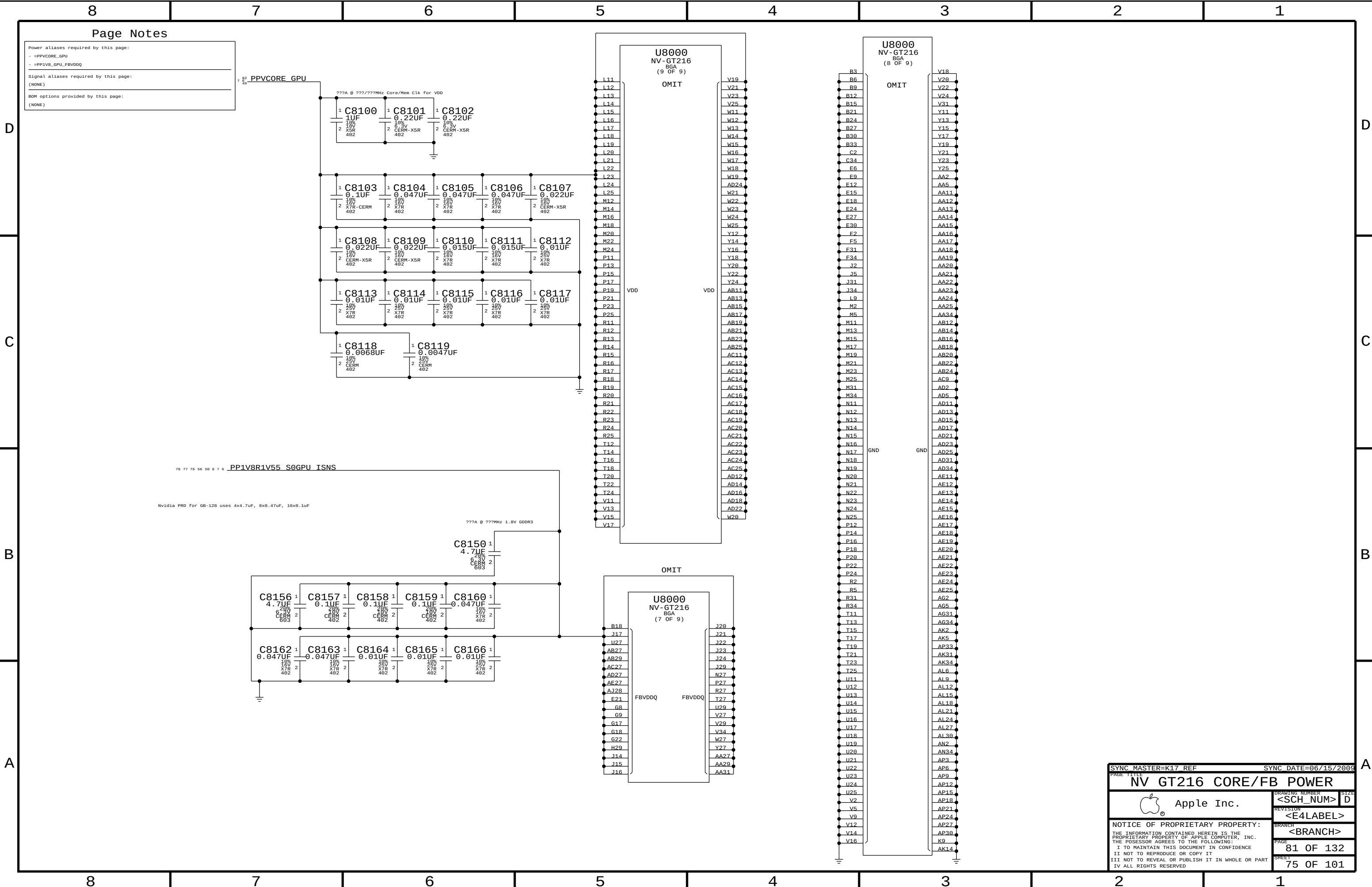
CRITICAL

WLAN Enable Generation

```
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
```

NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.





Page Notes

Power aliases required by this page:

- =PPVCORE_GPU
- =PP1V8_GPU_FBVDDQ

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

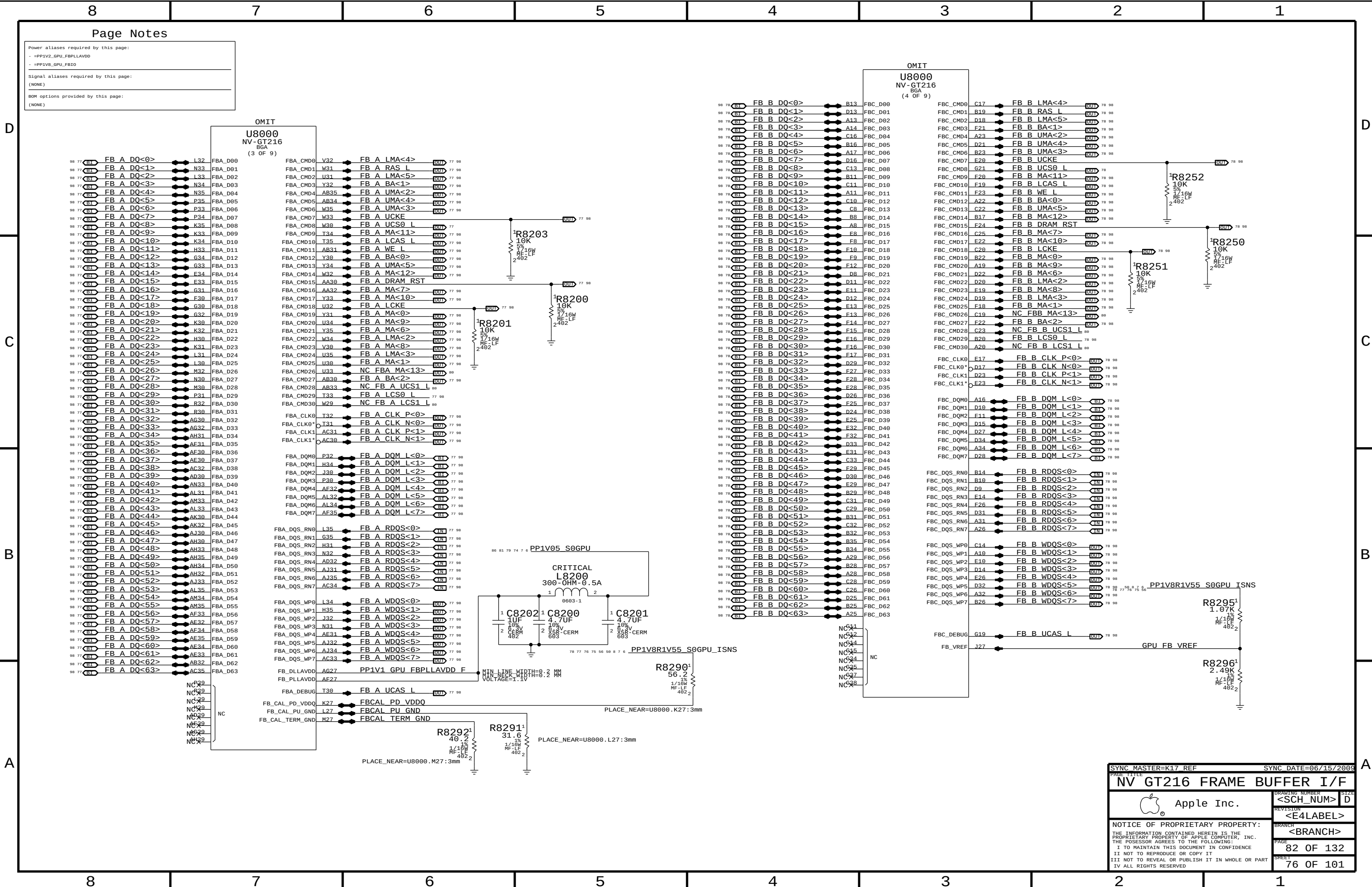
(NONE)

PP1V8R1V55 S0GPU ISNS

Nvidia PRD for GB-128 uses 4x4.7uF, 8x0.47uF, 16x0.1uF

???A @ ???MHz 1.8V GDDR3

SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
NV GT216 CORE/FB POWER			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	81 OF 132
		SHEET	75 OF 101



Power aliases required by this page:

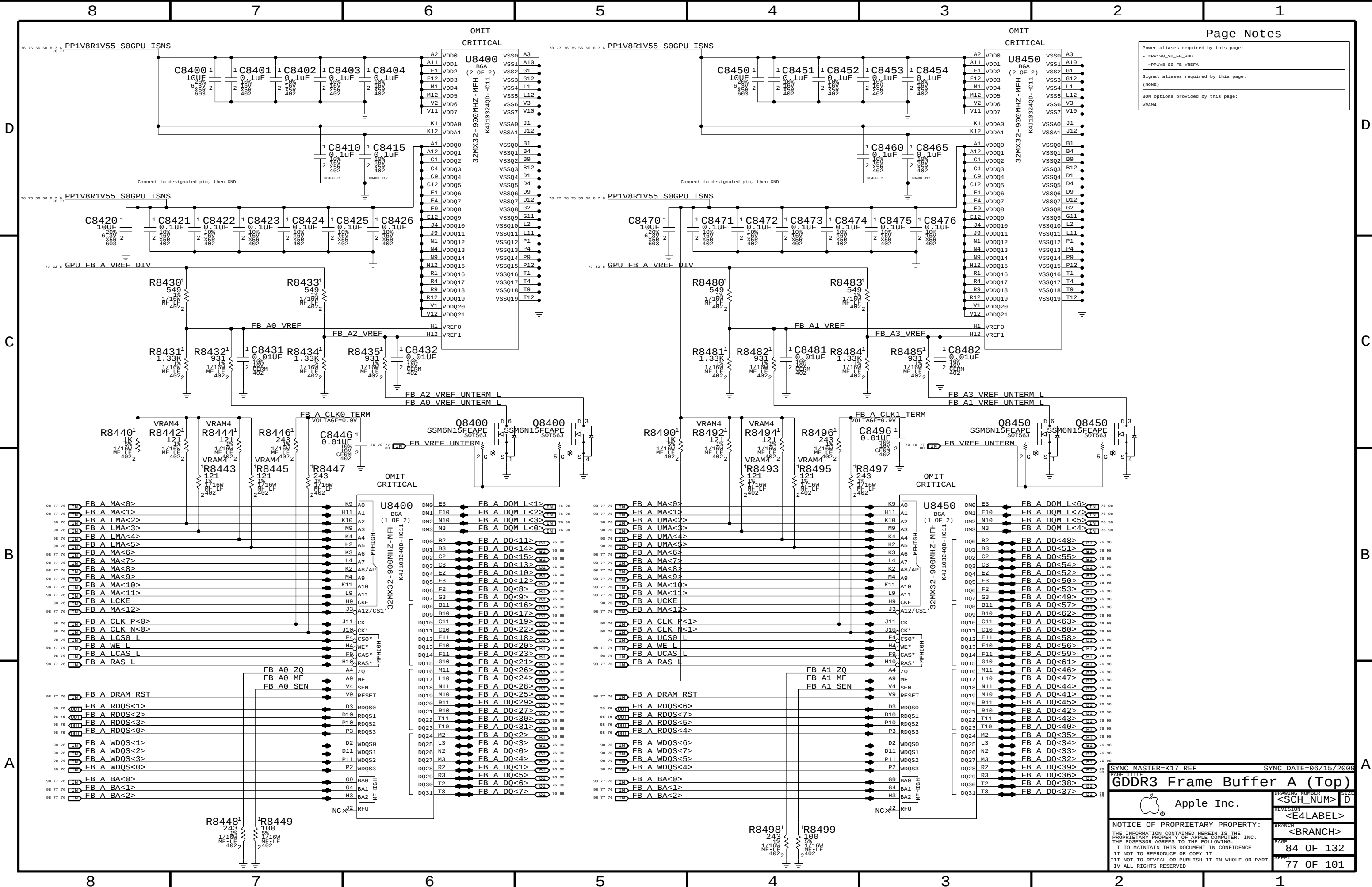
- PP1V2_GPU_FBLLAVDD
- PP1V8_GPU_FBIO

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)



Page Notes

Power aliases required by this page:

- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VREFA

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

VRAM4

SYNC MASTER=K17 REF

SYNC DATE=06/15/2009

GDDR3 Frame Buffer A (Top)

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

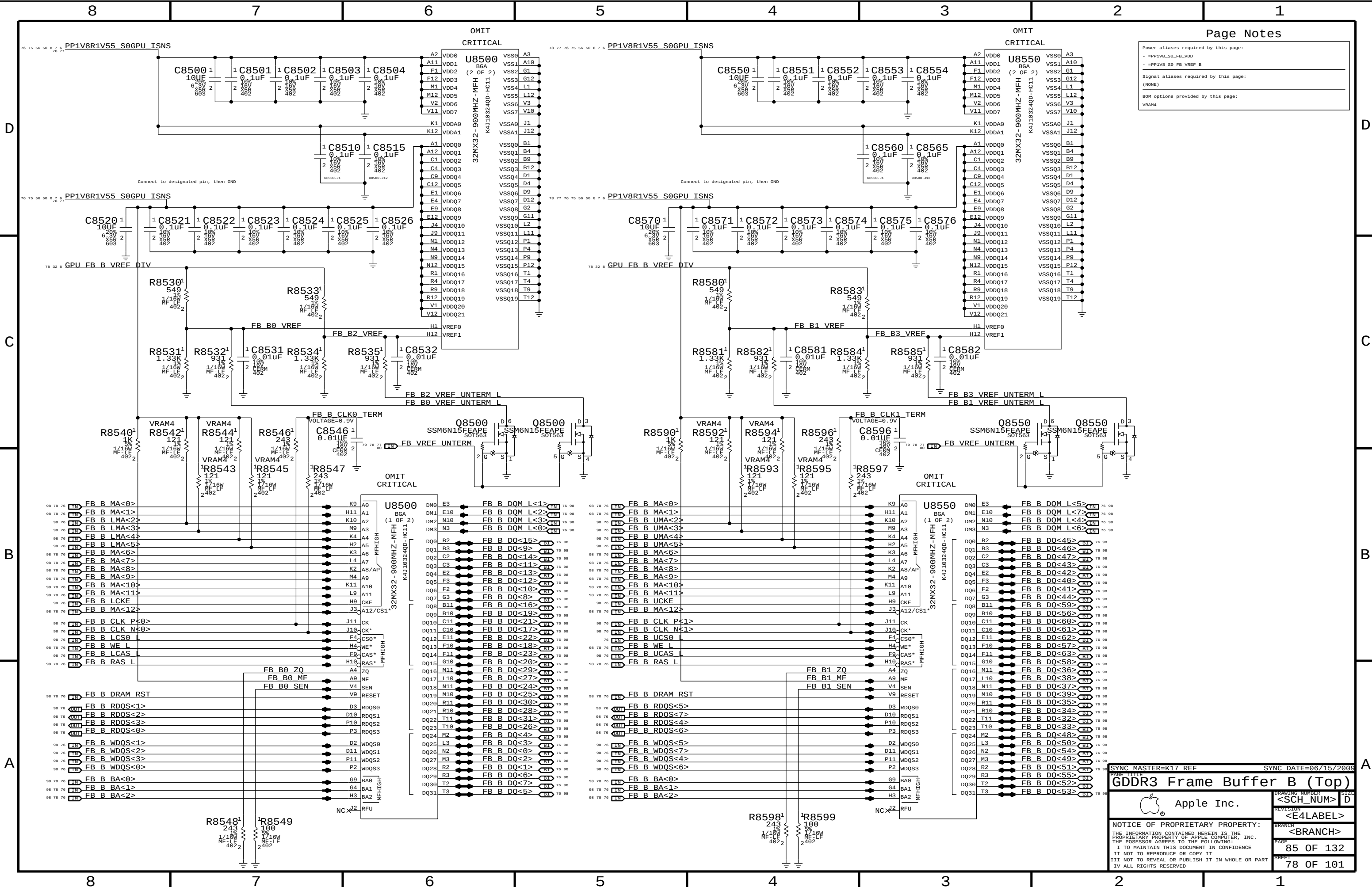
<SCH_NUM>

<E4LABEL>

<BRANCH>

84 OF 132

77 OF 101



Page Notes

Power aliases required by this page:

- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VREF_B

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

VRAM4

SYNC MASTER=K17 REF

SYNC DATE=06/15/2009

PAGE TITLE

GDDR3 Frame Buffer B (Top)

Apple Inc.

<SCH_NUM> D

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

<E4LABEL>

REVISION

<BRANCH>

PAGE

85 OF 132

SHEET

78 OF 101

Page Notes

Power aliases required by this page:

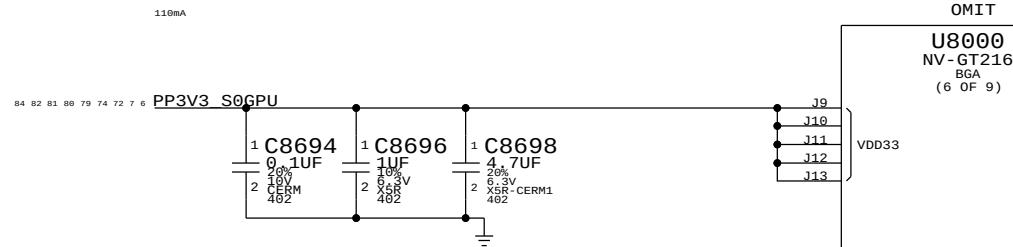
- =PP3V3_GPU_VDD33
- =PP3V3_GPI_MIO
- =PP1V2_GPU_PLLVDD
- =PP1V2_GPU_H_PLLVDD
- =PP1V2_GPU_VID_PLLVDD

Signal aliases required by this page:

(NONE)

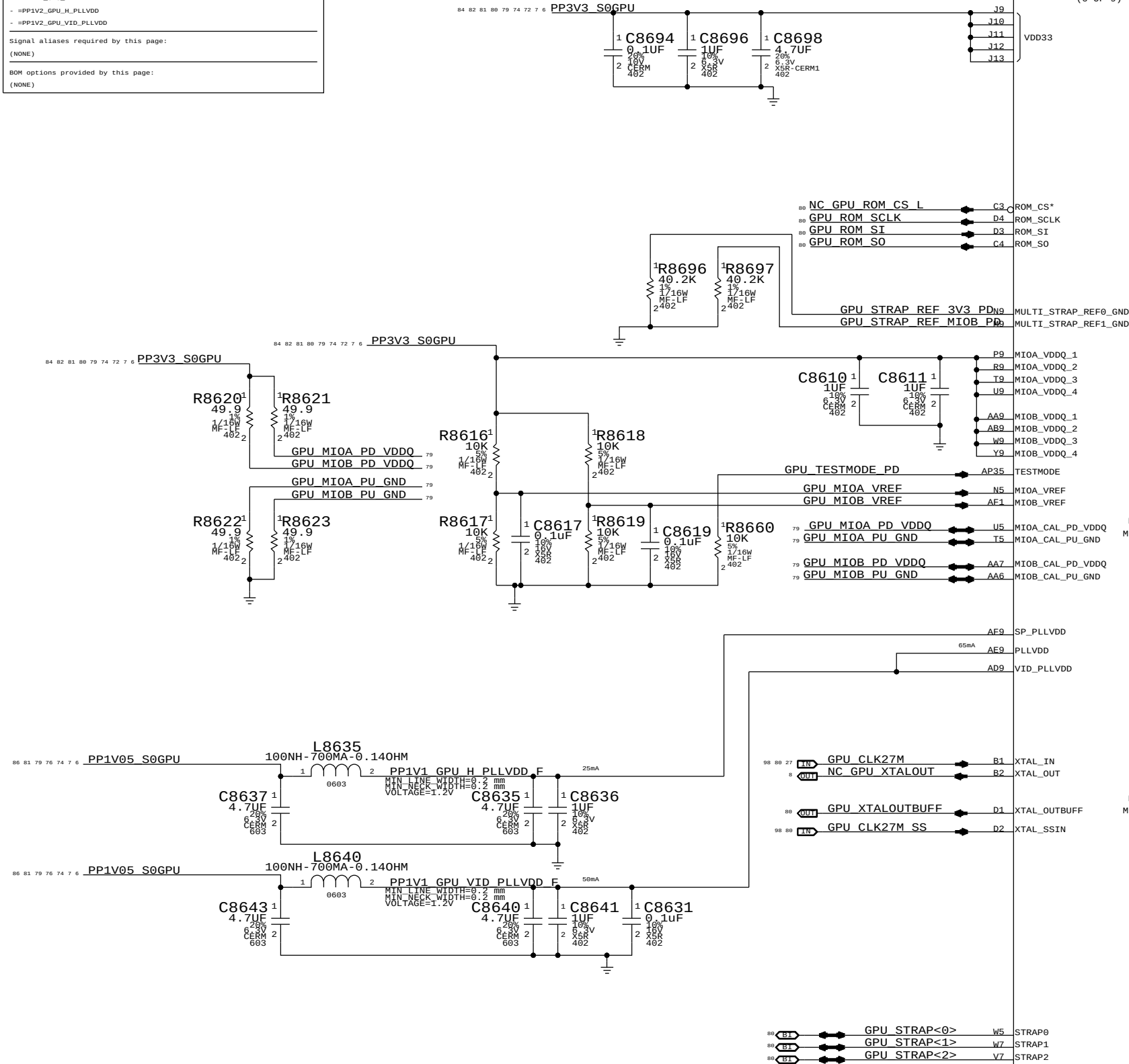
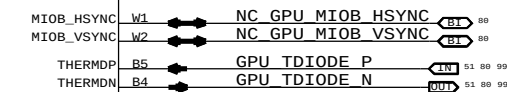
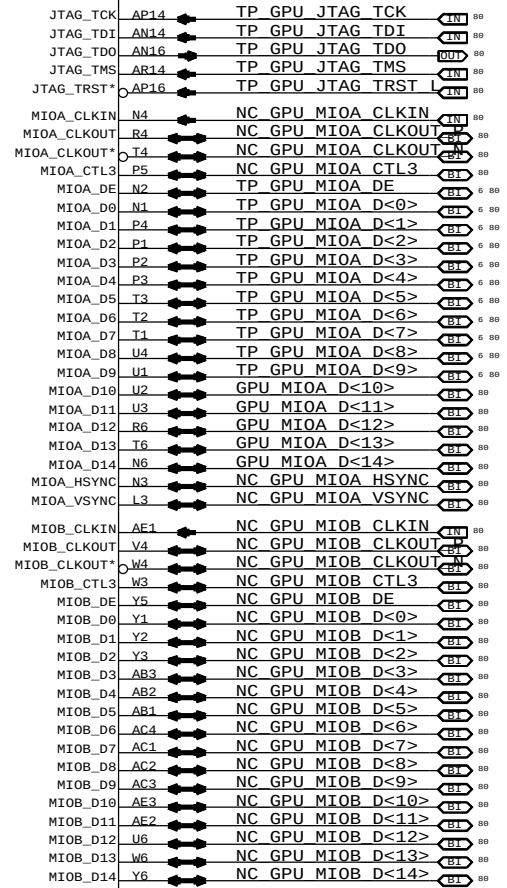
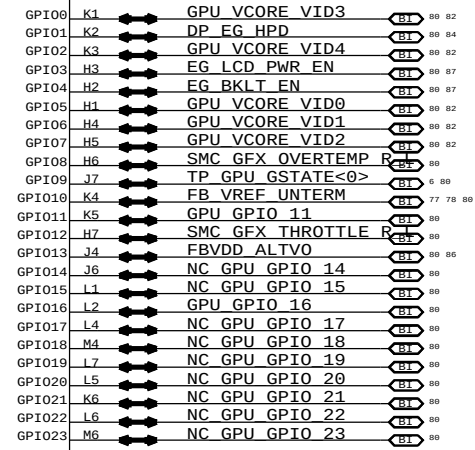
BOM options provided by this page:

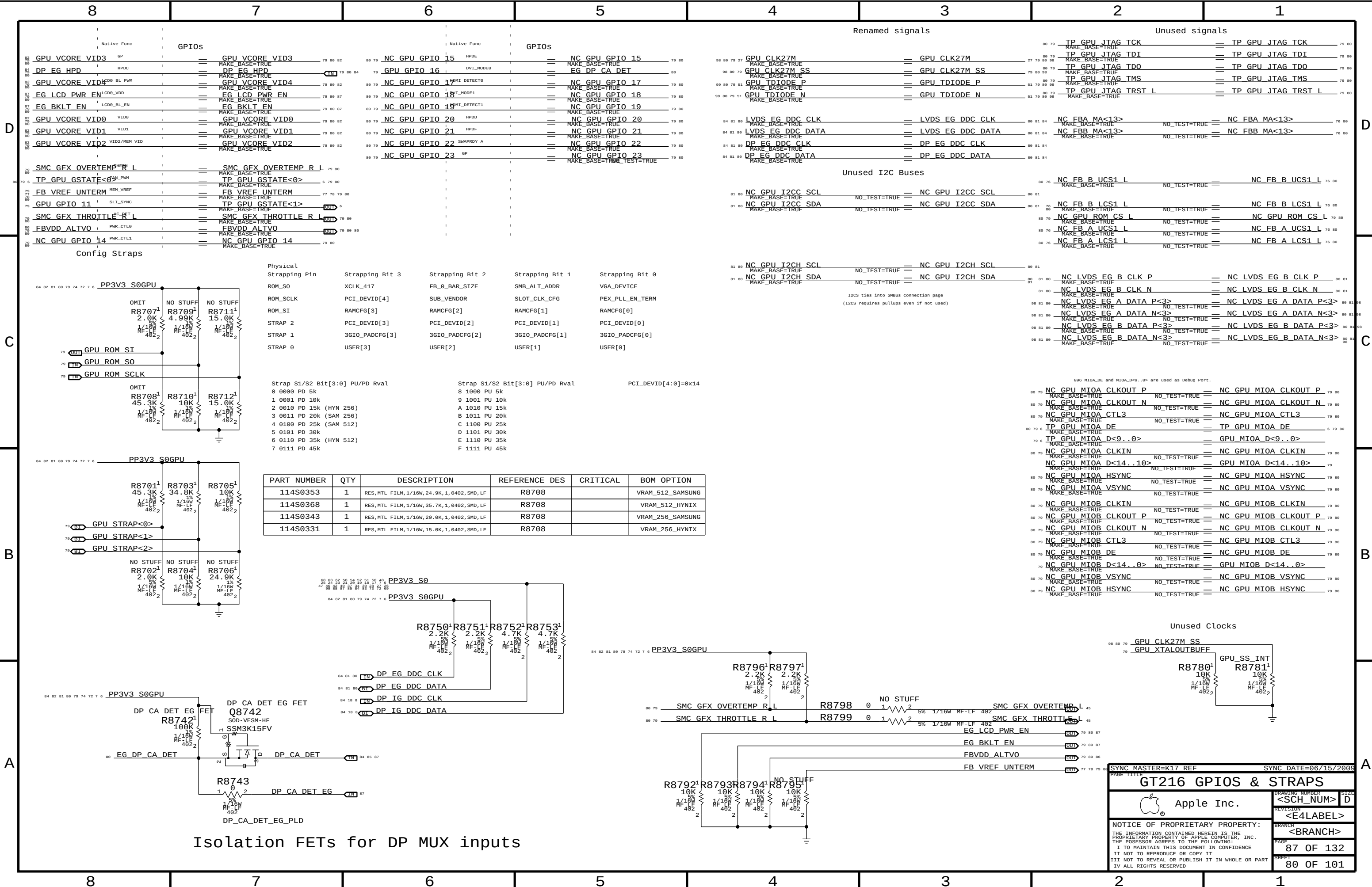
(NONE)



OMIT

U8000
V-GT21
BGA
(6 OF 9)



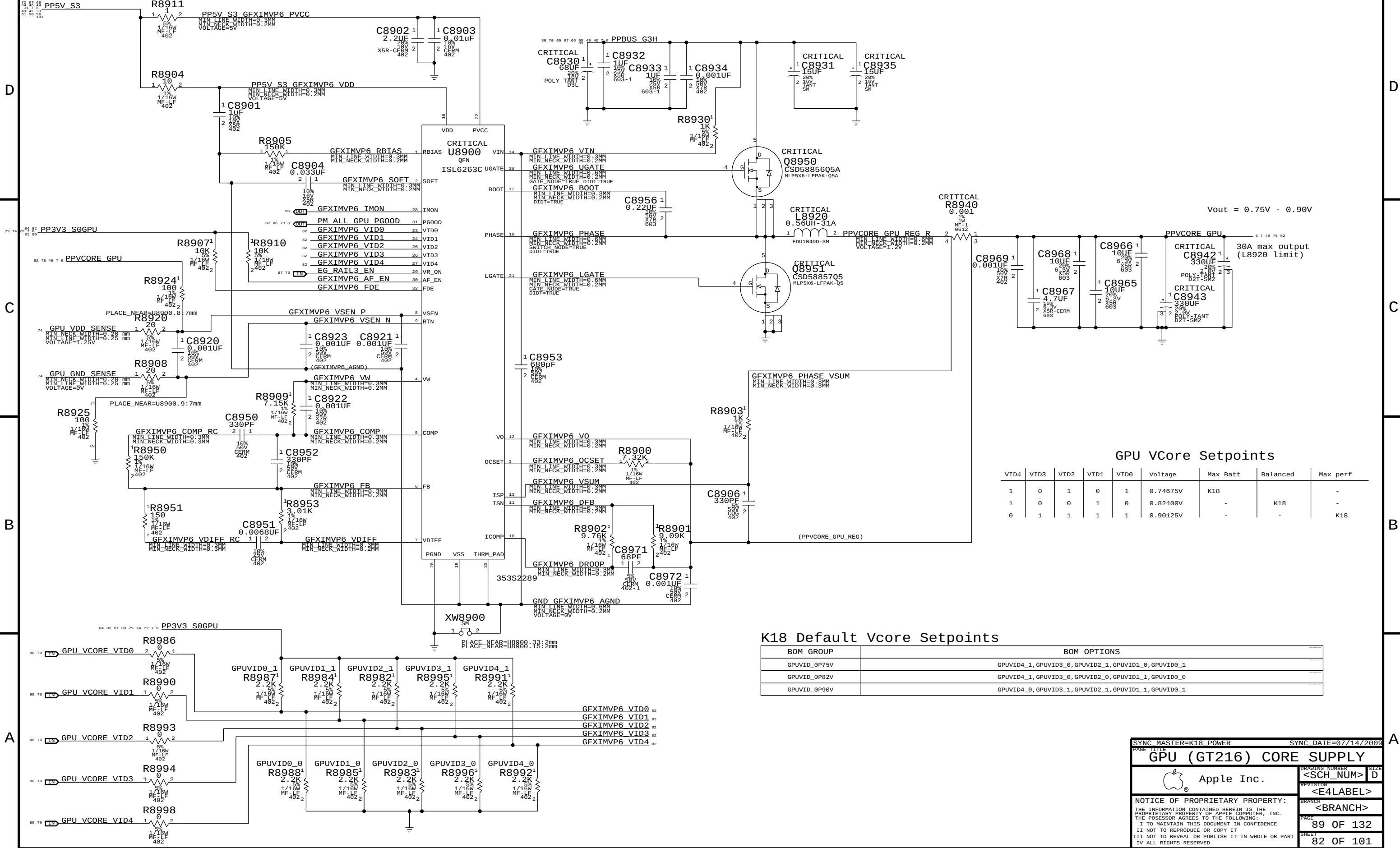


Strapping Bit 3	Strapping Bit 2	Strapping Bit 1	Strapping Bit 0
XCLK_417	FB_0_BAR_SIZE	SMB_ALT_ADDR	VGA_DEVICE
PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
RAMCFG[3]	RAMCFG[2]	RAMCFG[1]	RAMCFG[0]
PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
3GIO_PADCFG[3]	3GIO_PADCFG[2]	3GIO_PADCFG[1]	3GIO_PADCFG[0]
USER[3]	USER[2]	USER[1]	USER[0]

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0353	1	RES,MTL FILM,1/16W,24.9K,1,0402,SMD,LF	R8708		VRAM_512_SAMSUNG
114S0368	1	RES,MTL FILM,1/16W,35.7K,1,0402,SMD,LF	R8708		VRAM_512_HYNIX
114S0343	1	RES,MTL FILM,1/16W,20.0K,1,0402,SMD,LF	R8708		VRAM_256_SAMSUNG
114S0331	1	RES,MTL FILM,1/16W,15.0K,1,0402,SMD,LF	R8708		VRAM_256_HYNIX

PAGE TITLE		SYNC DATE=06/15/2009	
GT216 GPIOs & STRAPS		DRAWING NUMBER	SIZE
Apple Inc.		<SCH_NUM>	D
		REVISION	<E4LABEL>
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	<BRANCH>
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	87 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	80 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

GPU VCore Regulator



GPU VCore Setpoints

VID4	VID3	VID2	VID1	VID0	Voltage	Max Batt	Balanced	Max perf
1	0	1	0	1	0.74675V	K18		-
1	0	0	1	0	0.82400V	-	K18	-
0	1	1	1	1	0.90125V	-	-	K18

K18 Default Vcore Setpoints

BOM GROUP	BOM OPTIONS
GPUVID_0P75V	GPUVID4_1, GPUVID3_0, GPUVID2_1, GPUVID1_0, GPUVID0_1
GPUVID_0P82V	GPUVID4_1, GPUVID3_0, GPUVID2_0, GPUVID1_1, GPUVID0_0
GPUVID_0P90V	GPUVID4_0, GPUVID3_1, GPUVID2_1, GPUVID1_1, GPUVID0_1

SYNC MASTER=K18 POWER

SYNC DATE=07/14/2009

GPU (GT216) CORE SUPPLY

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

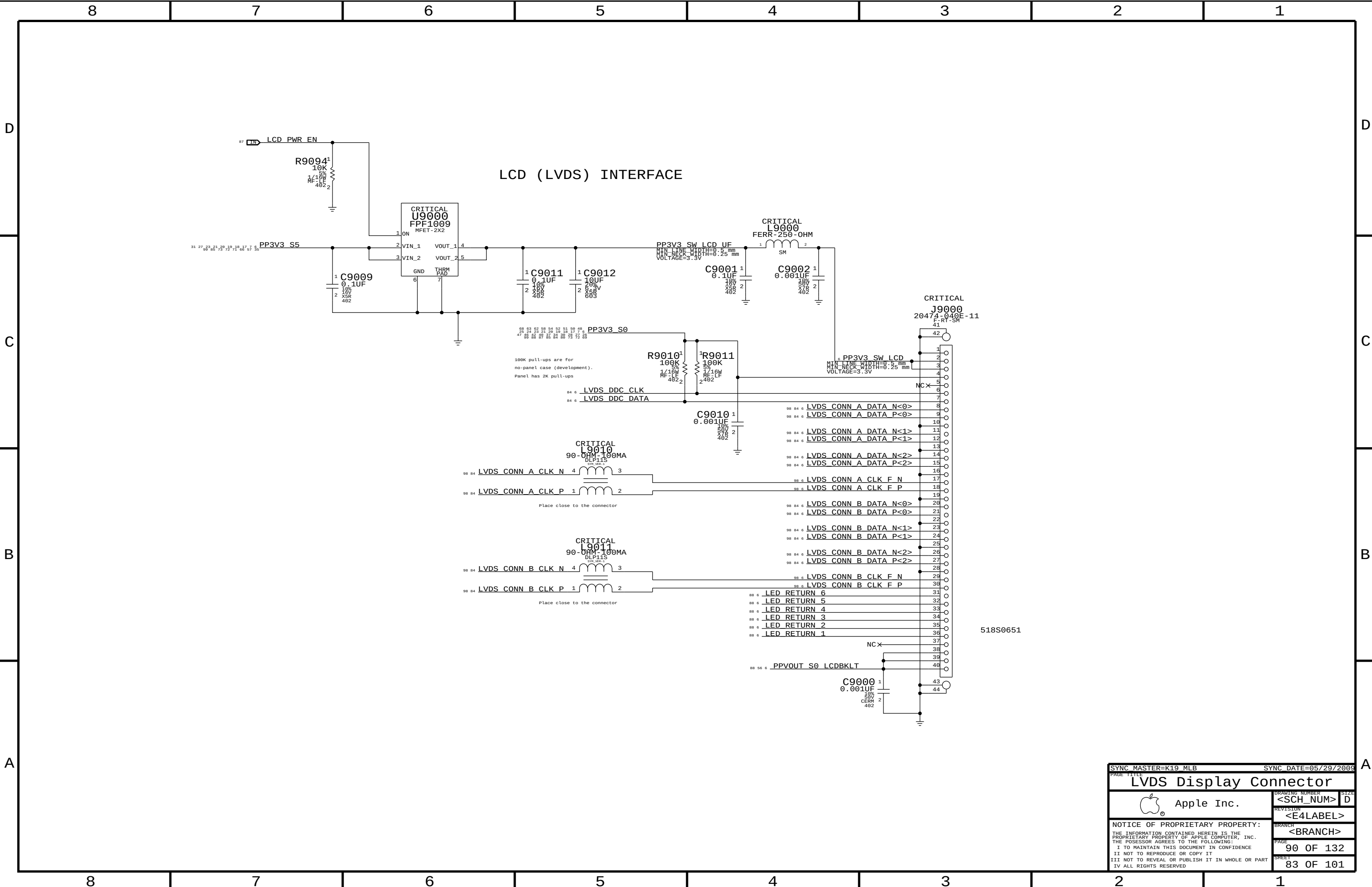
DRAWING NUMBER
<SCH_NUM> D

REVISION
<E4LABEL>

BRANCH
<BRANCH>

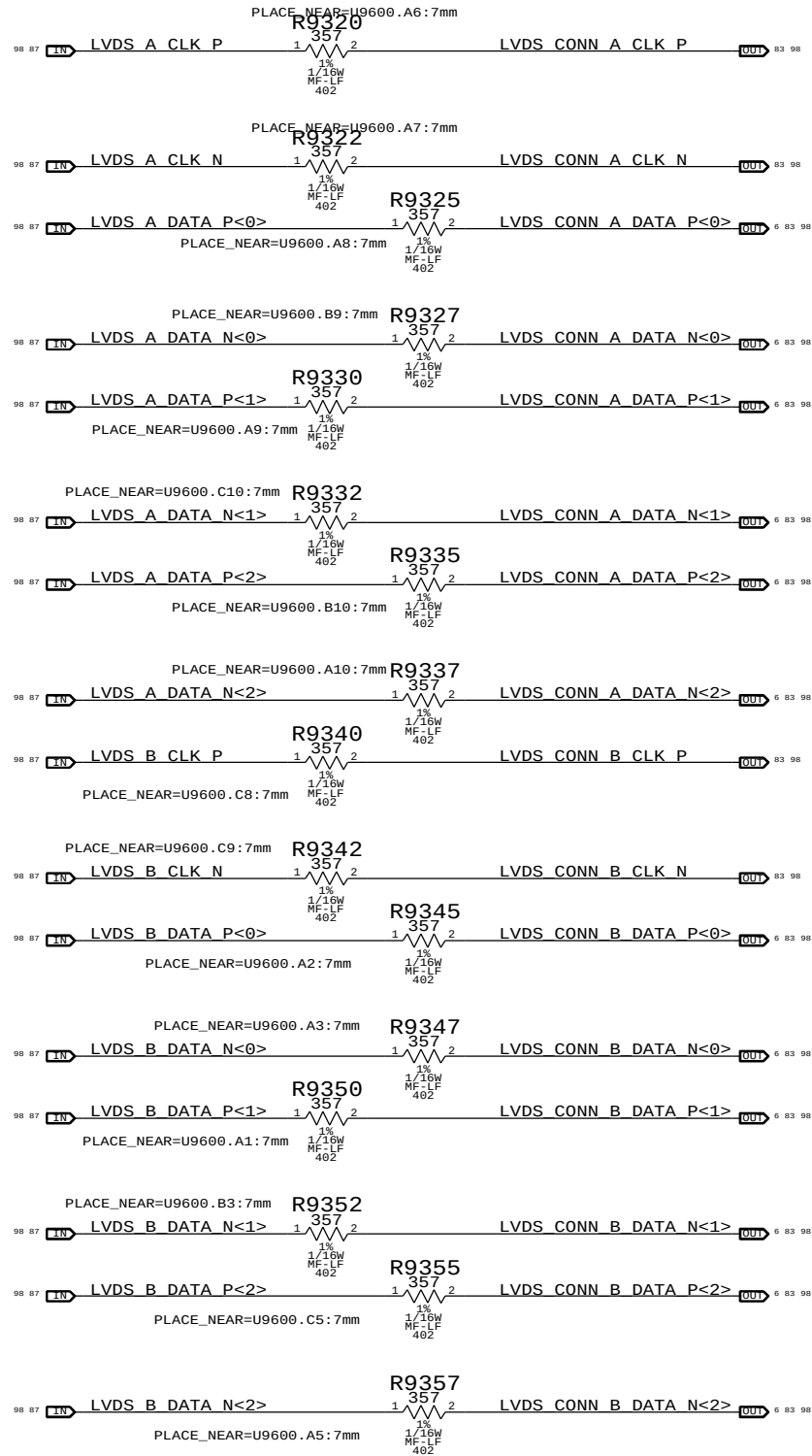
PAGE
89 OF 132

SHEET
82 OF 101

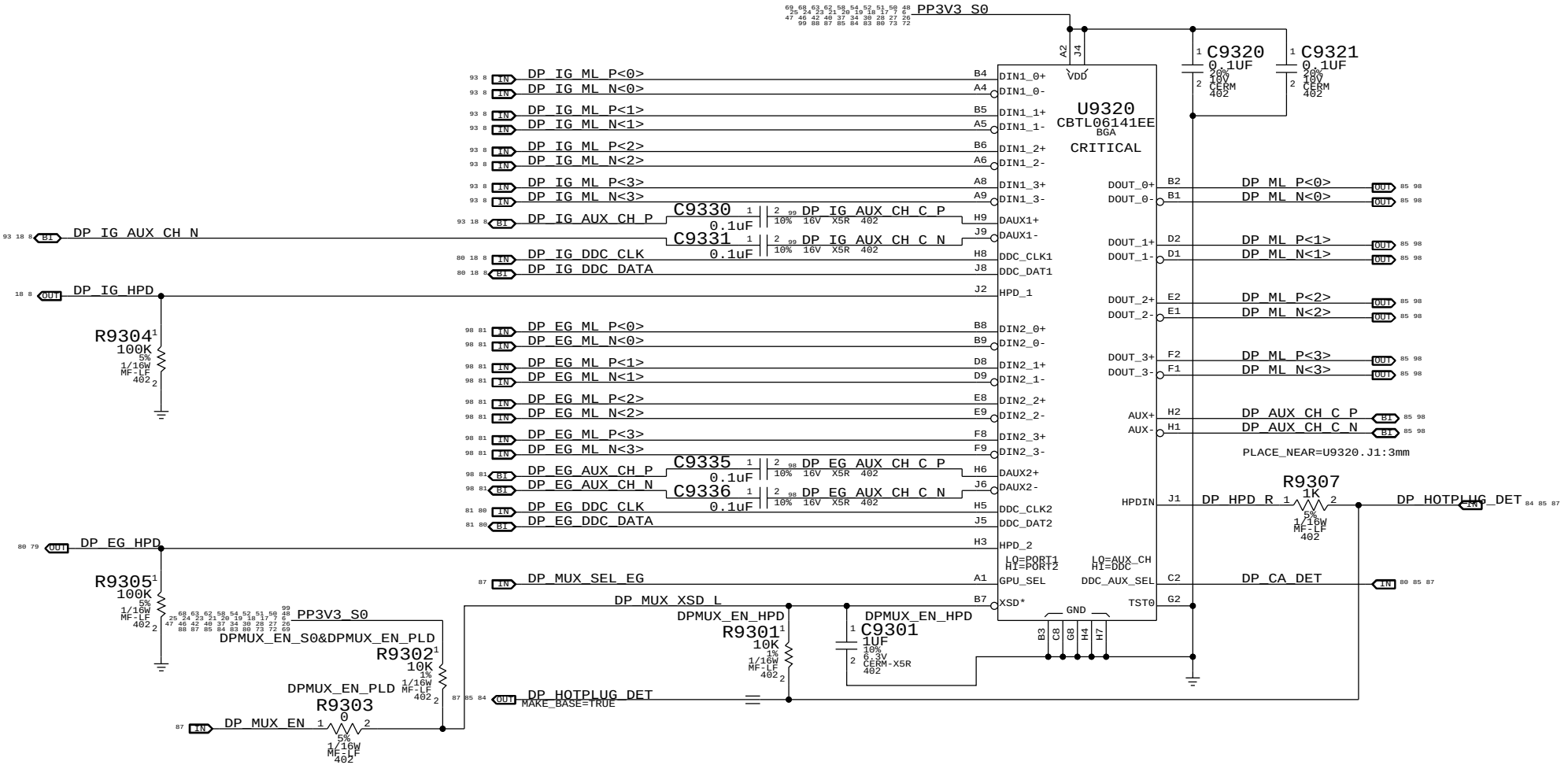


LVDS Transmitter Termination

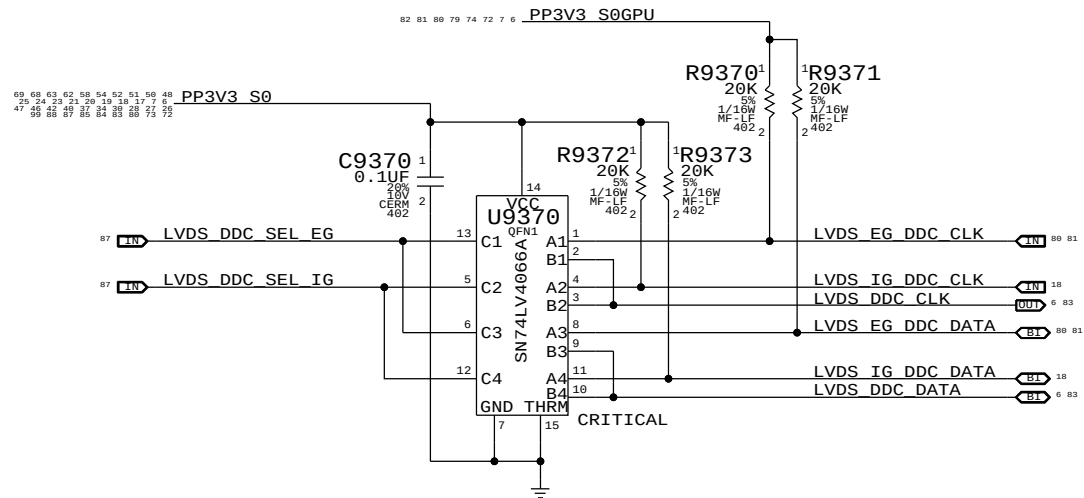
All emulated LVDS outputs require this termination



DisplayPort Mux

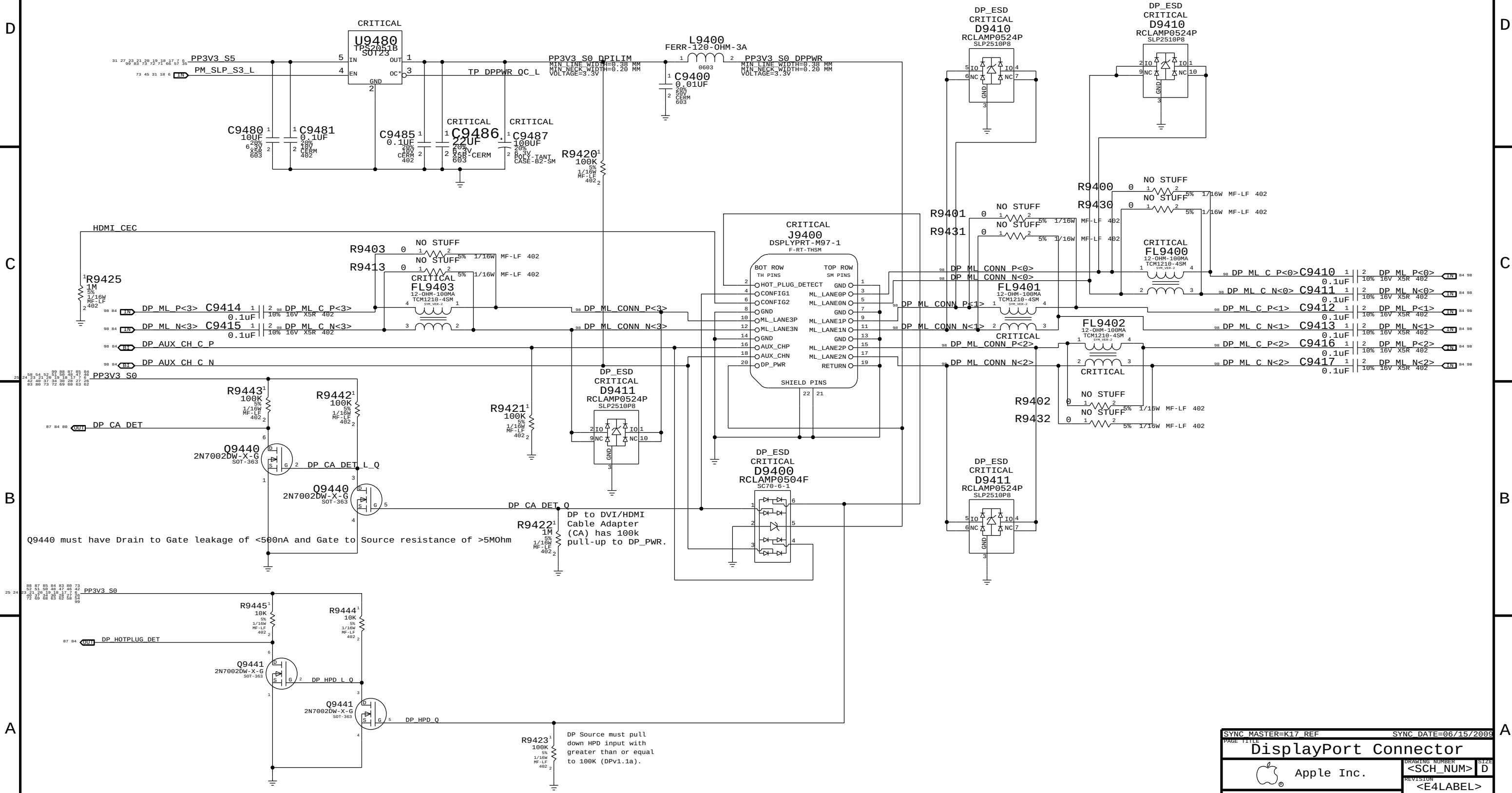


LVDS DDC MUX

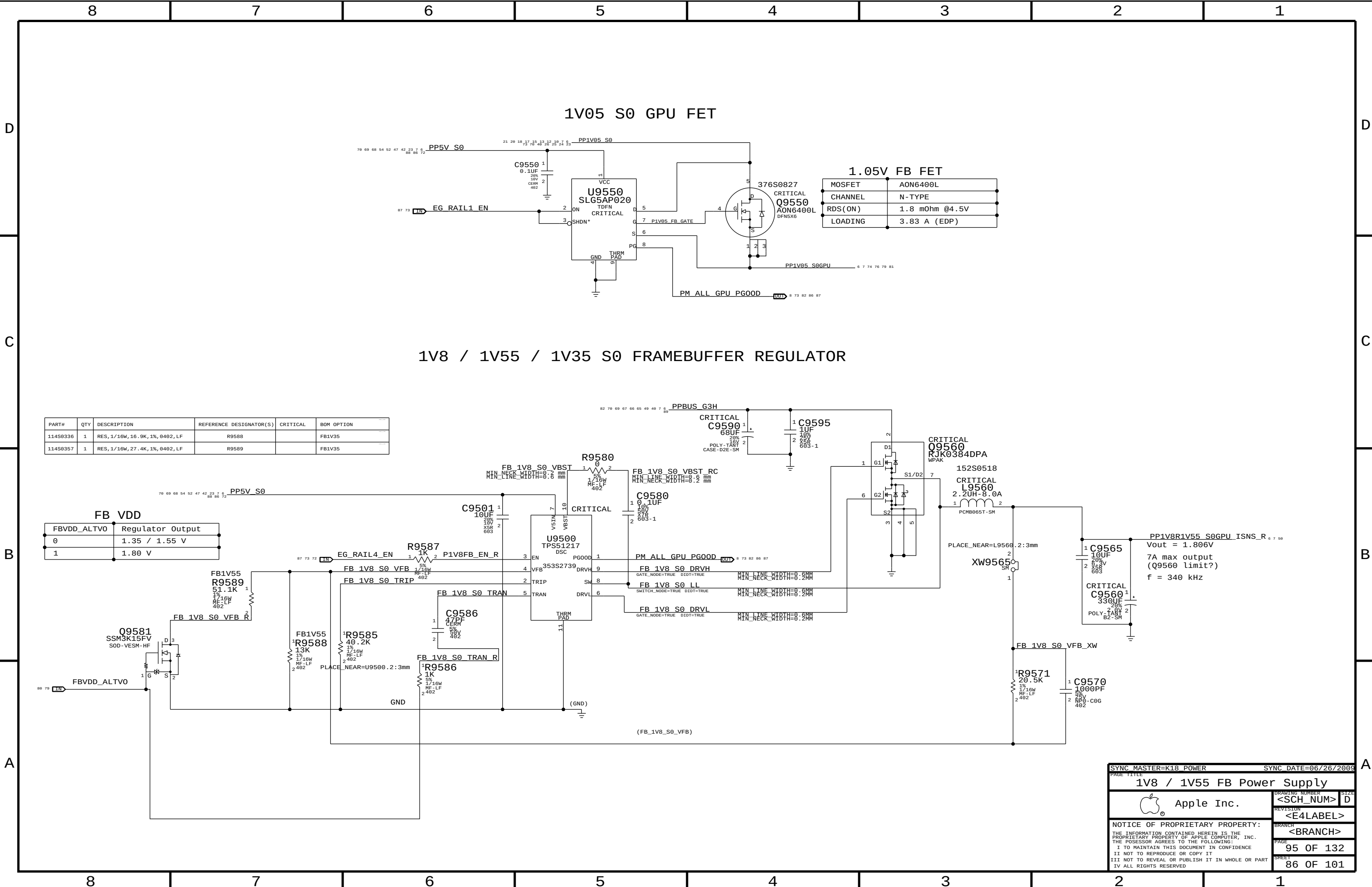


PAGE TITLE		SYNC DATE=06/15/2009	
Muxed Graphics Support		DRAWING NUMBER	
Apple Inc.		<SCH_NUM>	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		<E4LABEL>	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		<BRANCH>	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		PAGE	
IV ALL RIGHTS RESERVED		93 OF 132	
		SHEET	
		84 OF 101	

Port Power Switch



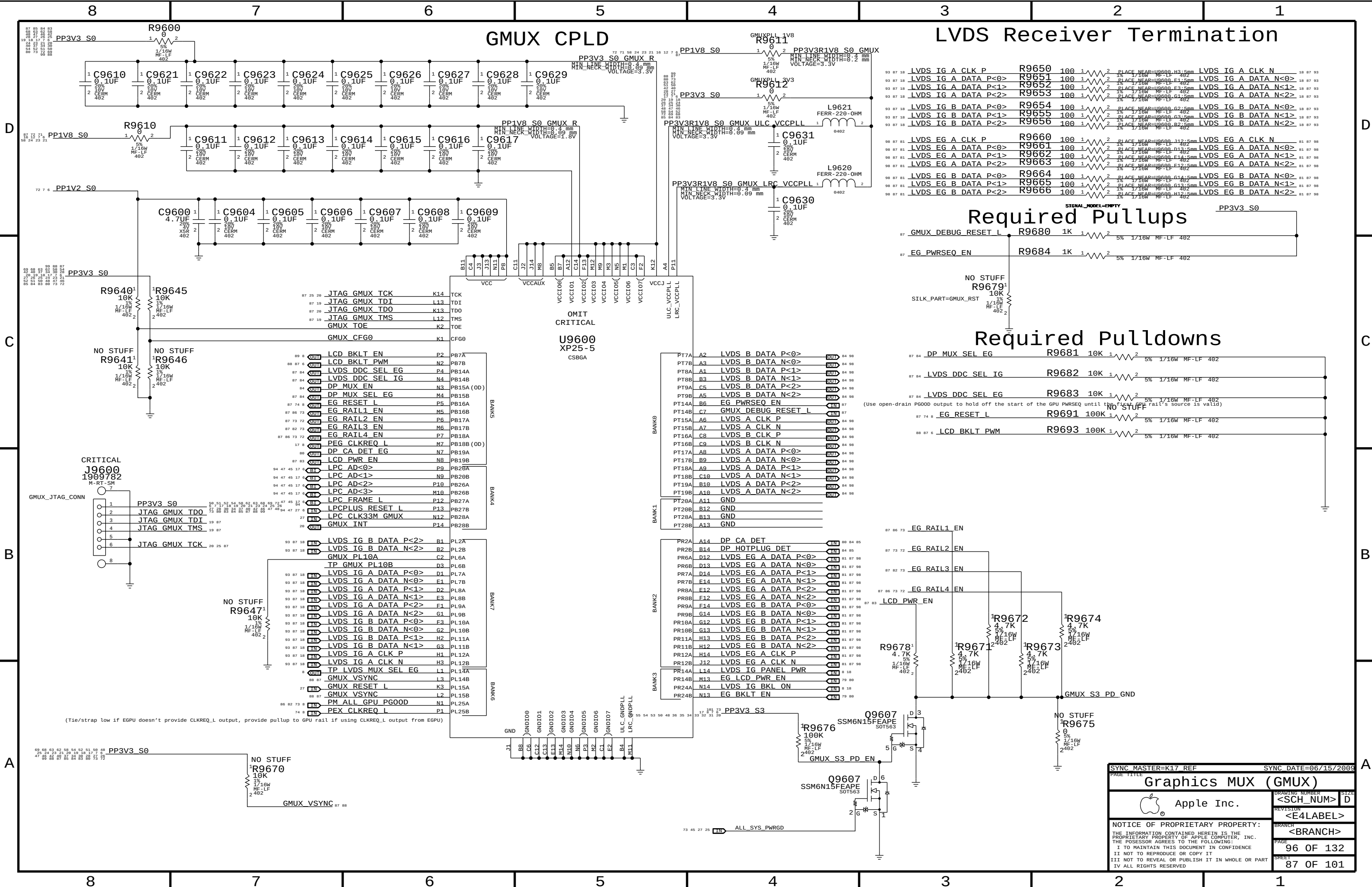
SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
DisplayPort Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	94 OF 132
		SHEET	85 OF 101



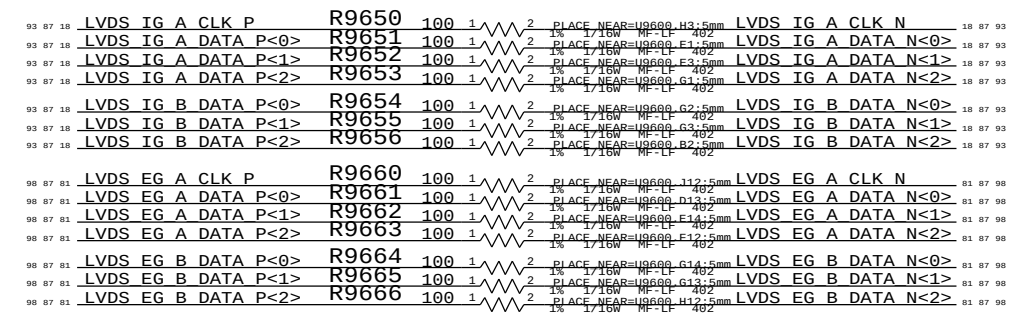
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
114S0336	1	RES, 1/16W, 16.9K, 1%, 0402, LF	R9588		FB1V35
114S0357	1	RES, 1/16W, 27.4K, 1%, 0402, LF	R9589		FB1V35

FB VDD	
FBVDD_ALTV0	Regulator Output
0	1.35 / 1.55 V
1	1.80 V

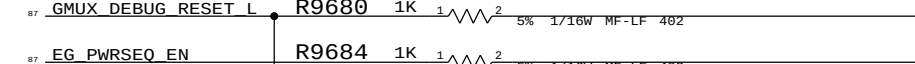
SYNC MASTER=K18 POWER		SYNC DATE=06/26/2009	
PAGE TITLE			
1V8 / 1V55 FB Power Supply			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	95 OF 132
		SHEET	86 OF 101
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			



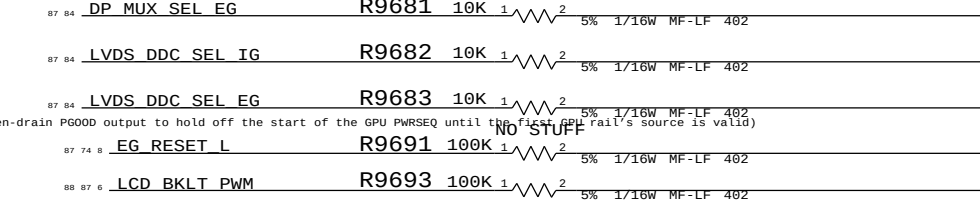
LVDS Receiver Termination



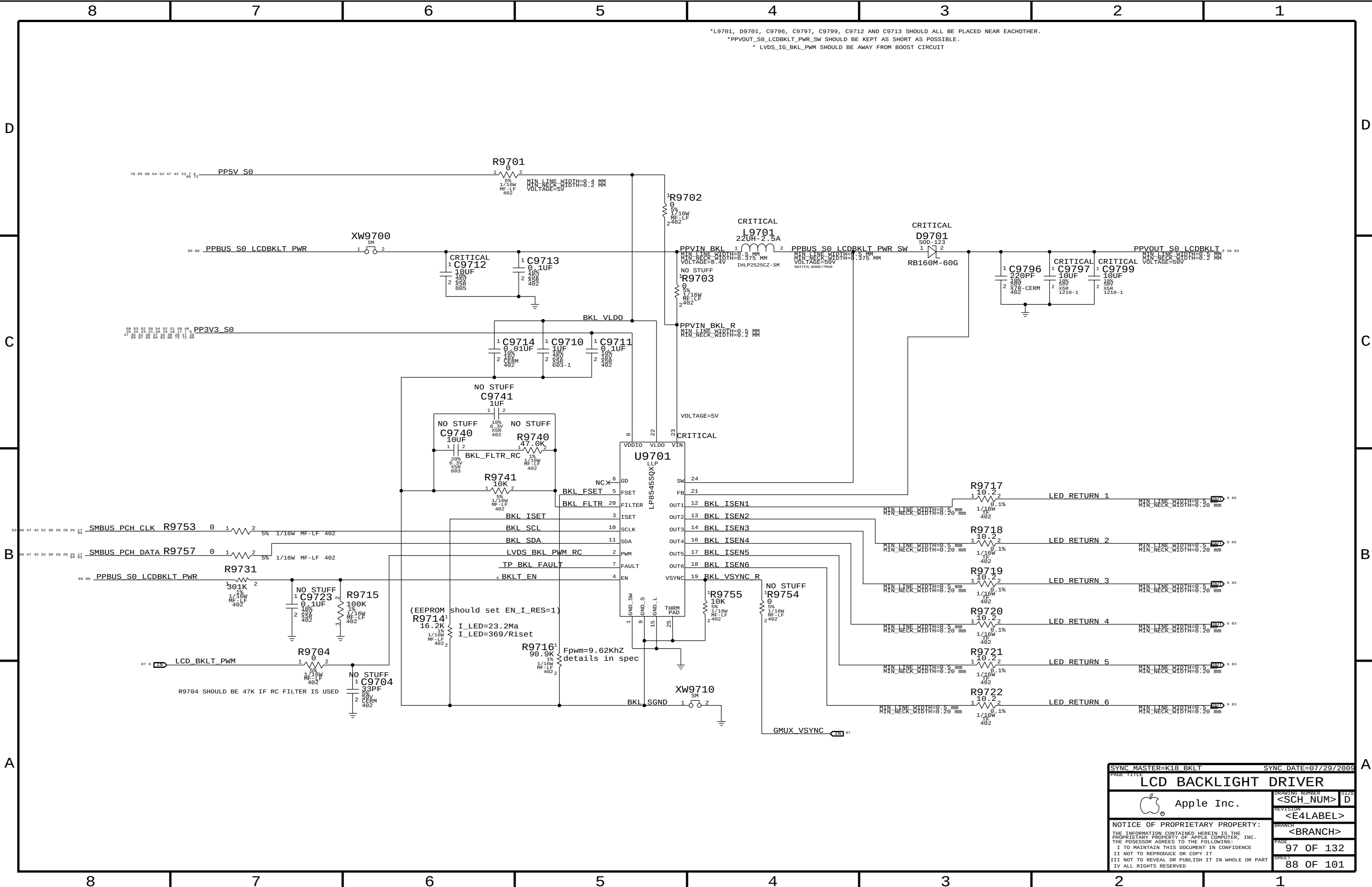
Required Pullups



Required Pulldowns

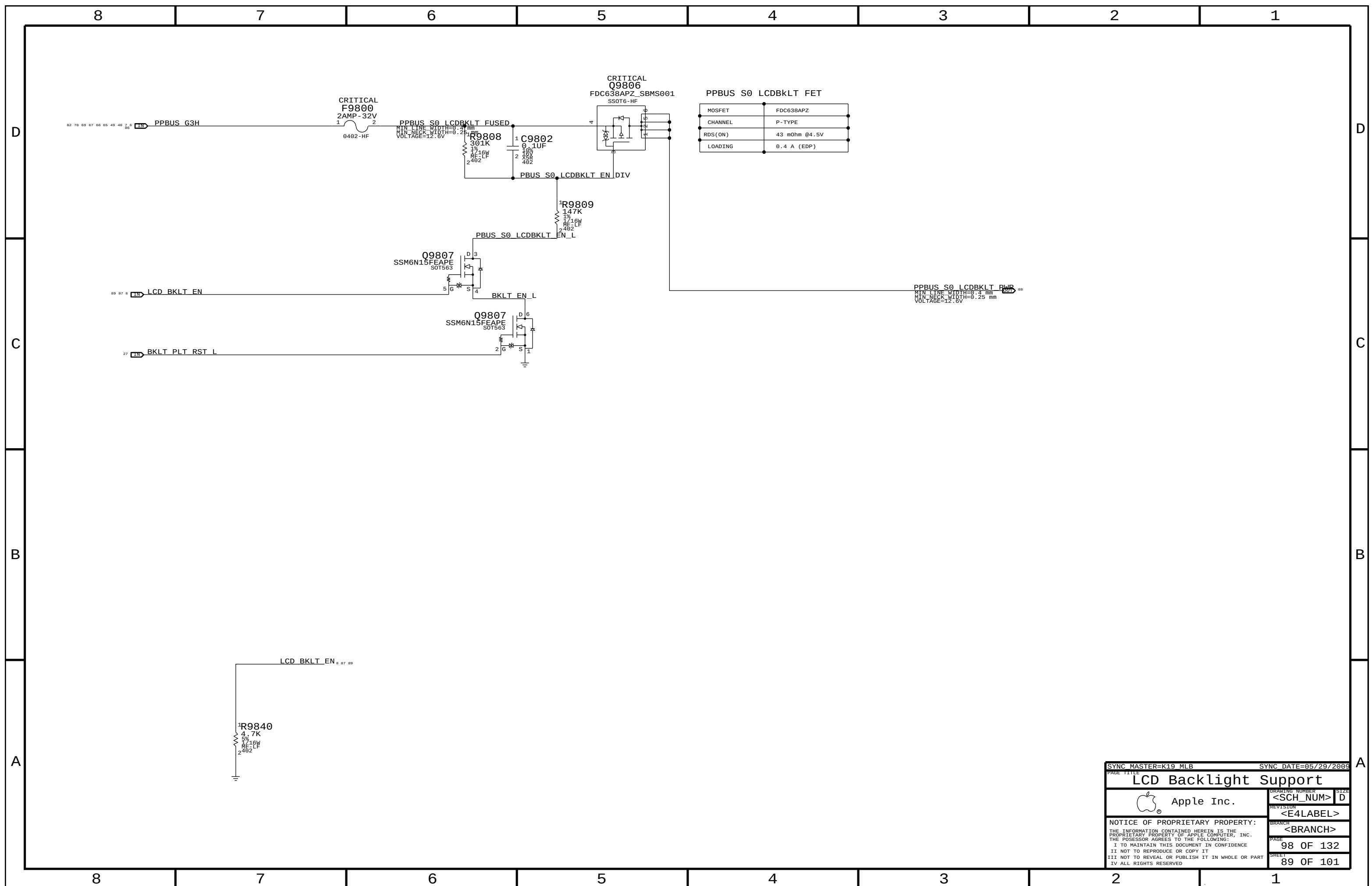


SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE		GRAPHICS MUX (GMUX)	
Apple Inc.		<SCH NUMBER> D	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		<E4LABEL>	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		<BRANCH>	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		PAGE	
IV ALL RIGHTS RESERVED		96 OF 132	
		SHEET	
		87 OF 101	



*L9701, D9701, C9796, C9797, C9799, C9712 AND C9713 SHOULD ALL BE PLACED NEAR EACHOTHER.
*PPVOUT_S0_LCDBKLT_PWR_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
* LVDS_IG_BKL_PWM SHOULD BE AWAY FROM BOOST CIRCUIT

SYNC MASTER=K18 BKLT		SYNC DATE=07/29/2009	
PAGE TITLE			
LCD BACKLIGHT DRIVER			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	97 OF 132
		SHEET	88 OF 101



8

7

6

5

4

3

2

1

D

D

C

C

B

B

A

A

Blank Page, was 1.2V/1.8V in K19

SYNC MASTER=K18 POWER		SYNC DATE=06/10/2009	
PAGE TITLE			
Misc Power Supplies			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
		SIZE D	
		REVISION <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
		BRANCH <BRANCH>	
		PAGE 99 OF 132	
		SHEET 90 OF 101	

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for IbeX Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for IbeX Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

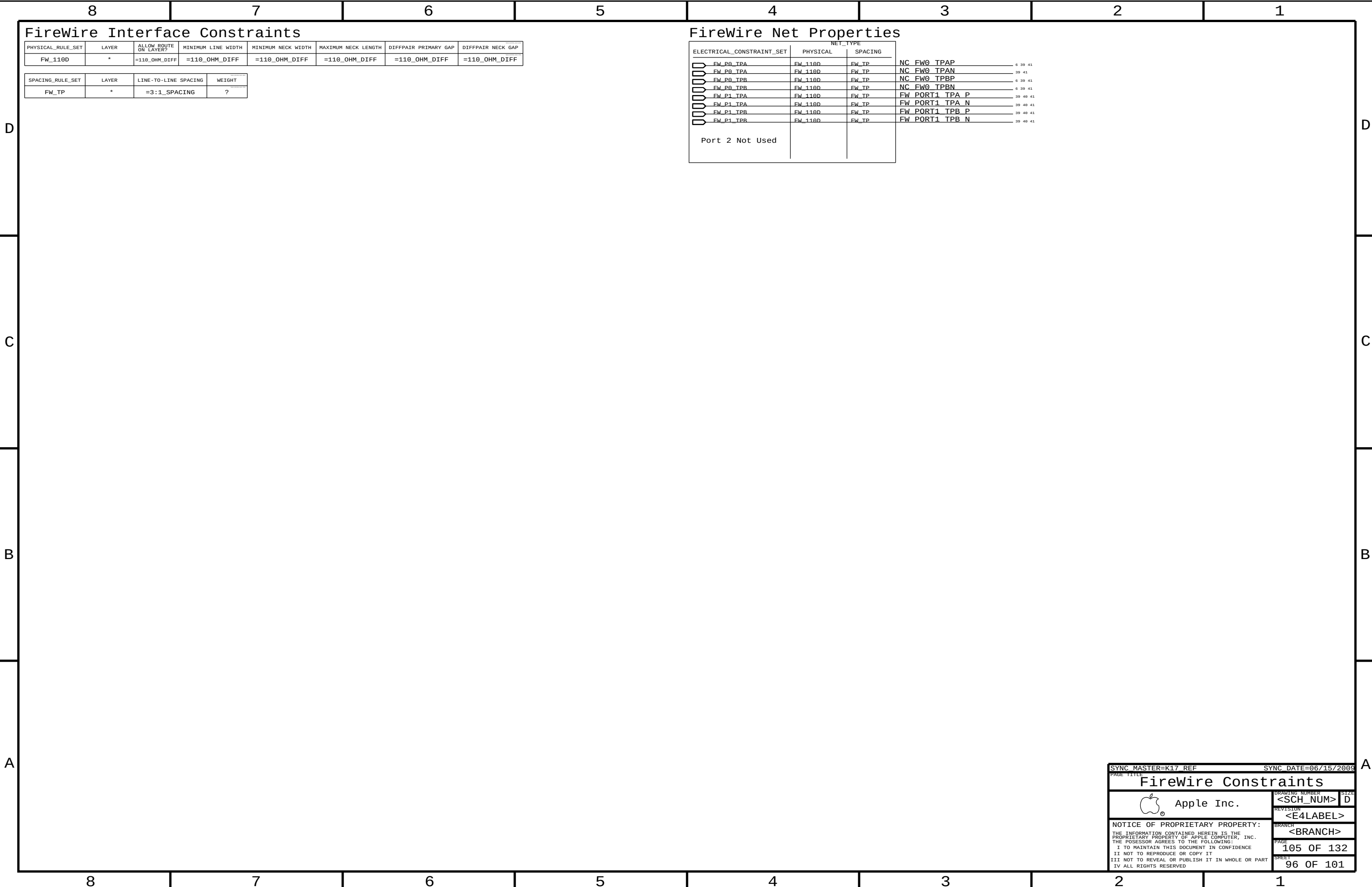
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

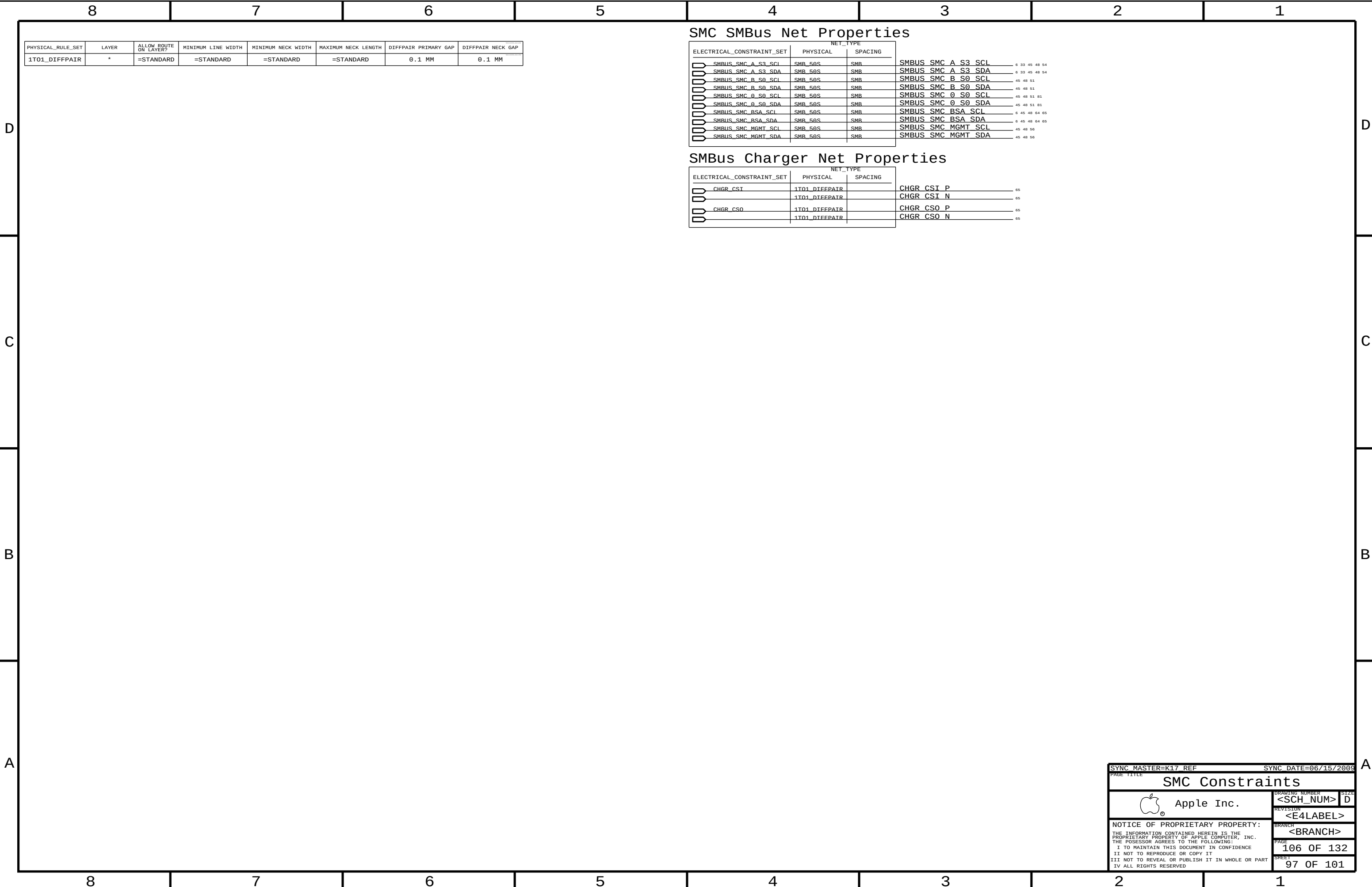
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	LPC_AD	LPC_50S	LPC	LPC_AD<3...0> 6 17 45 47 87
	LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L 6 17 45 47 87
	LPC_RESET_L	LPC_50S	LPC	LPCPLUS_RESET_L 6 27 47 87
	MCP_LPC_CLK0	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R 19 27
		CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC 27 45
		CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS 6 27 47
	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK 17 25 26 28 39 32 42 47 48 63
	SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA 17 25 26 28 39 32 42 47 48 63
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SML_PCH_0_CLK 17 48
	SMBUS_PCH_0_DATA	SMB_50S	SMB	SML_PCH_0_DATA 17 48
	SMBUS_PCH_1_CLK	SMB_50S	SMB	SML_PCH_1_CLK 17 48
	SMBUS_PCH_1_DATA	SMB_50S	SMB	SML_PCH_1_DATA 17 48
	HDA_BIT_CLK	HDA_50S	HDA	HDA_BIT_CLK 17 58
		HDA_50S	HDA	HDA_BIT_CLK_R 17
	HDA_SYNC	HDA_50S	HDA	HDA_SYNC 17 58
		HDA_50S	HDA	HDA_SYNC_R 17
	HDA_RST_L	HDA_50S	HDA	HDA_RST_R_L 17
		HDA_50S	HDA	HDA_RST_L 17 58
	HDA_SDTN0	HDA_50S	HDA	HDA_SDTN0 17 58
		HDA_50S	HDA	AUD_SDI_R 58
	HDA_SDOUT	HDA_50S	HDA	HDA_SDOUT 17 58
		HDA_50S	HDA	HDA_SDOUT_R 17
	PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK 18 46
	SPT_CLK	SPT_55S	SPT	SPT_CLK_R 17 47
		SPT_55S	SPT	SPT_CLK 47
	SPT_MOSI	SPT_55S	SPT	SPT_MOSI_R 17 47
		SPT_55S	SPT	SPT_MOSI 47
	SPT_MISO	SPT_55S	SPT	SPT_MISO 17 47
	SPT_CS0	SPT_55S	SPT	SPT_CS0_R_L 17 47
		SPT_55S	SPT	SPT_CS0_L 47
		PCIE_85D	PCIE	PCIE_ENET_R2D_P 37
		PCIE_85D	PCIE	PCIE_ENET_R2D_N 37
	PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P 17 37
		PCIE_85D	PCIE	PCIE_ENET_R2D_C_N 17 37
	PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_P 17 37
		PCIE_85D	PCIE	PCIE_ENET_D2R_N 17 37
		PCIE_85D	PCIE	PCIE_ENET_D2R_C_P 37
		PCIE_85D	PCIE	PCIE_ENET_D2R_C_N 37
		PCIE_85D	PCIE	PCIE_AP_R2D_P 6 33
		PCIE_85D	PCIE	PCIE_AP_R2D_N 6 33
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_C_P 17 33
		PCIE_85D	PCIE	PCIE_AP_R2D_C_N 17 33
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_D2R_P 6 17 33
		PCIE_85D	PCIE	PCIE_AP_D2R_N 6 17 33
		PCIE_85D	PCIE	PCIE_FW_R2D_P 39
		PCIE_85D	PCIE	PCIE_FW_R2D_N 39
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_P 17 39
		PCIE_85D	PCIE	PCIE_FW_R2D_C_N 17 39
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_P 17 39
		PCIE_85D	PCIE	PCIE_FW_D2R_N 17 39
		PCIE_85D	PCIE	PCIE_FW_D2R_C_P 39
		PCIE_85D	PCIE	PCIE_FW_D2R_C_N 39
	PCIE_AP_D2R	PCIE_85D	PCIE	CONN_PCIE_AP_D2R_P
		PCIE_85D	PCIE	CONN_PCIE_AP_D2R_N
	PCIE_AP_R2D	PCIE_85D	PCIE	CONN_PCIE_AP_R2D_P
		PCIE_85D	PCIE	CONN_PCIE_AP_R2D_N
	MCP_PE0_REFCLK	CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_P 17 74
		CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_N 17 74
	PCIE_CLK100M_ENET	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_P 17 37
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_N 17 37
	MCP_PE1_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_P 17 33
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_N 17 33
	MCP_PE2_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P 17 39
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N 17 39
	MCP_PE3_REFCLK	CLK_PCIE_90D	CLK_PCIE	NC_PCIE_CLK100M_EXCARD_P 6 17
		CLK_PCIE_90D	CLK_PCIE	NC_PCIE_CLK100M_EXCARD_N 6 17
10000		CPU1_27P4S	CPU1_COMP	PCH_VSS_NCTF<1> 6 20
10000		CPU1_27P4S	CPU1_COMP	PCH_VSS_NCTF<2> 6 20
10000		CPU1_27P4S	CPU1_COMP	PCH_VSS_NCT

SYNC_MASTER=K17 REF		SYNC_DATE=06/15/2008	
PAGE TITLE			
PCH Constraints 2			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
		REVISION <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH <BRANCH>	
		PAGE 103 OF 132	
		SHEET 94 OF 101	





GDDR3 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR3_40R55SE	*	=55_OHM_SE	=40_OHM_SE	0.095 MM	12.7 MM	=STANDARD	=STANDARD
GDDR3_40SE	*	=40_OHM_SE	=40_OHM_SE	0.095 MM	=40_OHM_SE	=STANDARD	=STANDARD
GDDR3_80D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.095 MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR3_CLK	*	=2.5:1_SPACING	?
GDDR3_CMD	*	=2.5:1_SPACING	?
GDDR3_DATA	*	=2.5:1_SPACING	?
GDDR3_DQS	*	=2.5:1_SPACING	?

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP, BOTTOM	=4x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?	LVDS	TOP, BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 0.127 mm. Pairs should be within 0.508mm of entire channel.
DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
Max length of LVDS/DisplayPort/TMDS traces: 13 inches.
SOURCE: Catpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.

GDDR3 FB A/B Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	FB_A_CLK	GDDR3_80D	GDDR3_CLK	FB A CLK P<0>
		GDDR3_80D	GDDR3_CLK	FB A CLK N<0>
	FB_B_CLK	GDDR3_80D	GDDR3_CLK	FB A CLK P<1>
		GDDR3_80D	GDDR3_CLK	FB A CLK N<1>
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A MA<1..0>
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A MA<12..6>
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A BA<2..0>
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A RAS_L
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A UCAS_L
	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A WE_L
	FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB A UCKE
	FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB A LCKE
	FB_AB_CS0	GDDR3_40R55SE	GDDR3_CMD	FB A LCS0_L
	FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB A DRAM_RST
	FB_A_CMD	GDDR3_40SF	GDDR3_CMD	FB A LMA<5..2>
	FB_B_CMD	GDDR3_40SF	GDDR3_CMD	FB A UMA<5..2>
	FB_A_WDQS0	GDDR3_40SF	GDDR3_DQS	FB A WDQS<0>
	FB_A_WDQS1	GDDR3_40SF	GDDR3_DQS	FB A WDQS<1>
	FB_A_WDQS2	GDDR3_40SF	GDDR3_DQS	FB A WDQS<2>
	FB_A_WDQS3	GDDR3_40SF	GDDR3_DQS	FB A WDQS<3>
	FB_A_RDQS0	GDDR3_40SF	GDDR3_DQS	FB A RDQS<0>
	FB_A_RDQS1	GDDR3_40SF	GDDR3_DQS	FB A RDQS<1>
	FB_A_RDQS2	GDDR3_40SF	GDDR3_DQS	FB A RDQS<2>
	FB_A_RDQS3	GDDR3_40SF	GDDR3_DQS	FB A RDQS<3>
	FB_A_DQ_BYTE0	GDDR3_40SF	GDDR3_DATA	FB A DQ<7..0>
	FB_A_DQ_BYTE1	GDDR3_40SF	GDDR3_DATA	FB A DQ<15..8>
	FB_A_DQ_BYTE2	GDDR3_40SF	GDDR3_DATA	FB A DQ<23..16>
	FB_A_DQ_BYTE3	GDDR3_40SF	GDDR3_DATA	FB A DQ<31..24>
	FB_A_DQM0	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<0>
	FB_A_DQM1	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<1>
	FB_A_DQM2	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<2>
	FB_A_DQM3	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<3>
	FB_B_WDQS0	GDDR3_40SF	GDDR3_DQS	FB A WDQS<4>
	FB_B_WDQS1	GDDR3_40SF	GDDR3_DQS	FB A WDQS<5>
	FB_B_WDQS2	GDDR3_40SF	GDDR3_DQS	FB A WDQS<6>
	FB_B_WDQS3	GDDR3_40SF	GDDR3_DQS	FB A WDQS<7>
	FB_B_RDQS0	GDDR3_40SF	GDDR3_DQS	FB A RDQS<4>
	FB_B_RDQS1	GDDR3_40SF	GDDR3_DQS	FB A RDQS<5>
	FB_B_RDQS2	GDDR3_40SF	GDDR3_DQS	FB A RDQS<6>
	FB_B_RDQS3	GDDR3_40SF	GDDR3_DQS	FB A RDQS<7>
	FB_B_DQ_BYTE0	GDDR3_40SF	GDDR3_DATA	FB A DQ<39..32>
	FB_B_DQ_BYTE1	GDDR3_40SF	GDDR3_DATA	FB A DQ<47..40>
	FB_B_DQ_BYTE2	GDDR3_40SF	GDDR3_DATA	FB A DQ<55..48>
	FB_B_DQ_BYTE3	GDDR3_40SF	GDDR3_DATA	FB A DQ<63..56>
	FB_B_DQM0	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<4>
	FB_B_DQM1	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<5>
	FB_B_DQM2	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<6>
	FB_B_DQM3	GDDR3_40SF	GDDR3_DATA	FB A DQM_L<7>

GDDR3 FB C/D Net Properties

ELECTRICAL CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACINGS	
	FB_C_CLK	GDDR3_80D	GDDR3_CLK	FB B CLK P<0>
		GDDR3_80D	GDDR3_CLK	FB B CLK N<0>
	FB_D_CLK	GDDR3_80D	GDDR3_CLK	FB B CLK P<1>
		GDDR3_80D	GDDR3_CLK	FB B CLK N<1>
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B MA<1..0>
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B MA<12..6>
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B BA<2..0>
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B RAS_L
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B UCAS_L
	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B WE_L
	FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB B UCKE
	FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB B LCKE
	FB_CD_CS0	GDDR3_40R55SE	GDDR3_CMD	FB B LCS0_L
	FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB B DRAM_RST
	FB_C_CMD	GDDR3_40SE	GDDR3_CMD	FB B LMA<5..2>
	FB_D_CMD	GDDR3_40SE	GDDR3_CMD	FB B UMA<5..2>
	FB_C_WDQS0	GDDR3_40SE	GDDR3_DQS	FB B WDQS<0>
	FB_C_WDQS1	GDDR3_40SE	GDDR3_DQS	FB B WDQS<1>
	FB_C_WDQS2	GDDR3_40SE	GDDR3_DQS	FB B WDQS<2>
	FB_C_WDQS3	GDDR3_40SE	GDDR3_DQS	FB B WDQS<3>
	FB_C_RDQS0	GDDR3_40SE	GDDR3_DQS	FB B RDQS<0>
	FB_C_RDQS1	GDDR3_40SE	GDDR3_DQS	FB B RDQS<1>
	FB_C_RDQS2	GDDR3_40SE	GDDR3_DQS	FB B RDQS<2>
	FB_C_RDQS3	GDDR3_40SE	GDDR3_DQS	FB B RDQS<3>
	FB_C_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB B DQ<7..0>
	FB_C_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB B DQ<15..8>
	FB_C_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB B DQ<23..16>
	FB_C_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB B DQ<31..24>
	FB_C_DQM0	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<0>
	FB_C_DQM1	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<1>
	FB_C_DQM2	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<2>
	FB_C_DQM3	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<3>
	FB_D_WDQS0	GDDR3_40SE	GDDR3_DQS	FB B WDQS<4>
	FB_D_WDQS1	GDDR3_40SE	GDDR3_DQS	FB B WDQS<5>
	FB_D_WDQS2	GDDR3_40SE	GDDR3_DQS	FB B WDQS<6>
	FB_D_WDQS3	GDDR3_40SE	GDDR3_DQS	FB B WDQS<7>
	FB_D_RDQS0	GDDR3_40SE	GDDR3_DQS	FB B RDQS<4>
	FB_D_RDQS1	GDDR3_40SE	GDDR3_DQS	FB B RDQS<5>
	FB_D_RDQS2	GDDR3_40SE	GDDR3_DQS	FB B RDQS<6>
	FB_D_RDQS3	GDDR3_40SE	GDDR3_DQS	FB B RDQS<7>
	FB_D_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB B DQ<39..32>
	FB_D_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB B DQ<47..40>
	FB_D_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB B DQ<55..48>
	FB_D_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB B DQ<63..56>
	FB_D_DQM0	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<4>
	FB_D_DQM1	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<5>
	FB_D_DQM2	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<6>
	FB_D_DQM3	GDDR3_40SE	GDDR3_DATA	FB B DQM_L<7>
IF20	FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB A LCAS_L
IF20	FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB B LCAS_L

MUXGFX Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
R145	LVDS_A_CLK	LVDS_85D	LVDS	LVDS A CLK P	84 87
R149	LVDS_A_CLK	LVDS_85D	LVDS	LVDS A CLK N	84 87
R150	LVDS_A_DATA	LVDS_85D	LVDS	LVDS A DATA P<2..0>	84 87
R156	LVDS_A_DATA	LVDS_85D	LVDS	LVDS A DATA N<2..0>	84 87
R160	LVDS_B_CLK	LVDS_85D	LVDS	LVDS B CLK P	84 87
R163	LVDS_B_CLK	LVDS_85D	LVDS	LVDS B CLK N	84 87
R201	LVDS_B_DATA	LVDS_85D	LVDS	LVDS B DATA P<2..0>	84 87
R206	LVDS_B_DATA	LVDS_85D	LVDS	LVDS B DATA N<2..0>	84 87
R343	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A CLK F P	6 83
R382	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A CLK F N	6 83
R384	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B CLK F P	6 83
R478	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B CLK F N	6 83
R479	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A CLK P	83 84
R531	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A CLK N	83 84
R532	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A DATA P<2..0>	6 83 84
R533	LVDS_85D	LVDS_85D	LVDS	LVDS CONN A DATA N<2..0>	6 83 84
R534	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B CLK P	83 84
R535	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B CLK N	83 84
R536	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B DATA P<2..0>	6 83 84
R537	LVDS_85D	LVDS_85D	LVDS	LVDS CONN B DATA N<2..0>	6 83 84
R545	DP_ML	DP_85D	DISPLAYPORT	DP ML C P<3..0>	85
R546		DP_85D	DISPLAYPORT	DP ML C N<3..0>	85
R550	DP_ML	DP_85D	DISPLAYPORT	DP ML P<3..0>	84 85
R551		DP_85D	DISPLAYPORT	DP ML N<3..0>	84 85
R552	DP_ML	DP_85D	DISPLAYPORT	DP ML CONN P<3..0>	85
R553		DP_85D	DISPLAYPORT	DP ML CONN N<3..0>	85
R554	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_AUX_CH C P	84 85
R555	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_AUX_CH C N	84 85

G96 Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	(CARGO_D0T38)	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M 27 79 80
	CK505_CLK27MSS	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M_SS 79 80
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_P 81 87
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_N 81 87
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA_P<2..0> 81 87
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA_N<2..0> 81 87
1250	LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS_EG_A_DATA_P<3> 80 81
1250	LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS_EG_A_DATA_N<3> 80 81
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA_P<2..0> 81 87
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA_N<2..0> 81 87
1250	LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS_EG_B_DATA_P<3> 80 81
1250	LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS_EG_B_DATA_N<3> 80 81
1250	DP_MI	DP_85D	DISPLAYPORT	DP_EG_ML_P<3..0> 81 84
1250	DP_MI	DP_85D	DISPLAYPORT	DP_EG_ML_N<3..0> 81 84
1250	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EG_AUX_CH_P 81 84
1250	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EG_AUX_CH_N 81 84
1250		DP_85D	DISPLAYPORT	DP_EG_AUX_CH_C_P 84
1250		DP_85D	DISPLAYPORT	DP_EG_AUX_CH_C_N 84

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_I101_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_I101_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 mm OVERRIDE	OVERRIDE	OVERRIDE
USB_85D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

Graphics ,SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_85D	BGA	LVDS_85D
DP_85D	BGA	100_DIFF_BGA
SATA_90D	BGA	100_DIFF_BGA
CLK_PCIE_90D	BGA	100_DIFF_BGA

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.1 MM	6.35 MM		

K18 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
		ENET_180D	ENETCONN	ENETCONN P<3...0>	38
		ENET_180D	ENETCONN	ENETCONN N<3...0>	38
		SATA_90D	SATA	SATA_ODD_R2D_UF_P	42
		SATA_90D	SATA	SATA_ODD_R2D_UF_N	42
		SATA_90D	SATA	SATA_ODD_D2R_UF_P	42
		SATA_90D	SATA	SATA_ODD_D2R_UF_N	42
		SATA_90D	SATA	SATA_HDD_D2R_UF_P	42
		SATA_90D	SATA	SATA_HDD_D2R_UF_N	42
		SATA_90D	SATA	SATA_HDD_R2D_UF_P	42
		SATA_90D	SATA	SATA_HDD_R2D_UF_N	42
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	CPUTHMSNS_D2_P	51
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	CPUTHMSNS_D2_N	51
H287	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	CPU_THERMD_P	9 51
H288	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	CPU_THERMD_N	9 51
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	GPUTHMSNS_D_P	51
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	GPUTHMSNS_D_N	51
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	GPU_TDIODE_P	51 79 88
	SENSE_DIFFEPAIR	THERM_1701_55S	THERM	GPU_TDIODE_N	51 79 88
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	CPUVTTISNS_R_N	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	CPUVTTISNS_R_P	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	CPUVTTIS0_CS_N	50 78
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	CPUVTTIS0_CS_P	50 78
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	DDRISNS_R_N	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	DDRISNS_R_P	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFXIMVP_CS_N	69
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFXIMVP_CS_P	69
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFXIMVP_CS_R_N	58 69
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFXIMVP_CS_R_P	58 69
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFX_ISNS_R_N	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GFX_ISNS_R_P	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GPUISENS_N	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	GPUISENS_P	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_1V5_S3_N	50 67
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_1V5_S3_P	50 67
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_N	99
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_N	99
H240	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_P	99
H241	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_P	99
H242	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_R_N	56
H243	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_AIRPORT_R_P	56
H244	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_CPU_N	49
H245	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_CPU_P	49
H246	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_HDD_N	56
H247	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_HDD_P	56
H248	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_HDD_R_N	56
H249	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_HDD_R_P	56
H250	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_LCDBKLT_N	56
H251	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_LCDBKLT_P	56
H252	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_ODD_N	56
H253	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_ODD_P	56
H254	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_ODD_R_N	56
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_ODD_R_P	56
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_P1V8GPU_N	50
	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_P1V8GPU_P	50
H257	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_P1V8GPU_R_N	50
H258	SENSE_DIFFEPAIR	SENSE_1701_55S	SENSE	ISNS_P1V8GPU_R_P	50

K18 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING	
PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE		PCIE_CLK100M_AP_CONN_P 6 33
	CLK_PCIE_90D	CLK_PCIE		PCIE_CLK100M_AP_CONN_N 6 33
	1T01_DIEFPAIR			CHGR_CSI_R_P 65
	1T01_DIEFPAIR			CHGR_CSI_R_N 65
	1T01_DIEFPAIR			CHGR_CS0_R_P 49 65
	1T01_DIEFPAIR			CHGR_CS0_R_N 49 65
(USB_FXTA)	USB_R50	USB		USB2_EXTM_MUXED_P 43
(USB_FXTA)	USB_R50	USB		USB2_EXTM_MUXED_N 43
(USB_FXTA)	USB_R50	USB		USB2_LT1_P 6 43
(USB_FXTA)	USB_R50	USB		USB2_LT1_N 6 43
	USB_R50	USB		CONN_USB2_BT_P 6 33
	USB_R50	USB		CONN_USB2_BT_N 6 33
	USB_R50	USB		USB_LT2_P 6 43
	USB_R50	USB		USB_LT2_N 6 43
	DP_R50	DISPLAYPORT		DP_IG_AUX_CH_C_P 84
	DP_R50	DISPLAYPORT		DP_IG_AUX_CH_C_N 84
SPK_OUT	DIEFPAIR	AUDIO		SPKRCONN_L_OUT_P 6 61 62
	DIEFPAIR	AUDIO		SPKRCONN_L_OUT_N 6 61 62
SPK_OUT	DIEFPAIR	AUDIO		SPKRCONN_R_OUT_P 6 61 62
	DIEFPAIR	AUDIO		SPKRCONN_R_OUT_N 6 61 62
SPK_OUT	DIEFPAIR	AUDIO		SPKRCONN_S_OUT_P 6 61 62
	DIEFPAIR	AUDIO		SPKRCONN_S_OUT_N 6 61 62
	USB_R50	USB		USB_TPAD_R_P 53
	USB_R50	USB		USB_TPAD_R_N 53
		SB_POWER		PP3V3_S5 6 7 17 18 19 20 21 23 27 31 35 57 66 71 72 73 83 85
		SB_POWER		PP3V3_S0 6 7 17 18 19 20 21 23 50 51 52
		SB_POWER		PP1V5_S3RS0 6 7 28 30 34 37 46 42 46 47 48 50 52 56 57
		GND		GND 89 88 85

SYNC MASTER=K17 REF		SYNC DATE=06/15/2009	
PAGE TITLE			
Project Specific Constraints			
 Apple Inc.		DRAWING NUMBER <SCH NUM>	
		SIZE <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		REVISION <E4LABEL>	
		BRANCH <BRANCH>	
		PAGE 108 OF 132	
		SHEET 99 OF 101	

8	7	6	5	4	3	2	1
K18 Board-Specific Spacing & Physical Constraints							
BOARD LAYERS				BOARD AREAS		BOARD UNITS (ALL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA		MM	15.5.1
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.185 MM	0.095 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL9, ISL10	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL9, ISL10	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL9, ISL10	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL9, ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DEFAULT	*	0.1 MM	?	*	*	BGA	BGA_P1MM
STANDARD	*	=DEFAULT	?	MEM_CLK	*	BGA	BGA_P2MM
BGA_P1MM	*	=DEFAULT	?	CLK_PCIE	*	BGA	BGA_P2MM
BGA_P2MM	*	=DEFAULT	?	CLK_SLOW	*	BGA	BGA_P2MM
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?	2X_DIELECTRIC	*	0.140 MM	?
2:1_SPACING	*	0.2 MM	?	3X_DIELECTRIC	*	0.210 MM	?
2.5:1_SPACING	*	0.25 MM	?	4X_DIELECTRIC	*	0.280 MM	?
3:1_SPACING	*	0.3 MM	?				
4:1_SPACING	*	0.4 MM	?				
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.							
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL9, ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.							
8	7	6	5	4	3	2	1

SYNC MASTER=K17_REF

SYNC DATE=06/15/2009

PCB Rule Definitions

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
<SCH_NUM>

REVISION
<E4LABEL>

BRANCH
<BRANCH>

PAGE
109 OF 132

SHEET
100 OF 101

